CHAPTER 12

Optical phenomena and properties of matter

12.1 Introduction ESCQH

Many people are using solar power as a source of energy for their homes. Solar power can be used to heat water or to supply electricity. Have you ever noticed solar panels on homes and buildings? Have you ever wondered how solar energy is converted to electrical energy? In this chapter, we examine the process that is used to achieve this energy conversion.
See video: 27Z3 at www.everythingscience.co.za

Figure 12.1: The use of solar cells to supply electricity.

Key linked concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Electronic configuration — Physical Sciences, Grade 10, The atom
- Electromagnetic radiation — Physical Sciences, Grade 10, Electromagnetic radiation

12.2 The photoelectric effect ESCQJ

Around the turn of the twentieth century, it was observed by a number of physicists (including Hertz, Thomson and Von Lenard) that when light was shone onto a metal plate, electrons were emitted by the metal. This is called the photoelectric effect. (*photo* for light, *electric* for the electron.)

The characteristics of the photoelectric effect were a surprise and a very important development in modern Physics. To understand why it was a surprise we need to look at the history to understand what physicists were expecting to happen and then understand the implications for Physics going forward.

History and expectations as to the photoelectric effect ESCQK

In 1887, Heinrich Hertz (a German physicist) noticed that ultraviolet light incident on a metal plate could cause sparks. Metals were known to be good conductors of electricity, because the electrons are more able to move. They should be able to be dislodged if energy were added through the incident light. The problem was that different metals requred different minimum frequencies of light.

The expectation at the time was that electrons would be emitted for any frequency of light, after a delay (for low intensities) during which the electrons absorbed sufficient energy to escape from the metal surface. The higher the intensity the shorter the delay was be as they would absorb energy faster. This was based on the idea that light was a wave continuously delivering energy to the electrons.

It is important to remember that higher frequency light corresponds to higher energy.

The next piece of the puzzle came from Philipp Lenard (a Hungarian physicist) in 1902 when he discovered that the maximum velocity with which electrons are ejected by ultraviolet light is entirely independent of the intensity of light.

His expectation was that at high intensities the electrons would absorb more energy and so would have a greater velocity.

A **paradox** existed as the expectations and the observations did not match.

Albert Einstein (a German physicist) solved this paradox by proposing that light is made up of packets of energy called *quanta* (now called photons) which interacted with the electrons in the metal like particles instead of waves. Each incident photon would transfer all its energy to one electron in the metal.

DEFINITION: *The photoelectric effect*

The photoelectric effect is the process whereby an electron is emitted by a substance when light shines on it.

Einstein received the 1921 Nobel Prize for his contribution to understanding the photoelectric effect. His explanation wasn't very popular and took a while to be accepted, in fact, some scientists at the time felt that is was a big mistake.

In the motivation letter for Einstein to be accepted into the Prussian Academy of Science it was specifically mentioned as a mistake:

> *In sum, one can say that there is hardly one among the great problems in which modern physics is so rich to which Einstein has not made a remarkable contribution. That he may sometimes have missed the targeting his speculations, as, for example, in his hypothesis of light-quanta, cannot really be held too much against him, for it is not possible to introduce really new ideas even in the most exact sciences without sometimes taking a risk – A. Pais, "Subtle is the Lord: The Science and the Life of Albert Einstein," New York: Oxford University Press, 1982, p. 382*

Implications of Einstein's model ESCQM

Einstein's model is consistent with the observation that the electrons were emitted immediately when light was shone on the metal and that the *intensity* of the light made *no difference* to the maximum kinetic energy of the emitted electrons.

The energy needed to knock an electron out of the substance is called the **work function** (symbol W_0) of the substance. This is a characteristic of the substance. If the energy of the photon is less than the work function then no electron can be emitted, no matter how many photons strike the substance. We know that the frequency of light is related to the energy, that is why there is a minimum frequency of light that can eject electrons. This minimum frequency we call the cut-off frequency, f_0. For a specific colour of light (i.e. a certain frequency or wavelength), the energy of the photons is given by $E = hf = hc/\lambda$, where h is Planck's constant. This tells us that the $W_0 = hf_0$.

DEFINITION: *Work function*

The minimum energy needed to knock an electron out of a metal is called the **work function** (symbol W_0) of the metal. As it is energy, it measured in joules (J).

Energy is conserved so if the photon has a higher energy than W_0 then the excess energy goes into the kinetic energy E_k of the electron that was emitted from the substance.

The excess over and above the binding energy is actually the **maximum** kinetic energy the emitted electron can have. This is because not all electrons are on the surface of the substance. For electrons below the surface there is additional energy required to eject the electron from the material which then cannot contribute to the kinetic energy of the electron.

$$E = W_0 + E_{k\,\max}$$
$$E_{k\,\max} = hf - W_0$$

This equation is known as the photoelectric equation.

The last piece of the puzzle is now clear, the question was 'why does increasing the intensity of the light not affect the maximum kinetic energy of the emitted photons?'. The answer is that each emitted electron has absorbed **one** photon, increasing the intensity just increases the number of photons (we expect more electrons but we don't expect their maximum kinetic energy to change).

The discovery and understanding of the photoelectric effect was one of the major breakthroughs in science in the twentieth century as it provided concrete evidence of the particle nature of light. It overturned previously held views that light was composed purely of a continuous transverse wave. On the one hand, the wave nature is a good description of phenomena such as diffraction and interference for light, and on the other hand, the photoelectric effect demonstrates the particle nature of light. This is now known as the 'dual-nature' of light. (dual means two)

Einstein won the 1921 Nobel Prize for Physics for this quantum theory and his explanation of the photoelectric effect.

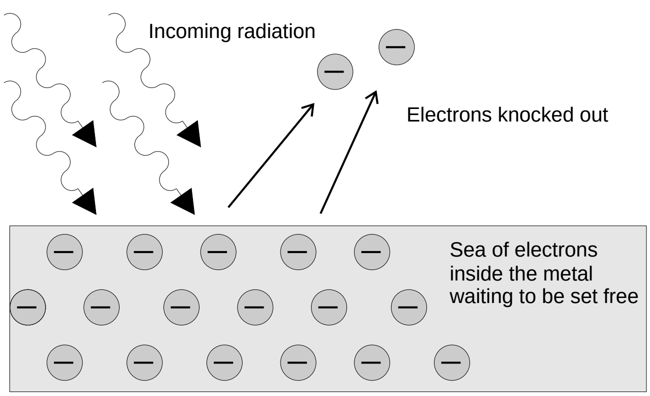


Figure 12.2: The photoelectric effect: Incoming photons on the left hit the electrons inside the metal surface. The electrons absorb the energy from the photons, and are ejected from the metal surface.

We can observe this effect in the following practical demonstration of photoelectric emission. A zinc plate is charged negatively and placed onto the cap of an electroscope. In Figure 12.3, red light is shone onto the zinc plate. There is no change observed even if the intensity (brightness) of the red light is increased. In Figure 12.4, ultraviolet light of low intensity is shone onto the zinc and it is observed that the leaf of the electroscope collapse. This allows us to conclude that the negative charge on the plate decreased as electrons were ejected from the metal when the ultraviolet light was incident on the plate.

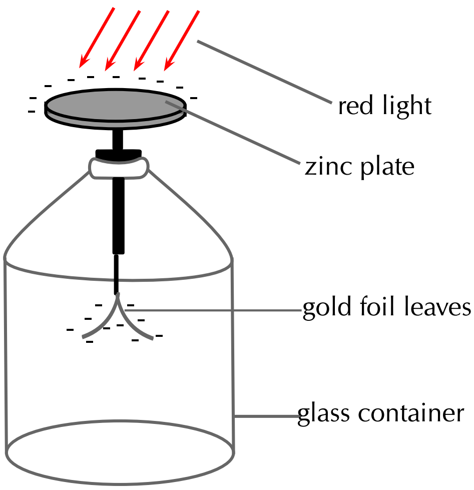


Figure 12.3: Red light incident on the zinc plate of an electroscope.

See video: 27Z4 at www.everythingscience.co.za

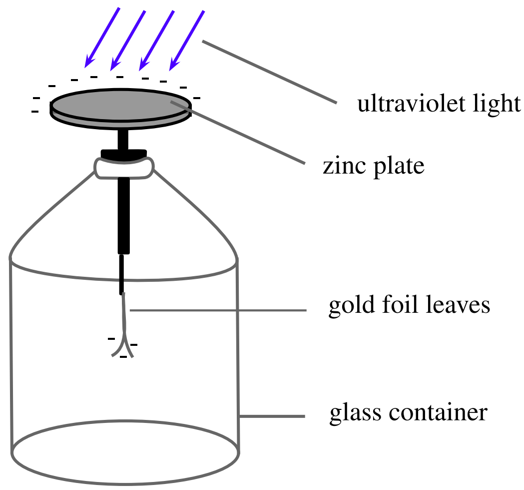


Figure 12.4: Ultraviolet light incident on the zinc plate of an electroscope.

See video: 27Z5 at www.everythingscience.co.za

The work function is different for different elements. The smaller the work function, the easier it is for electrons to be emitted from the metal. Metals with low work functions make good conductors. This is because the electrons are attached less strongly to their surroundings and can move more easily through these materials. This reduces the resistance of the material to the flow of current i.e. it conducts well. Table 12.1 shows the work functions for a range of elements.

Element	Work Function (J)
Aluminium	$6,9 \times 10^{-19}$
Beryllium	$8,0 \times 10^{-19}$
Calcium	$4,6 \times 10^{-19}$
Copper	$7,5 \times 10^{-19}$
Gold	$8,2 \times 10^{-19}$
Lead	$6,9 \times 10^{-19}$
Silicon	$1,8 \times 10^{-19}$
Silver	$6,9 \times 10^{-19}$
Sodium	$3,7 \times 10^{-19}$

Table 12.1: Work functions of selected elements determined from the photoelectric effect. (From the *Handbook of Chemistry and Physics.*)

Units of energy ESCQN

When dealing with calculations at a small scale (like at the level of electrons) it is more convenient to use different units for energy rather than the joule (J). We define a unit called the electron-volt (eV) as the kinetic energy gained by an electron passing through a potential difference of one volt. $E = q \times V$ where q is the charge of the electron and V is the potential difference applied. The charge of 1 electron is $1,6 \times 10^{-19}$ C, so 1 eV is calculated to be: $1\text{ eV} = \left(1{,}6 \times 10^{-19}\text{ C} \times 1\text{ V}\right) = 1{,}6 \times 10^{-19}\text{ J}$. You can see that $1{,}6 \times 10^{-19}$ J is a very small amount of energy and so using electron-volts (eV) at this level is easier. Hence, $1\text{ eV} = 1{,}6 \times 10^{-19}$ J which means that $1\text{ J} = 6{,}241 \times 10^{18}$ eV.

Activity: Demonstration of the photoelectric effect

We can set up an experiment similar to the one used originally to study the photoelectric effect. The experiment allows us to measure the number of electrons emitted and the maximum kinetic energy of the ejected electrons.

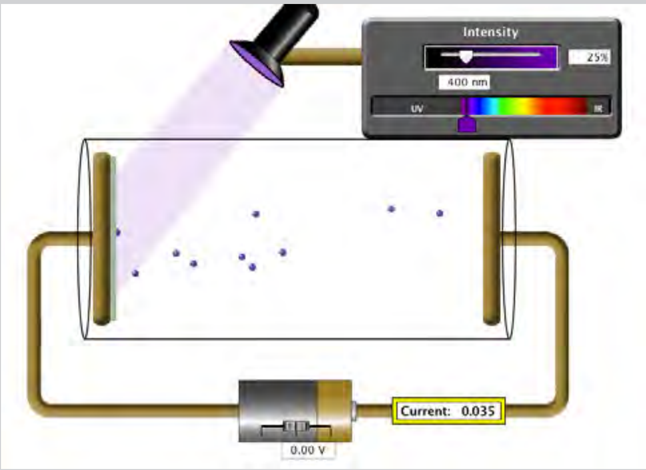


Figure 12.5: Photoelectric effect apparatus

In the diagram of the Photoelectric Effect Apparatus, an ammeter allows for a current to be measured. Measuring the current allows us to deduce information about the number of electrons emitted and the kinetic energy of the ejected electrons.

In the diagram, notice that the potential difference supplied by the battery is zero and yet a current is still measured on the ammeter. This is due to the incoming photons having sufficient frequency and hence energy greater than the work function to eject electrons. The ejected electrons travel across the evacuated space and allow for a current to be measured in the circuit.

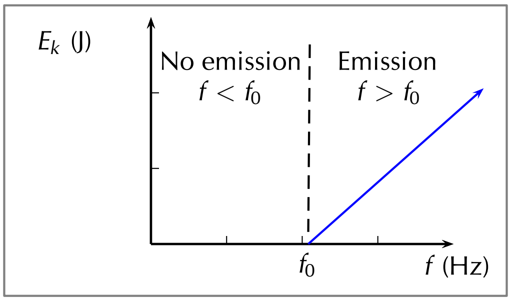
Remember, photon energy is related to frequency while intensity is related to the number of photons.

See simulation: 27Z6 at www.everythingscience.co.za

It is useful to observe the photoelectric effect equation represented graphically. It can be seen from the graph that E_k is plotted on the $y-$axis and f is plotted on the $x-$axis. Using the straight line equation, $y = mx + c$, we can identify

$$E_{k\ \text{max}} = hf - W_0$$

$$\underbrace{E_{k\ \text{max}}}_{y} = h\underbrace{f}_{x} - hf_0$$



This allows us to conclude that the slope of the graph m is Planck's constant h. Also, the x intercept is the cut-off frequency f_0.

Worked example 1: The photoelectric effect using silver

QUESTION

Ultraviolet radiation with a wavelength of 250 nm is incident on a silver foil (work function $W_0 = 6{,}9 \times 10^{-19}$ J). What is the maximum kinetic energy of the emitted electrons?

SOLUTION

Step 1: Determine what is required and how to approach the problem

We need to determine the maximum kinetic energy of an electron ejected from a silver foil by ultraviolet radiation.

The photoelectric effect tells us that:

$$E_{k\ \text{max}} = E_{\text{photon}} - W_0$$

$$E_{k\ \text{max}} = h\frac{c}{\lambda} - W_0$$

We also have:

- Work function of silver: $W_{0\ \text{silver}} = 6{,}9 \times 10^{-19}$ J
- UV radiation wavelength = 250 nm = 250×10^{-9} m = $2{,}50 \times 10^{-7}$ m
- Planck's constant: $h = 6{,}63 \times 10^{-34}\ \text{m}^2\cdot\text{kg}\cdot\text{s}^{-1}$
- Speed of light: $c = 3 \times 10^{8}\ \text{m}\cdot\text{s}^{-1}$

Step 2: Solve the problem

$$\begin{aligned} E_k &= \frac{hc}{\lambda} - W_{0\ \text{silver}} \\ &= \left[6{,}63 \times 10^{-34} \times \frac{3 \times 10^{8}}{2{,}5 \times 10^{-7}}\right] - 6{,}9 \times 10^{-19} \\ &= 1{,}06 \times 10^{-19}\ \text{J} \end{aligned}$$

The maximum kinetic energy of the emitted electron will be $1{,}06 \times 10^{-19}$ J.

Worked example 2: The photoelectric effect using gold

QUESTION

If we were to shine the same ultraviolet radiation ($f = 1{,}2 \times 10^{15}$ Hz) on a gold foil (work function $= 8{,}2 \times 10^{-19}$ J) would any electrons be emitted from the surface of the gold foil?

SOLUTION

Step 1: Calculate the energy of the incident photons

For the electrons to be emitted from the surface, the energy of each photon needs to be *greater* than the work function of the material.

$$\begin{aligned} E_{\text{photon}} &= hf \\ &= 6{,}63 \times 10^{-34} \times 1{,}2 \times 10^{15} \\ &= 7{,}96 \times 10^{-19}\ \text{J} \end{aligned}$$

Therefore each photon of ultraviolet light has an energy of $7{,}96 \times 10^{-19}$ J.

Step 2: Write down the work function for gold.

$$W_{0\ \text{gold}} = 8{,}92 \times 10^{-19}\ \text{J}$$

Step 3: Is the energy of the photons greater or smaller than the work function?

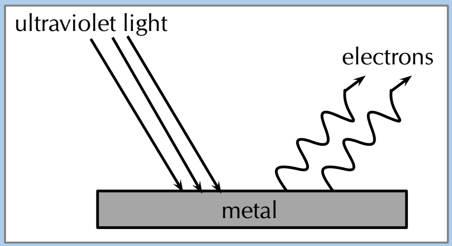
$$\begin{gathered} 7{,}96 \times 10^{-19}\ \text{J} < 8{,}92 \times 10^{-19}\ \text{J} \\ E_{\text{photons}} < W_{0\ \text{gold}} \end{gathered}$$

Since the energy of each photon is less than the work function of gold, the photons do not have enough energy to knock electrons out of the gold. No electrons would be emitted from the gold foil.

Worked example 3: Photoelectric effect [NSC 2011 Paper 1]

QUESTION

A metal surface is illuminated with ultraviolet light of wavelength 330 nm. Electrons are emitted from the metal surface. The minimum amount of energy required to emit an electron from the surface of this metal is $3{,}5 \times 10^{-19}$ J.



1. Name the phenomenon illustrated above. (1 mark)
2. Give ONE word or term for the underlined sentence in the above paragraph. (1 mark)
3. Calculate the frequency of the ultraviolet light. (4 marks)
4. Calculate the kinetic energy of a photoelectron emitted from the surface of the metal when the ultraviolet light shines on it. (4 marks)
5. The intensity of the ultraviolet light illuminating the metal is now increased. What effect will this change have on the following:
   a) Kinetic energy of the emitted photoelectrons (Write down only INCREASES, DECREASES or REMAINS THE SAME.) (1 mark)

b) Number of photoelectrons emitted per second (Write down only INCREASES, DECREASES or REMAINS THE SAME.) (1 mark)

6. Overexposure to sunlight causes damage to skin cells.

a) Which type of radiation in sunlight is said to be primarily responsible for this damage? (1 mark)

b) Name the property of this radiation responsible for the damage. (1 mark)

[TOTAL: 14 marks]

SOLUTION

Question 1: Photo-electric effect (1 mark)

Question 2: Work function (1 mark)

Question 3

$$\begin{aligned} c &= f\lambda \\ 3 \times 10^8 &= f(330 \times 10^{-9}) \\ \therefore f &= 9{,}09 \times 10^{14} \text{ Hz} \end{aligned}$$

(4 marks)

OR

$$\begin{aligned} E &= \frac{hc}{\lambda} \\ &= \frac{(6{,}63 \times 10^{-34})(3 \times 10^8)}{(330 \times 10^{-9})} \\ &= 6{,}03 \times 10^{-19} \text{ J} \end{aligned}$$

$$\begin{aligned} E &= hf \\ 6{,}03 \times 10^{-19} &= (6{,}63 \times 10^{-34})f \\ \therefore f &= 9{,}09 \times 10^{14} \text{ Hz} \end{aligned}$$

Question 4

Option 1:

$$\begin{aligned} E &= W_o + K \\ \frac{hc}{\lambda} &= W_o + K \\ \therefore \frac{(6{,}63 \times 10^{-34})(3 \times 10^8)}{330 \times 10^{-9}} &= 3{,}5 \times 10^{-19} + K \\ \therefore K &= 2{,}53 \times 10^{-19} \text{ J} \end{aligned}$$

Option 2:

$$\begin{aligned} E &= W_o + K \\ hf &= W_o + K \\ \therefore (6{,}63 \times 10^{-34})(9{,}09 \times 10^{14} &= 3{,}5 \times 10^{-19} + K \\ \therefore K &= 2{,}53 \times 10^{-19} \text{ J} \end{aligned}$$

(4 marks)

Question 5.1: Remains the same. (1 mark)

Question 5.2: Increases (1 mark)

Question 6.1: Ultraviolet radiation (1 mark)

Question 6.2: High energy or high frequency (1 mark)

[TOTAL: 12 marks]

Solar cells

The generation of electricity using solar cells isn't due to the photoelectric effect but is similar in nature. Photons in sunlight hit the solar panel and are absorbed by semiconducting materials, such as silicon.

Electrons (negatively charged) are knocked loose from their atoms, allowing them to flow through the material to produce electricity. This is called the photovoltaic effect. The photovoltaic effect was first observed by French physicist Antoine E. Becquerel in 1839.

Due to the special composition of solar cells, the electrons are only allowed to move in a single direction. An array of solar cells converts solar energy into a usable amount of direct current (DC) electricity.

Exercise 12 – 1: The photoelectric effect

1. Describe the photoelectric effect.
2. List two reasons why the observation of the photoelectric effect was significant.
3. Refer to 12.1: If I shine ultraviolet light with a wavelength of 288 nm onto some aluminium foil, what would the kinetic energy of the emitted electrons be?
4. I shine a light of an unknown wavelength onto some silver foil. The light has only enough energy to eject electrons from the silver foil but not enough to give them kinetic energy. (Refer to 12.1 when answering the questions below:)
   a) If I shine the same light onto some copper foil, would electrons be ejected?
   b) If I shine the same light onto some silicon, would electrons be ejected?
   c) If I increase the intensity of the light shining on the silver foil, what happens?
   d) If I increase the frequency of the light shining on the silver foil, what happens?
5. The following results were obtained from a photoelectric effect experiment.

f ($\times 10^{15}$ Hz)	E_k ($\times 10^{-19}$ J)
0.60	0.24
0.80	1.59
1.00	2.89
1.20	4.20
1.40	5.55
1.60	6.89

   a) Plot a graph of E_k on the y axis and f on the x axis
   b) Calculate the gradient of the graph.
   c) The metal used in the experiment is Sodium which has a work function of $3,7 \times 10^{-19}$. Calculate the cut-off frequency for sodium.
   d) Determine the x intercept. Compare your x intercept value with the cut-off frequency you calculated.
   e) If the sodium metal was replaced with another metal with double the work function, sketch on the same graph for sodium, the result you would expect to obtain.

6. More questions. Sign in at Everything Science on-line and click 'Practice Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27Z7 2. 27Z8 3. 27Z9 4. 27ZB 5. 27ZC

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12.3 Emission and absorption spectra ESCQR

Emission spectra ESCQS

You have learnt previously about the structure of an atom. The electrons surrounding the atomic nucleus are arranged in a series of levels of increasing energy. Each element has a unique number of electrons in a unique configuration therefore each element has its own distinct set of energy levels. This arrangement of energy levels serves as the atom's unique fingerprint.

In the early 1900s, scientists found that a liquid or solid heated to high temperatures would give off a broad range of colours of light. However, a gas heated to similar temperatures would emit light only at certain specific wavelengths (colours). The reason for this observation was not understood at the time.

Scientists studied this effect using a discharge tube.

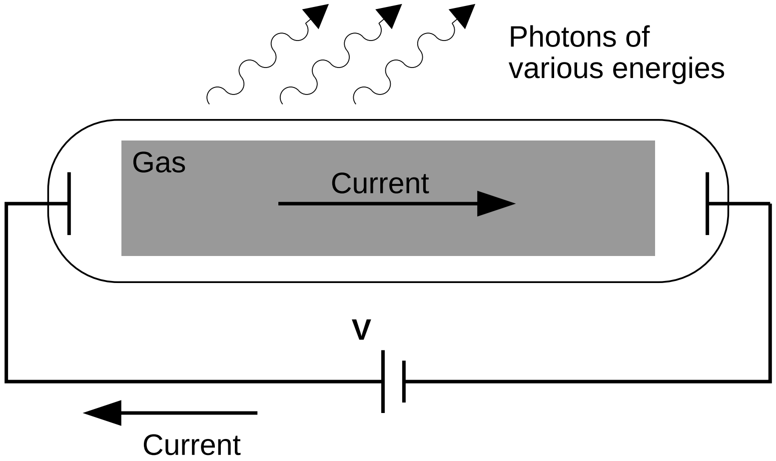


Figure 12.6: Diagram of a discharge tube. The tube is filled with a gas. When a high enough voltage is applied across the tube, the gas ionises and acts like a conductor, allowing a current to flow through the circuit. The current excites the atoms of the ionised gas. When the atoms fall back to their ground state, they emit photons to carry off the excess energy.

A discharge tube (shown in Figure 12.6) is a gas-filled, glass tube with a metal plate at both ends. If a large enough voltage difference is applied between the two metal plates, the gas atoms inside the tube will absorb enough energy to make some of their electrons come off, i.e. the gas atoms are ionised. These electrons start moving through the gas and create a current, which raises some electrons in other atoms to higher energy levels. Then as the electrons in the atoms fall back down, they emit electromagnetic radiation (light). The amount of light emitted at different wavelengths, called the **emission spectrum**, is shown for a discharge tube filled with hydrogen gas in 12.7 below. Only certain wavelengths (i.e. colours) of light are seen, as shown by the lines in the picture.

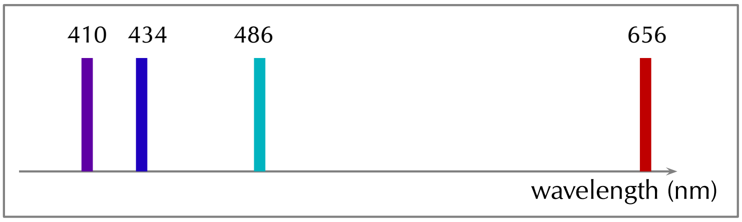
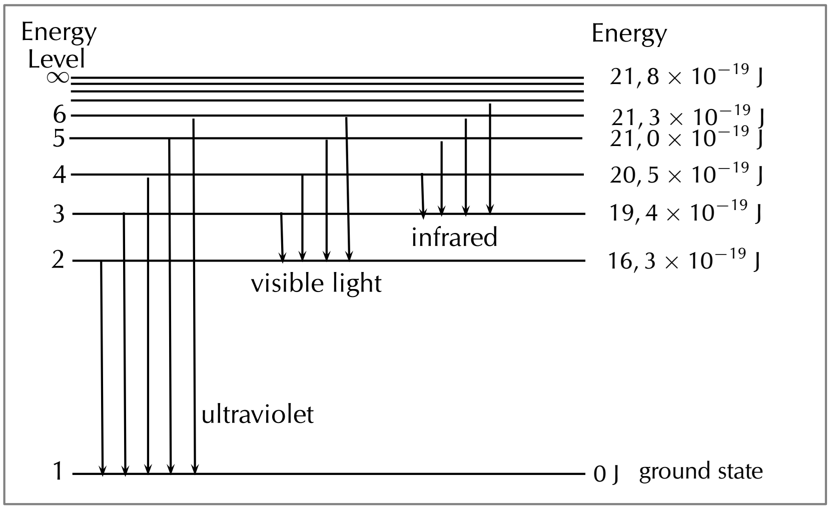


Figure 12.7: Diagram of the emission spectrum of hydrogen in the visible spectrum. Four lines are visible, and are labelled with their wavelengths. The three lines in the 400–500 nm range are in the blue part of the spectrum, while the higher line (656 nm) is in the red/orange part.

Eventually, scientists realised that these lines come from photons of a specific energy, emitted by electrons making transitions between specific energy levels of the atom. Figure 12.8 shows an example of this happening. When an electron in an atom falls from a higher energy level to a lower energy level, it emits a photon to carry off the extra energy. This photon's energy is equal to the energy difference between the two energy levels (ΔE).

$$\Delta E_{\text{electron}} = E_f - E_i$$

As we previously discussed, the frequency of a photon is related to its energy through the equation $E = hf$. Since a specific photon frequency (or wavelength) gives us a specific colour, we can see how each coloured line is associated with a specific transition.



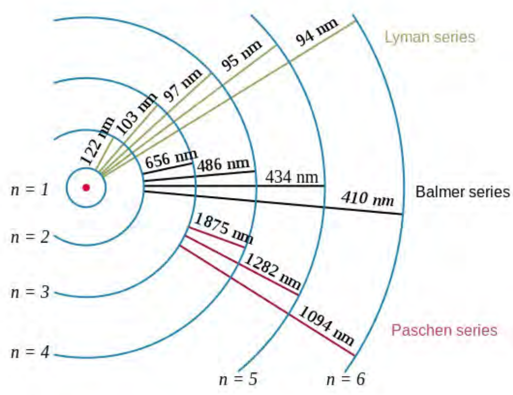


Figure 12.8: In the first diagram are shown some of the electron energy levels for the hydrogen atom. The arrows show the electron transitions from higher energy levels to lower energy levels. The energies of the emitted photons are the same as the energy difference between two energy levels. You can think of absorption as the opposite process. The arrows would point upwards and the electrons would jump up to higher levels when they absorb a photon of the right energy. The second representation shows the wavelengths of the light that is emitted for the the various transitions. The transistions are grouped into a series based on the lowest level involved in the transition.

Visible light is not the only kind of electromagnetic radiation emitted. More energetic or less energetic transitions can produce ultraviolet or infrared radiation. However, because each atom has its own distinct set of energy levels (its fingerprint!), each atom has its own distinct emission spectrum.

Absorption spectra ESCQT

Atoms do not only emit photons; they also absorb photons. If a photon hits an atom and the energy of the photon is the same as the gap between two electron energy levels in the atom, then the electron in the lower energy level can absorb the photon and jump up to the higher energy level. If the photon energy does not correspond to the difference between two energy levels then the photon will not be absorbed (it can still be scattered).

Using this effect, if we have a source of photons of various energies we can obtain the **absorption spectra** for different materials. To get an absorption spectrum, just shine white light on a sample of the material that you are interested in. White light is made up of all the different wavelengths of visible light put together. In the absorption spectrum there will be gaps. The gaps correspond to energies (wavelengths) for which there is a corresponding difference in energy levels for the particular element.

The absorbed photons show up as black lines because the photons of these wavelengths have been absorbed and do not show up. Because of this, the absorption spectrum is the exact *inverse* of the emission spectrum. Look at the two figures below. In Figure 12.9 you can see the line emission spectrum of hydrogen. Figure 12.10 shows the absorption spectrum. It is the exact opposite of the emission spectrum! Both emission and absorption techniques can be used to get the same information about the energy levels of an atom.

Figure 12.9: *Emission* spectrum of Hydrogen.

Figure 12.10: *Absorption* spectrum of Hydrogen.

The dark lines correspond to the frequencies of light that have been absorbed by the gas. As the photons of light are absorbed by electrons, the electrons move into higher energy levels. This is the opposite process of emission.

The dark lines, absorption lines, correspond to the frequencies of the emission spectrum of the same element. The amount of energy absorbed by the electron to move into a higher level is the same as the amount of energy released when returning to the original energy level.

Worked example 4: Absorption

QUESTION

I have an unknown gas in a glass container. I shine a bright white light through one side of the container and measure the spectrum of transmitted light. I notice that there is a black line (*absorption* line) in the middle of the visible red band at 642 nm. I have a hunch that the gas might be hydrogen. If I am correct, between which 2 energy levels does this transition occur? (Hint: look at Figure 12.8 and the transitions which are in the visible part of the spectrum.)

SOLUTION

Step 1: What is given and what needs to be done?

We have an absorption line at 642 nm. This means that the substance in the glass container absorbed photons with a wavelength of 642 nm. We need to calculate which 2 energy levels of hydrogen this transition would correspond to. Therefore we need to know what energy the absorbed photons had.

Step 2: Calculate the energy of the absorbed photons

$$\begin{aligned} E &= \frac{hc}{\lambda} \\ &= \frac{\left(6{,}63 \times 10^{-34}\right) \times \left(3 \times 10^{8}\right)}{642 \times 10^{-9}} \\ &= 3{,}1 \times 10^{-19}\ \text{J} \end{aligned}$$

The absorbed photons had an energy of $3{,}1 \times 10^{-19}$ J.

Step 3: Find the energy of the transitions resulting in radiation at visible wavelengths

Figure 12.8 shows various energy level transitions. The transitions related to visible wavelengths are marked as the transitions beginning or ending on Energy Level 2. Let us find the energy of those transitions and compare with the energy of the absorbed photons we have just calculated.

Energy of transition (absorption) from Energy Level 2 to Energy Level 3:

$$\begin{aligned}\Delta E_{electron} &= E_{2,3} = E_2 - E_3 \\ &= 16{,}3 \times 10^{-19} - 19{,}4 \times 10^{-19} \\ &= -3{,}1 \times 10^{-19}\ \text{J}\end{aligned}$$

Therefore the energy of the photon that an electron must absorb to jump from Energy Level 2 to Energy Level 3 is $3{,}1 \times 10^{-19}$ J. (NOTE: The minus sign means that *absorption* is occurring.)

This is the same energy as the photons which were absorbed by the gas in the container! Therefore, since the transitions of all elements are unique, we can say that the gas in the container is hydrogen. The transition is absorption of a photon between Energy Level 2 and Energy Level 3.

IMPORTANT!

The energy of the photon does not correspond to the energy of an energy level, it corresponds to the **difference** in energy between two energy levels.

Applications of emission and absorption spectra ESCQV

The study of spectra from stars and galaxies in astronomy is i called *spectroscopy*. Spectroscopy is a tool widely used in astronomy to learn different things about astronomical objects.

Identifying elements in astronomical objects using their spectra

Measuring the spectrum of light from a star can tell astronomers what the star is made of. Since each element emits or absorbs light only at particular wavelengths, astronomers can identify what elements are in the stars from the lines in their spectra. From studying the spectra of many stars we know that there are many different types of stars which contain different elements and in different amounts.

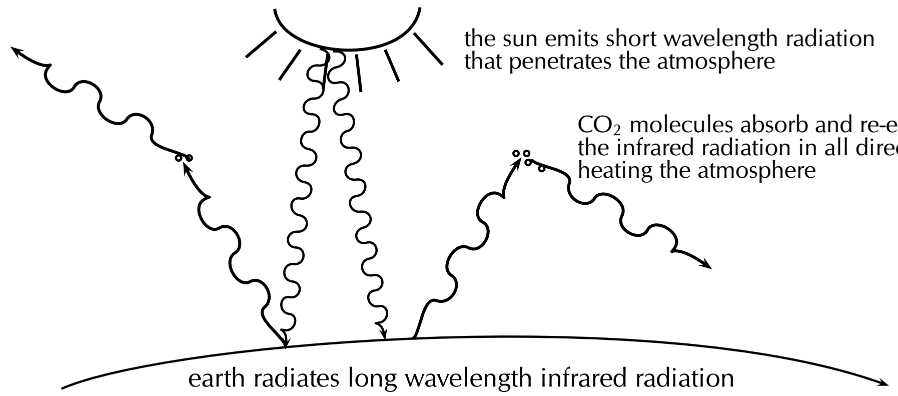
Determining velocities of galaxies using spectroscopy

You have already learnt in Chapter 6 about the Doppler effect and how the frequency (and wavelength) of sound waves changes depending on whether the object emitting the sound is moving *towards* or *away* from you. The same thing happens to electromagnetic radiation (light). If the object emitting the light is moving *towards* us, then the wavelength of the light appears shorter (called **blueshifted**). If the object is moving *away* from us, then the wavelength of its light appears stretched out (called **redshifted**).

The Doppler effect affects the spectra of objects in space depending on their motion relative to us on the earth. For example, the light from a distant galaxy that is moving away from us at some velocity will appear redshifted. This means that the emission and absorption lines in the galaxy's spectrum will be shifted to a longer wavelength (lower frequency). Knowing where each line in the spectrum would normally be if the galaxy was not moving and comparing it to the redshifted position, allows astronomers to precisely measure the velocity of the galaxy relative to the Earth.

Global warming and greenhouse gases

The sun emits radiation (light) over a range of wavelengths that are mainly in the visible part of the spectrum. Radiation at these wavelengths passes through the gases of the atmosphere to warm the land and the oceans below. The warm earth then radiates this heat at longer infrared wavelengths. Carbon dioxide (one of the main greenhouse gases) in the atmosphere has energy levels that correspond to the infrared wavelengths that allow it to absorb the infrared radiation. It then also emits at infrared wavelengths in all directions. This effect stops a large amount of the infrared radiation from getting out of the atmosphere, which causes the atmosphere and the earth to heat up. More radiation is coming in than is getting back out.



So increasing the amount of greenhouse gases in the atmosphere increases the amount of trapped infrared radiation and therefore the overall temperature of the earth. The earth is a very sensitive and complicated system upon which life depends and changing the delicate balances of temperature and atmospheric gas content may have disastrous consequences if we are not careful.

Exercise 12 – 2: Emission and absorption spectra

1. Explain how atomic emission spectra arise and how they relate to each element on the periodic table.
2. How do the lines on the atomic spectrum relate to electron transitions between energy levels?
3. Explain the difference between atomic absorption and emission spectra.
4. Describe how the absorption and emission spectra of the gases in the atmosphere give rise to the Greenhouse Effect.
5. What colour is the light emitted by hydrogen when an electron makes the transition from energy level 5 down to energy level 2? (Use 12.8 to find the energy of the released photon.)
6. I have a glass tube filled with hydrogen gas. I shine white light onto the tube. The spectrum I then measure has an absorption line at a wavelength of 474 nm. Between which two energy levels did the transition occur? (Use 12.8 in solving the problem.)
7. More questions. Sign in at Everything Science on-line and click 'Practice Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27ZD 2. 27ZF 3. 27ZG 4. 27ZH 5. 27ZK
7. 27ZM

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12.4 Chapter summary ESCQW

See presentation: 27ZN at www.everythingscience.co.za

- The photoelectric effect is the process whereby an electron is emitted by a substance when light shines on it.
- A substance has a work function which is the minimum energy needed to emit an electron from the metal. The frequency of light whose photons correspond exactly to the work function is known as the cut-off frequency.

$$E = W_0 + E_{k\ \text{max}}$$
$$E_{k\ \text{max}} = hf - W_0$$

- The number of electrons ejected increases with the intensity of the incident light.
- The photoelectric effect illustrates the particle nature of light and establishes the quantum theory.
- Emission spectra are formed when certain frquencies of light are emitted by a gas, as a result of electrons in the atmoms dropping from higher to lower energy levels. The pattern of the spectra is charactersistic of the specific gas.
- Absorption spectra are formed when certain frequencies of light are absorbed by a material. These photons are absorbed when their energy is exactly the correct amount to raise an electron from one energy level to another.

Physical Quantities		
Quantity	**Unit name**	**Unit symbol**
Energy (E)	joule	J
Work function (W_0)	joule	J
Frequency (f)	hertz	Hz
Wavelength (λ)	metre	m

Table 12.2: Units used in **optical phenomena and properties of matter**.

Exercise 12 – 3:

1. Calculate the energy of a photon of red light with a wavelength of 400 nm.
2. Will ultraviolet light with a wavelength of 990 nm be able to emit electrons from a sheet of calcium with a work function of 2,9 eV?
3. More questions. Sign in at Everything Science on-line and click 'practise'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27ZP 2. 27ZQ

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CHAPTER 13

Electrochemical reactions

We use batteries throughout our day-to-day lives. Cell phones use lithium-ion batteries, cars use lead-acid batteries (Figure 13.1 (left)), while silver-oxide batteries (Figure 13.1 (top right)) are used in watches. Some batteries are rechargeable (Figure 13.1 (bottom right)), while others cannot be recharged and have to be thrown away.

Figure 13.1: A car battery, watch batteries and rechargeable AA batteries.

A battery consists of multiple **electrochemical cells**. And within each cell there are **electrochemical reactions** taking place. This can be seen in the *lemon battery* experiment shown in Figure 13.2. Each lemon is a cell in the battery, which consists of three lemon cells. The reactions the copper and zinc undergo in the lemons are electrochemical reactions, and a current is produced.

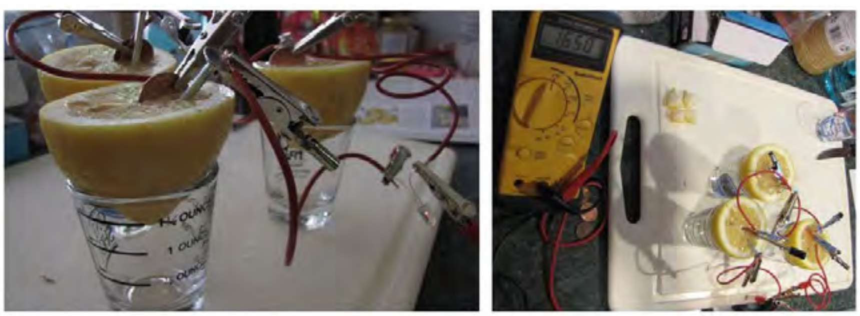

Figure 13.2: A current is produced by connecting lemons with zinc and copper metal. The reactions taking place here are electrochemical reactions.

Electrochemical reactions, and electrochemical cells are covered in this chapter. Before going into any more detail however, it is important to revise oxidation and reduction, as well as redox reactions and how to balance them, as these concepts are very important in electrochemistry.

13.1 Revision of oxidation and reduction ESCQX

You should remember the terms *oxidation* and *reduction* from Grade 11:

- **Oxidation** involves a **loss of electrons**
- **Reduction** involves a **gain of electrons**.

An easy way to remember this is:

In both oxidation and reduction a **transfer of electrons** is involved resulting in a change in the oxidation state of the elements.

OIL
Oxidation **I**s **L**oss

RIG
Reduction **I**s **G**ain

- An element or compound that **loses** electrons is **oxidised**.
  e.g. $\mathbf{Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}}$
  As it loses electrons it *gives* them away to another element or compound and the element or compound it gives the electrons to is **reduced**.
  This makes the compound or element which loses electrons a **reducing agent**.
- An element or compound that **gains** electrons is **reduced**.
  e.g. $\mathbf{Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)}$
  As it gains electrons it *takes* them away from another element or compound and the element or compound it takes them from is **oxidised**.
  This makes the compound or element which gains electrons an **oxidising agent**.

Exercise 13 – 1: Oxidation and reduction

1. Define the following terms:
   a) oxidation b) reduction c) oxidising agent d) reducing agent
2. In each of the following reactions say whether the reactant **iron species** (Fe, Fe^{2+}, Fe^{3+}) is oxidised or reduced.
   a) $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^{-}$
   b) $Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$
   c) $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$
   d) $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^{-}$
   e) $Fe_2O_3(s) + 2Al(s) \rightarrow Al_2O_3(s) + 2Fe(s)$
3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27ZR 2. 27ZS

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13.2 Writing redox and half-reactions ESCQY

Redox reactions and half-reactions ESCQZ

Remember from Grade 11 that oxidation and reduction occur simultaneously in a **redox** reaction. The reactions taking place in electrochemical cells are redox reactions. Two questions should be asked to determine if a reaction is a redox reaction:

- Is there a compound or atom being oxidised?
- Is there a compound or atom being reduced?

If the answer to both of these questions is yes, then the reaction is a redox reaction. For example, this reaction is a redox reaction:

$2Fe^{3+}(aq) + Sn^{2+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{4+}(aq)$

You can write a redox reaction as two **half-reactions**, one showing the reduction process, and one showing the oxidation process. Fe^{3+} is gaining an electron to become Fe^{2+}. **Iron** is therefore being **reduced** and **tin** is the **reducing agent** (causing iron to be reduced). The **reduction half-reaction** is:

$\mathbf{Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)}$

Sn^{2+} is losing two electrons to become Sn^{4+}. **Tin** is therefore being **oxidised** and **iron** is the **oxidising agent** (causing tin to be oxidised). The **oxidation half-reaction** is:

$\mathbf{Sn^{2+}(aq) \rightarrow Sn^{4+}(aq) + 2e^-}$

Notice that in the **overall** reaction the reduction half-reaction is multiplied by two. This is so that the number of electrons gained in the reduction half-reaction match the number of electrons lost in the oxidation half-reaction.

Balancing redox reactions ESCR2

Half-reactions can be used to balance redox reactions. We are going to use some worked examples to help explain the method.

Worked example 1: Balancing redox reactions

QUESTION

Chlorine gas oxidises Fe^{2+} ions to Fe^{3+} ions. In the process, chlorine is reduced to chloride ions. Write a balanced equation for this reaction.

SOLUTION

Step 1: Write down the unbalanced oxidation half-reaction

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq)$

Step 2: Balance the number of atoms on both sides of the equation

There is one iron atom on the left and one on the right, so no additional atoms need to be added.

Step 3: Once the atoms are balanced, check that the *charges* balance

The charge on the left of the equation is $+2$, but the charge on the right is $+3$. Therefore, one electron must be added to the right hand side so that the charges balance. The half-reaction is now:

$Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

Step 4: Repeat steps 1 - 3 with the reduction half-reaction

The unbalanced reduction half-reaction is:

$Cl_2(g) \rightarrow Cl^-(aq)$

The atoms don't balance, so we need to multiply the right hand side by two to fix this.

$Cl_2(g) \rightarrow 2Cl^-(aq)$

Two electrons must be added to the left hand side to balance the charges.

$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$

Step 5: Compare the number of electrons in each equation

Multiply each half-reaction by a suitable number so that the number of electrons released (oxidation) is equal to the number of electrons accepted (reduction).

oxidation half-reaction: $\times 2$: $2Fe^{2+}(aq) \rightarrow 2Fe^{3+}(aq) + \mathbf{2e^-}$
reduction half-reaction: $\times 1$: $Cl_2(g) + \mathbf{2e^-} \rightarrow 2Cl^-(aq)$

Step 6: Combine the two half-reactions to get a final equation for the overall reaction

$2Fe^{2+}(aq) + Cl_2(g) \rightarrow 2Fe^{3+}(aq) + 2Cl^-(aq)$

Step 7: Do a final check to make sure that the equation is balanced

We check the number of atoms and the charges and find that the equation is balanced.

Worked example 2: Balancing redox reactions in an acid medium

QUESTION

The following reaction takes place in an acid medium:

$Cr_2O_7^{2-}(aq) + H_2S(g) \rightarrow Cr^{3+}(aq) + S(s)$

Write a balanced equation for this reaction.

SOLUTION

Step 1: Write down the unbalanced reduction half-reaction

In $Cr_2O_7^{2-}$ chromium exists as Cr^{6+}. It becomes Cr^{3+}. Therefore electrons are gained, this is the reduction half-reaction:

$$Cr_2O_7^{2-}(aq) \rightarrow Cr^{3+}(aq)$$

Step 2: Balance the number of atoms on both sides of the equation

We need to multiply the right side by two so that the number of Cr atoms will balance. In an acid medium there are water molecules and H^+ ions in the solution, so these can be used to balance the equation.

To balance the oxygen atoms, we will need to add water molecules to the right hand side:

$$Cr_2O_7^{2-}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

Now the oxygen atoms balance but the hydrogens don't. Because the reaction takes place in an acid medium, we can add hydrogen ions to the left side.

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

Step 3: Once the atoms are balanced, check that the *charges* balance

The charge on the left of the equation is $(-2 + 14) = +12$, but the charge on the right is $+6$. Therefore, six electrons must be added to the left hand side so that the charges balance. This makes sense as electrons are gained in the reduction half-reaction. The half-reaction is now:

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$$

Step 4: Repeat steps 1 - 3 with the oxidation half-reaction

The unbalanced oxidation half-reaction is: $H_2S(g) \rightarrow S(s)$

However, you can ignore the H^+ in H_2S as they are accounted for by the acid medium and the reduction half-reaction. $S^{2-}(aq) \rightarrow S(s)$

The atoms balance, however the charges do not. Two electrons must be added to the right hand side of the equation. $S^{2-}(aq) \rightarrow S(s) + 2e^-$

Step 5: Compare the number of electrons in each equation

Multiply each half-reaction by a suitable number so that the number of electrons released (oxidation) is equal to the number of electrons accepted (reduction).

oxidation half-reaction: $\times\mathbf{3}$: $3S^{2-}(aq) \rightarrow 3S(s) + \mathbf{6e^-}$

reduction half-reaction: $\times\mathbf{1}$: $Cr_2O_7^{2-}(aq) + 14H^+(aq) + \mathbf{6e^-} \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$

Step 6: Combine the two half-reactions to get a final equation for the overall reaction

$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 3S^{2-}(aq) \rightarrow 3S(s) + 2Cr^{3+}(aq) + 7H_2O(\ell)$

However, $H_2S(g)$ is the reactant, so it would be better to write:

$Cr_2O_7^{2-}(aq) + 8H^+(aq) + 3H_2S(g) \rightarrow 3S(s) + 2Cr^{3+}(aq) + 7H_2O(\ell)$

Step 7: Do a final check to make sure that the equation is balanced

We check the number of atoms and the charges and find that the equation is balanced.

Worked example 3: Balancing redox reactions in an alkaline medium

QUESTION

The complex ion hexaamminecobalt(II) ($Co(NH_3)_6^{2+}$) is oxidised by hydrogen peroxide to form the hexaamminecobalt(III) ion ($Co(NH_3)_6^{3+}$). Write a balanced equation for this reaction.

SOLUTION

Step 1: Write down the unbalanced oxidation half-reaction

Ammonia (NH_3) has an oxidation number of 0. Therefore, in $Co(NH_3)_6^{2+}$ cobalt exists as Co^{2+}. In $Co(NH_3)_6^{3+}$ cobalt exists as Co^{3+}. Electrons are lost and this is the oxidation half-reaction:

$$Co^{2+}(aq) \rightarrow Co^{3+}(aq)$$

Step 2: Balance the number of atoms on both sides of the equation

The number of atoms are the same on both sides.

Step 3: Once the atoms are balanced, check that the *charges* balance

The charge on the left of the equation is $+2$, but the charge on the right is $+3$. One electron must be added to the right hand side to balance the charges in the equation:

$$Co^{2+} \rightarrow Co^{3+} + e^-$$

Step 4: Repeat steps 1 - 3 with the reduction half-reaction

Cobalt is oxidised by hydrogen peroxide, therefore hydrogen peroxide is reduced. Reduction means a gain of electrons. The product of the reduction of H_2O_2 in an alkaline medium is OH^-:

$$H_2O_2(\ell) \rightarrow OH^-(aq)$$

Next you need to balance the atoms: $H_2O_2(\ell) \rightarrow 2OH^-(aq)$

Then you need to balance the charges: $H_2O_2(\ell) + 2e^- \rightarrow 2OH^-(aq)$

Step 5: Compare the number of electrons in each equation

Multiply each half-reaction by a suitable number so that the number of electrons released (oxidation) is equal to the number of electrons accepted (reduction):

oxidation half-reaction: $\times\mathbf{2}$: $2Co^{2+}(aq) \rightarrow 2Co^{3+}(aq) + \mathbf{2e^-}$

reduction half-reaction: $\times\mathbf{1}$: $H_2O_2(\ell) + \mathbf{2e^-} \rightarrow 2OH^-(aq)$

Step 6: Combine the two half-reactions, and add in the spectator ions, to get a final equation for the overall reaction

$2Co(NH_3)_6^{2+}(aq) + H_2O_2(\ell) \rightarrow 2Co(NH_3)_6^{3+}(aq) + 2OH^-(aq)$

Step 7: Do a final check to make sure that the equation is balanced

We check the number of atoms and the charges and find that the equation is balanced.

Exercise 13 – 2: Balancing redox reactions

1. Balance the following equations:
   a) $HNO_3(\ell) + PbS(s) \rightarrow PbSO_4(s) + NO_2(g) + H_2O(\ell)$
   b) $NaI(aq) + Fe_2(SO_4)_3(aq) \rightarrow I_2(s) + FeSO_4(aq) + Na_2SO_4(aq)$
2. Permanganate(VII) ions (MnO_4^-) oxidise hydrogen peroxide (H_2O_2) to oxygen gas. The reaction is done in an acid medium. During the reaction, the permanganate(VII) ions are reduced to manganese(II) ions (Mn^{2+}). Write a balanced equation for the reaction.
3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1a. 27ZT 1b. 27ZV 2. 27ZW

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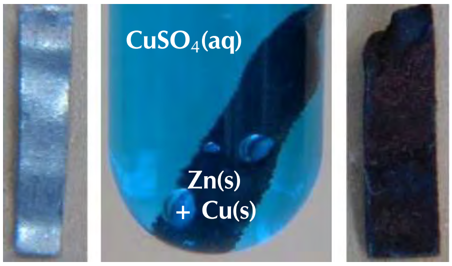
13.3 Galvanic and electrolytic cells ESCR3

Electrochemical reactions ESCR4

In Grade 11, you carried out an experiment to see what happens when zinc granules are added to a solution of copper(II) sulfate.

Zn(s) before



Zn(s) + Cu(s) layer after

Figure 13.3: When a sheet of zinc is placed in an aqueous solution of copper(II) sulfate, solid copper forms on the zinc sheet. (Screenshots taken from a video by Aaron Huggard on Youtube)

See video: 27ZX at www.everythingscience.co.za

In the experiment, the Cu^{2+} ions from the **blue copper(II) sulfate** solution were reduced (gained electrons) to copper metal, which was then deposited as a layer on the solid zinc. The zinc atoms were oxidised (lost electrons) to form Zn^{2+} ions in the solution. $Zn^{2+}(aq)$ is colourless, therefore the blue solution lost colour. As discussed in Grade 11, the half-reactions are as follows:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ (reduction half-reaction)

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ (oxidation half-reaction)

The overall redox reaction is: $Cu^{2+}(aq) + Zn(s) \rightarrow Cu(s) + Zn^{2+}(aq)$

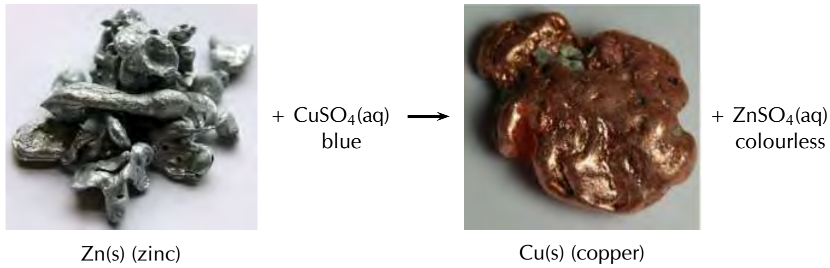


Figure 13.4: Solid zinc loses two electrons to form zinc ions (Zn^{2+}) in an aqueous solution of copper(II) sulfate. The copper ions (Cu^{2+}) gain two electrons and deposit as solid copper. (Photos by benjah-bmm27 and Jurii on wikipedia)

Remember that there was an increase in the temperature of the reaction when you carried out this experiment (it was exothermic). An exothermic reaction **releases energy**. This raises a few questions:

- Is it possible that this heat energy could be converted into electrical energy?
- Can we use a chemical reaction with an exchange of electrons, to produce electricity?
- If we supplied an electrical current could we *cause* some type of chemical reaction to take place?

The answers to these questions are the focus of this chapter:

- The energy of a chemical reaction can be converted to electrical potential energy, which forms an electric current.
- The transfer of electrons in a chemical reaction can cause electrical current to flow.
- If you supply an electric current it can cause a chemical reaction to take place, by supplying the electrons (and potential energy) necessary for the reactions taking place within the cell.

These types of reactions are called **electrochemical reactions**. An *electrochemical reaction* is a reaction where:

- a chemical reaction creates an *electrical potential difference*, and therefore an electric current in the external conducting wires

  or
- an electric current provides electrical potential energy and electrons, and therefore a *chemical reaction takes place*

DEFINITION: *Electrochemical reaction*

An electrochemical reaction involves a transfer of electrons. There is a conversion of chemical potential energy to electrical potential energy, **or** electrical potential energy to chemical potential energy.

Electrochemistry is the branch of chemistry that studies these electrochemical reactions. An **electrochemical cell** is a device in which electrochemical reactions take place.

DEFINITION: *Electrochemical cell*

A device where electrochemical reactions take place.

Exercise 13 – 3: Electrochemical reactions

1. In each of the following equations, say which elements in the reactants are oxidised and which are reduced.
   a) $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(g)$
   b) $2NO(g) + 2CO(g) \rightarrow N_2(g) + 2CO_2(g)$
   c) $Mg(s) + FeSO_4(aq) \rightarrow MgSO_4(aq) + Fe(s)$
   d) $Zn(s) + 2AgNO_3(aq) \rightarrow 2Ag(s) + Zn(NO_3)_2(aq)$
2. Which one of the substances listed below acts as the oxidising agent in the following reaction?
   $3SO_2(g) + Cr_2O_7^{2-}(aq) + 2H^+(aq) \rightarrow 3SO_4^{2-}(aq) + 2Cr^{3+}(aq) + H_2O(\ell)$
   a) H^+ b) Cr^{3+} c) SO_2 d) $Cr_2O_7^{2-}$
3. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 27ZY   2. 27ZZ

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TIP
Oxidation is the **loss of electrons**.
Reduction is the **gain of electrons**.

There are two types of electrochemical cells we will be looking more closely in this chapter: **galvanic** and **electrolytic cells**. Before we go into detail on galvanic and electrolytic cells you'll need to know a few definitions:

DEFINITION: *Electrode*

An electrode is an electrical conductor that connects the electrochemical species from its solution to the external electrical circuit of the cell.

There are two types of electrodes in an electrochemical cell, the **anode** and the **cathode**.

Oxidation always occurs at the **anode** while reduction always occurs at the **cathode**. So when trying to determine which electrode you are looking at first determine whether oxidation or reduction is occurring there. An easy way to remember this is:

An Ox
Oxidation is **L**oss at the **A**node

Red Cat
Reduction is gain at the **Cat**hode

Oxidation **is** **l**oss of electrons	**OIL**
Reduction **is** **g**ain of electrons	**RIG**
Oxidation is **l**oss of electrons at the **a**node	**An Ox**
Reduction is gain of electrons at the **cat**hode	**Red Cat**

Table 13.1: A summary of phrases to help you remember the oxidation and reduction rules.

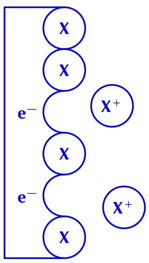
The electrode is placed in an **electrolyte** solution within the cell. If the cell is made up of two compartments, those compartments will be connected by a **salt bridge**.

DEFINITION: *Electrolyte*

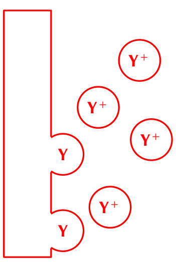
An electrolyte is a solution that contains free ions, and which therefore behaves as a conductor of charges (electrical conductor) in solution.

TIP
Remember that cells are not 2-D, although when asked to sketch a cell you should draw it as shown in Figure 13.5 or Figure 13.6.

TIP
The electrons released in the oxidation of X remain on the anode, X^+ moves into solution.



The Y^+ ions are being reduced at the cathode and forming solid Y.



DEFINITION: *Salt bridge*

A salt bridge is a material which contains electrolytic solution and acts as a connection between two half-cells (completes the circuit). It maintains electrical neutrality in and between the electrolytes in the half-cell compartments.

Galvanic cells ESCR5

A **galvanic cell** (which is also sometimes referred to as a **voltaic** or **wet** cell) consists of two half-cells, which convert chemical potential energy into electrical potential energy.

DEFINITION: *Galvanic cell*

A galvanic cell is an electrochemical cell which converts chemical potential energy to electrical potential energy through a spontaneous chemical reaction.

In a galvanic cell there are two half-cells. Each half-cell contains an electrode in an electrolyte. The separation is necessary to prevent direct chemical contact of the oxidation and reduction reactions, creating a potential difference. The electrons released in the oxidation reaction travel through an external circuit (and do work) before being used by the reduction reaction.

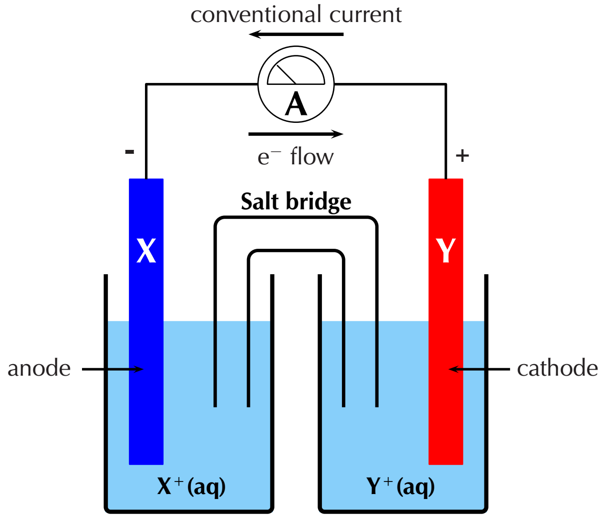


Figure 13.5: A sketch of a galvanic cell.

In a galvanic cell (for example the cell shown in Figure 13.5):

- The metal at the anode is **X**. **Ox**idation is loss of electrons at the **An**ode.
  - The **anode half-reaction** is $\mathbf{X(s) \rightarrow X^+(aq) + e^-}$
  - This half-reaction occurs in the half-cell containing the X(s) anode and the X^+(aq) electrolyte solution.
  - The electrons released in the **oxidation** of the metal remain on the **anode**, while the metal cations formed move into solution.
- The metal at the cathode is Y. **Red**uction is gain of electrons at the **Cat**hode.
  - The **cathode half-reaction** is $\mathbf{Y^+(aq) + e^- \rightarrow Y(s)}$
  - This half-reaction occurs in the half-cell containing the Y(s) cathode and the Y^+(aq) electrolyte solution.
  - At the **cathode** metal ions in the solution are being **reduced** (accepting electrons) and deposited on the electrode.
- There are **more electrons** at the anode than at the cathode.
- Electrons will flow from areas of high concentration to areas of low concentration, therefore the electrons move **from the anode**, through the external circuit, **to the cathode**
- Conventional current is measured as a flow of positive charge and so is in the opposite direction (from the cathode to the anode)

- The overall reaction is: **X**(**s**) + $\mathbf{Y}^+$(**aq**) → $\mathbf{X}^+$(**aq**) + **Y**(**s**).
- To represent this reaction using **standard cell notation** we write the following:
  X(s)|$\mathbf{X}^+$(aq)||$\mathbf{Y}^+$(aq)|Y(s)
  By convention:
  - The **anode** is always written on the **left**.
  - The **cathode** is always written on the **right**.
  - The anode and cathode half-cells are divided by || representing the salt bridge.
  - The different phases within each half-cell (solid (s) and aqueous (aq) here) are separated by |.
- The electrodes in each half-cell are connected through a wire in the external circuit. There is also a salt bridge between the individual half-cells.

FACT
A spontaneous reaction is one that will occur without the need for external energy. Refer to the subsection on spontaneity for more information.

A galvanic cell uses the reactions that take place at at the two electrodes to produce electrical energy, i.e. the reaction occurs without the need to add energy.

The zinc-copper reaction you performed in Grade 11 can be modified to make a galvanic cell. Bars of zinc and copper are used as **electrodes**, with zinc(II) sulfate and copper(II) sulfate solutions as the **electrolytes**.

Experiment: A galvanic cell

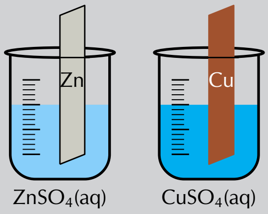
Aim:
To investigate the reactions that take place in a galvanic cell.

Apparatus:

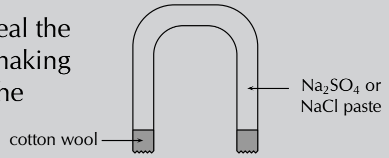
- Zinc plate, copper plate, zinc(II) sulfate ($ZnSO_4$) solution (1 mol.dm^{-3}), copper(II) sulfate ($CuSO_4$) solution (1 mol.dm^{-3}), NaCl paste
- Measuring balance, two 250 ml beakers, U-tube, cotton wool, zero-centered ammeter, connecting wire.
- Cleaning ethanol, ether (if available)

Method:

1. Weigh the copper and zinc plates and record their mass.
2. Pour 200 ml of the zinc sulfate solution into a beaker and place the zinc plate in the beaker.
3. Pour 200 ml of the copper(II) sulfate solution into the second beaker and place the copper plate in the beaker.

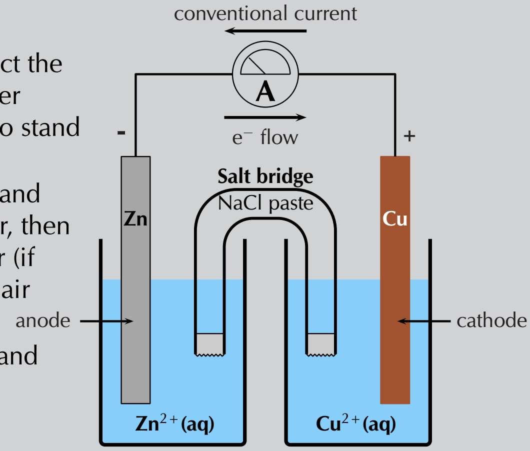


4. Fill the U-tube with the NaCl paste and seal the ends of the tubes with the cotton wool (making a salt-bridge). The cotton will help stop the paste from dissolving in the electrolyte.



5. Connect the zinc and copper plates to the zero-centered ammeter and observe the ammeter.
6. Place the U-tube so that one end is in the copper(II) sulfate solution and the other end is in the zinc sulfate solution. Observe the ammeter.

7. Take the ammeter away and connect the copper and zinc plates to each other directly using copper wire. Leave to stand for about one day.
8. After a day, remove the two plates and rinse them: first with distilled water, then with alcohol, and finally with ether (if available). Dry the plates using a hair dryer.
9. Weigh the zinc and copper plates and record their mass.



Note:

A voltmeter can also be used in place of the zero-centered ammeter. A zero-centered voltmeter will measure the potential difference across the cell (not the flow of electrons), while an ammeter will measure the current.

Discussion:

- Did the ammeter record a reading before the salt-bridge was placed in the solutions?
- Did the ammeter record a reading after the salt-bridge was placed in the solutions? If yes, in what direction does the current flow?
- Fill in the table below:

Plate	Initial mass	Final mass
Zinc		
Copper		

- How did the mass of the zinc and copper plates change?
- Based on what you know of oxidation and reduction, why did those mass changes take place?
- Which electrode is the anode and which is the cathode?

Results:

During the experiment, you should have noticed the following:

- When the salt bridge was absent, there was no reading on the ammeter.
- When the salt bridge was connected, a reading was recorded on the ammeter.
- The direction of electron flow is from the zinc plate towards the copper plate, meaning that conventional current flow is from the copper plate towards the zinc plate.
- After the plates had been connected directly to each other and left for a day, there was a change in their mass. The mass of the zinc plate decreased, while the mass of the copper plate increased.
- **O**xidation **i**s **l**oss of electrons, **R**eduction **i**s **G**ain of electrons.

  The zinc electrode lost mass. This implies that solid Zn metal atoms become ions and move into the electrolyte solution: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$. Oxidation occurs at the zinc electrode.

  The copper electrode gained mass. This implies that the Cu metal ions in the electrolyte solution become metal atoms and deposit on the electrode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$. Reduction occurs at the copper electrode.

- **Ox**idation is loss of electrons at the **an**ode. Oxidation occurs at the zinc electrode, therefore the zinc plate is the anode.

  Reduction is gain of electrons at the **cat**hode. Reduction occurs at the copper electrode, therefore the copper plate is the cathode.
- When $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}$ the electrons are deposited on the anode, which becomes negatively charged.

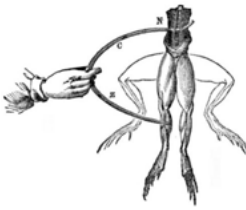
  When $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ the electrons are taken from the cathode, which becomes positively charged.

Conclusions:

When a zinc(II) sulfate solution containing a zinc plate is connected by a salt bridge to a copper(II) sulfate solution containing a copper plate, reactions occur in both solutions. The decrease in mass of the zinc plate suggests that the zinc metal electrode has been oxidised to form Zn^{2+} ions in solution. The increase in mass of the copper plate suggests that reduction of Cu^{2+} ions has occurred here to produce more copper metal.

FACT

It was the Italian physician and anatomist Luigi Galvani who marked the birth of electrochemistry by making a link between chemical reactions and electricity. In 1780, Galvani discovered that when two different metals (copper and zinc for example) were connected to each other and then both touched to different parts of a nerve of a frog leg at the same time, they made the leg contract. He called this 'animal electricity'.



The important thing to notice in this experiment is that:

- the chemical reactions that take place at the two electrodes cause an electric current to flow through the external circuit
- the overall reaction must be a spontaneous redox reaction
- **chemical energy** is converted to **electrical energy**
- the zinc-copper cell is one example of a **galvanic cell**

In the zinc-copper cell, the copper and zinc plates are the **electrodes**. The salt bridge plays a very important role in a galvanic cell:

- An electrolyte solution consists of metal cations and spectator anions.

  NaCl(aq) is $Na^{+}(aq)$ and $Cl^{-}(aq)$ in the paste.
- There is a build up of positive charge in the **anode half-cell compartment** as solid metal is **oxidised** and the positive ions move into solution. So there are more **postive metal ions** in the electrolyte than negative ions.

  $\mathbf{Zn(s)} \rightarrow \mathbf{Zn^{2+}(aq)}$, while the number of SO_4^{2-} ions remains the same.
- To balance the charge, negative ions from the salt bridge move into the anode half-cell compartment.

  Cl^{-} ions move from the salt bridge to the anode half-cell compartment.
- There is a decrease in positive charge in the **cathode half-cell compartment** as metal ions are **reduced** and form solid metal. So there are more **negative ions** in the electrolyte than positive metal ions.

  $\mathbf{Cu^{2+}}(aq) \rightarrow \mathbf{Cu(s)}$, while the number of SO_4^{2-} ions remains the same.
- To balance the charge, positive ions from the salt bridge move into the cathode half-cell compartment.

  Na^{+} ions move from the salt bridge to the cathode half-cell compartment.

The salt bridge acts as a transfer medium that allows ions to flow through without allowing the different solutions to mix and react directly. It allows a balancing of the charges in the electrolyte solutions, and allows the reactions in the cell to continue.

Without the salt bridge, the flow of electrons in the outer circuit stops completely. This is because the salt bridge is needed to complete the circuit.

FACT
Sometimes galvanic cells are just called electrochemical cells. While they are electrochemical cells, electrolytic cells are also electrochemical cells. Electrolytic and galvanic cells are not the same however.

Electrolytic cells ESCR6

In an **electrolytic cell** electrical potential energy is converted to chemical potential energy. An electrolytic cell uses an electric current to force a particular chemical reaction to occur, which would otherwise not take place.

DEFINITION: *Electrolytic cell*

An electrolytic cell is an electrochemical cell that converts electrical potential energy to chemical potential energy by using electricity to drive a non-spontaneous chemical reaction.

An electrolytic cell is activated by applying an electrical potential across the electrodes to force an internal chemical reaction between the electrodes and the ions that are in the electrolyte solution. This process is called **electrolysis**.

DEFINITION: *Electrolysis*

Electrolysis is a method of driving chemical reactions by passing an electric current through an electrolyte.

In an electrolytic cell (for example the cell shown in Figure 13.6):

- The electrolyte solution consists of the metal cations and spectator anions.
- The oxidation and reduction reactions occur in the same container but are non-spontaneous.They require the electrodes to be connected to an external power source to proceed.
- The electrodes in an electrolytic cell can be the same metal or different metals. The prinicple is the same. Let there be only one metal, and let that metal be Z.

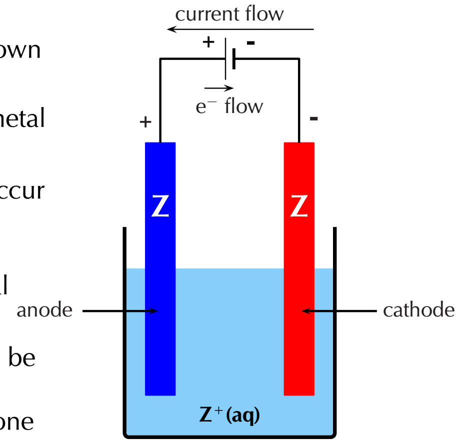


Figure 13.6: A sketch of an electrolytic cell.

- An electrode is connected to the **positive terminal** of the battery.
  - To balance the charge at the **positive electrode** metal atoms are **oxidised** to form metal ions. The ions move into solution, leaving their electrons on the electrode.
  - The following reaction takes place: $\textbf{Z(s)} \rightarrow \textbf{Z}^{+}\textbf{(aq)} + \textbf{e}^{-}$
  - **Ox**idation is loss at the **an**ode, therefore this electrode is the **anode**.
- An electrode is connected to the **negative terminal** of the battery.
  - When positive ions come in contact with the **negative electrode** the ions gain electrons and are **reduced**.
  - The following reaction takes place: $\textbf{Z}^{+}\textbf{(aq)} + \textbf{e}^{-} \rightarrow \textbf{Z(s)}$
  - **Red**uction is gain at the **cat**hode, therefore this electrode is the **cathode**.
- This means that the overall reaction is: $Z(s) + Z^{+}(aq) \rightarrow Z^{+}(aq) + Z(s)$. While this might seem trivial this is an important technique to purify metals (see Section 13.7).

Experiment: The movement of coloured ions under the effect of electrical charge

Aim:

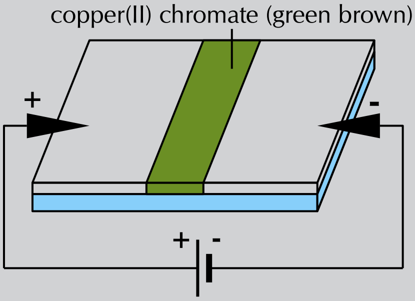
To demonstrate how ions migrate in solution towards oppositely charged electrodes.

Apparatus:

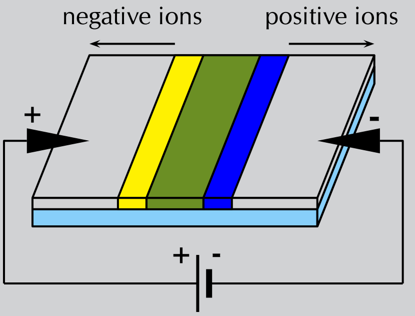
- Filter paper, glass slide, a 9V battery, two crocodile clips connected to wires, tape
- Ammonia ($NH_3(aq)$) and ammonium chloride (NH_4Cl) buffer solution, copper(II) chromate solution

Method:

1. Connect the wire from one crocodile clip to one end of the battery and secure with tape. Repeat with the other crocodile clip wire and the other end of the battery.
2. Soak a piece of filter paper in the ammonia and ammonium chloride buffer solution and place it on the glass slide.
3. Connect the filter paper to the battery using one of the crocodile clips, keep the other one nearby.
4. Place a line of copper(II) chromate solution at the centre of the filter paper. The colour of this solution is initially green-brown.
5. Attach the other crocodile clip opposite the first one (as shown in the diagram) and leave the experiment to run for about 20 minutes.



start of reaction



after 20 minutes

Results:

- After 20 minutes you should see that the central coloured band disappears and is replaced by two bands, one yellow and the other blue, which seem to have separated out from the first band of copper(II) chromate.
- The cell that is used to supply an electric current sets up a potential difference across the circuit, so that one of the electrodes is positive and the other is negative.
- The chromate (CrO_4^{2-}) ions in the copper(II) chromate solution are attracted to the positive electrode, this creates a yellow band. The Cu^{2+} ions are attracted to the negative electrode, this creates a blue band.

Conclusion:

The movement of ions occurs because the electric current in the external circuit provides a potential difference between the two electrodes.

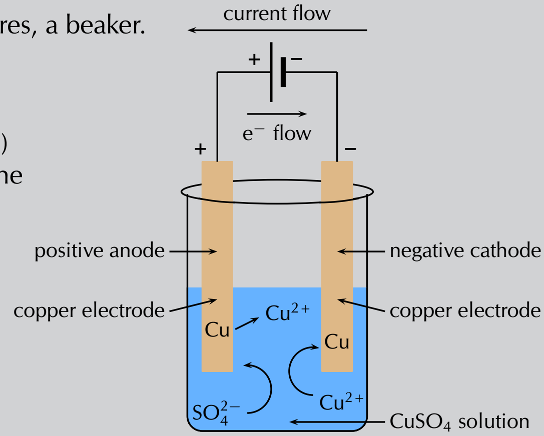
Experiment: An electrolytic cell

Aim:

To investigate the reactions that take place in an electrolytic cell.

Apparatus:

- Two copper plates (of equal size and mass), copper(II) sulfate ($CuSO_4$) solution (1 $mol.dm^{-3}$)
- A 9 V battery, two connecting wires, a beaker.



Method:

1. Half fill the beaker with copper(II) sulfate solution. What colour is the solution?
2. Weigh each copper electrode carefully and record the weight.
3. Place the two copper electrodes (of known mass) in the solution and make sure they are not touching each other.
4. Connect the electrodes to the battery as shown below and leave the experiment for a day. What colour is the solution after a day?

Discussion:

- What colour was the copper(II) sulfate solution before the experiment?
- What colour was the copper(II) sulfate solution after the experiment?
- Examine the two electrodes, what do you observe?
- What is the charge on each electrode?
- Which electrode is the anode and which is the cathode?

Observations:

- The initial blue colour of the solution remains unchanged throughout the experiment.
- It appears that copper has been *deposited* on one of the electrodes (it increased in mass) but *dissolved* from the other (it decreased in mass).
- The electrode connected to the negative terminal of the battery will have a negative charge. The electrode connected to the positive terminal of the battery will have a positive charge.
- When positively charged Cu^{2+} ions encounter the negatively charged electrode they gain electrons and are reduced to form copper metal. This metal is deposited on the electrode. The half-reaction that takes place is as follows:

  $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ **(reduction half-reaction)**

  Reduction occurs at the **cat**hode. Therefore, the electrode which increased in mass is the cathode.
- At the positive electrode, copper metal is oxidised to form Cu^{2+} ions, leaving electrons on the electrode. The half-reaction that takes place is as follows:

  $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$ **(oxidation half-reaction)**

  Oxidation occurs at the **an**ode. Therefore, the electrode which decreased in mass is the anode.
- The amount of copper that is *deposited* at one electrode is approximately the same as the amount of copper that is *dissolved* from the other. The number of

Cu^{2+} ions in the solution therefore remains almost the same, and the blue colour of the solution is unchanged.

Conclusion:

In this demonstration, the container held aqueous $CuSO_4$ (Cu^{2+}(aq) and SO_4^{2-}(aq)). The copper atoms of the electrode connected to the positive terminal (the anode) were oxidised and formed Cu^{2+}(aq) ions, causing a decrease in mass. The copper atoms of the electrode connected to the negative terminal (the cathode) were reduced to form solid copper, causing an increase in mass. This process is called *electrolysis*, and is very useful in the purification of metals.

Note that the cathode is negative and the anode is positive. Reduction still occurs at the cathode (Red Cat), and oxidation still occurs at the anode (An Ox).

TIP

FACT

August Wilhelm von Hofmann was a German chemist who invented the Hofmann cell, which uses a current to form H_2(g) and O_2(g) from water through electrolysis.

The electrolysis of water

Water can undergo electrolysis to form hydrogen gas and oxygen gas according to the following reaction:

$2H_2O(\ell) \rightarrow 2H_2(g) + O_2(g)$

This reaction is very important because hydrogen gas has the potential to be used as an energy source. The electrolytic cell for this reaction consists of two electrodes, submerged in an electrolyte and connected to a source of electric current (Figure 13.7).

The **oxidation half-reaction** is as follows:

$2H_2O(\ell) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$

The **reduction half-reaction** is as follows:

$2H^+(aq) + 2e^- \rightarrow H_2(g)$

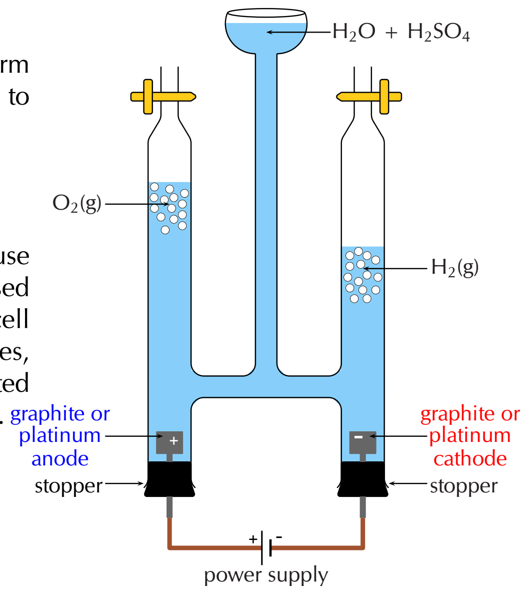


Figure 13.7: Hofmann apparatus for the electrolysis of water.

Informal experiment: The electrolysis of sodium iodide and water

Aim:

To study the electrolysis of water and sodium iodide.

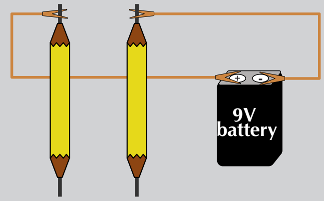
Apparatus:

- 4 pencils, copper wire attached to crocodile clips, 9 V battery, 2 beakers, pencil sharpener, spatula, glass rod
- distilled water, sodium iodide (NaI), phenolphthalein, 1 mol.dm^{-3} sulfuric acid (H_2SO_4) solution

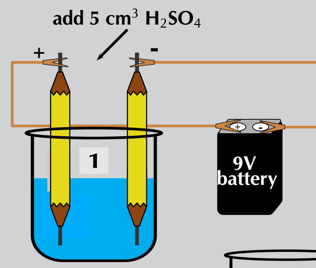
Method:

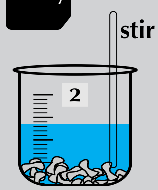
1. Label one beaker **1** and half fill it with distilled H_2O.

2. Sharpen both ends of two of the pencils. Strip away some of the wood to expose more of the graphite. (Graphite rods can be used instead of pencils if they are available).

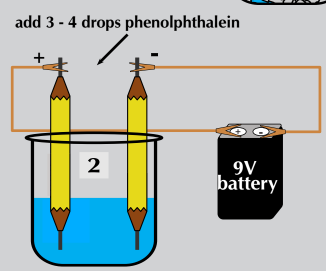


3. Attach one end of the crocodile clips to the pencils and the other end to the battery.
4. Place the pencils in the beaker, making sure they are not touching each other. Observe what happens.
5. Pour approximately 5 cm^3 of the H_2SO_4 solution into the beaker. Observe what happens.





6. Label a second beaker **2** and add 5 spatula tips of NaI to the beaker.
7. Half fill beaker 2 with distilled H_2O and stir with the glass rod until all the sodium iodide has dissolved.
8. Repeat steps 2 - 4 with the second set of pencils.
9. After a few minutes add 3 - 4 drops of the phenolphthalein to the beaker. Observe what happens.



Questions:

- In beaker **1**:
  1. What happened when the pencils were first put in the water?
  2. What happened when the sulfuric acid was added to the beaker?
  3. Why is the sulfuric acid necessary in this reaction?
  4. What is happening at the negative electrode (the pencil attached to the negative terminal of the battery)?
  5. What is happening at the positive electrode (the pencil attached to the positive terminal of the battery)?
  6. Which electrode is the anode and which is the cathode?
- In beaker **2**:
  1. What happened when the pencils were first put in the water?
  2. What is happening at the negative electrode?
  3. What is happening at the positive eletrode?
  4. Which electrode is the anode and which is the cathode?
  5. What happened when you added the phenolphthalein? Why did the change take place?

Results:

- In the electrolysis of water two H^+ ions each gain an electron (are reduced) and combine to form hydrogen gas ($H_2(g)$):

$2H^+(aq) + 2e^- \rightarrow H_2(g)$

Two O^{2-} ions each lose two electrons (are oxidised) and combine to form oxygen gas ($O_2(g)$):

$2O^{2-}(aq) \rightarrow O_2(g) + 4e^-$

- When the positive H^+ ions encounter the negative electrode they are reduced. Reduction is a gain of electrons at the cathode, therefore the negative electrode is the cathode.
- When the negative O^{2-} ions encounter the positive electrode they are oxidised. Oxidation is a loss of electrons at the anode, therefore the positive electrode is the anode.
- Remember that pure water does not conduct electricity. So, an electrolyte (such as sulfuric acid) is necessary for the reaction to take place.
- A salt dissolved in the water is also an electrolyte, so sulfuric acid is not necessary in the second beaker.
- The electrolysis of water occurs in beaker **2**, but there are other reactions taking place as well because of the presence of the Na^+ and I^- ions.
- When the negative I^- ions encounter the positive electrode they are oxidised:

  $2I^-(aq) \rightarrow I_2(s) + 2e^-$

  You should have observed solid iodine forming at the positive electrode.
- When the positive Na^+ ions encounter the negative electrode they are reduced:

  $Na^+(aq) + e^- \rightarrow Na(s)$

  Na(s) is very reactive with water:

  $2Na(s) + 2H_2O(\ell) \rightarrow 2NaOH(aq) + H_2(g)$.
- Remember that phenolphthalein turns pink in the presence of a base. So when the phenolphthalein is added, the water around the negative electrode should become pink due to the NaOH.

Conclusion:

The application of an electric current to water splits the water molecules and causes hydrogen and oxygen gas to form. This can only happen in the presence of an electrolyte (for example sulfuric acid). Sodium iodide dissolved in the water is also an electrolyte and enables the electrolysis of water. However, the cation and anion of the salt will also undergo a reaction. NaOH will form at the cathode, while solid iodine will form at the anode.

Exercise 13 – 4: Galvanic and electrolytic cells

1. An electrolytic cell consists of two electrodes in a silver chloride (AgCl) solution, connected to a source of current. A current is passed through the solution and Ag^+ ions are reduced to a silver metal deposit on one of the electrodes.
   a) What is the name of this process?
   b) Does reduction occur at the electrode where the deposit formed?
   c) Give the equation for the reduction half-reaction.
   d) Give the equation for the oxidation half-reaction.
2. A galvanic cell consists of two half-cells: a copper anode in a copper nitrate ($Cu(NO_3)_2(aq)$) solution, and a silver cathode in a silver nitrate ($AgNO_3(aq)$) solution.
   a) Give equations for the half-reactions that take place at the anode and cathode.

b) Write the overall reaction for this cell.

c) Give standard cell notation for this cell.

d) Draw a simple diagram of the galvanic cell.
On your diagram, show the direction in which current flows.

3. Electrolysis takes place in a solution of molten lead bromide ($PbBr_2$) to produce lead atoms.

a) Give equations for the half-reactions that take place at the anode and cathode.

b) Draw a simple diagram of the electrolytic cell.
On your diagram, show the direction in which current flows.

4. Fill in the table below to summarise the information on galvanic and electrolytic cells:

	Galvanic cells	**Electrolytic cells**
spontaneity		
type of energy		
anode		
cathode		
cell set-up		

5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 2822 2. 2823 3. 2824 4. 2825

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13.4 Processes in electrochemical cells ESCR7

Half-cells and half-reactions ESCR8

Galvanic cells are actually made up of two half-cells. One half-cell contains the anode and an electrolyte containing the same metal cations. The other half-cell contains the cathode and an electrolyte containing the same metal cations. These half-cells are connected by a salt-bridge and the electrodes are connected through an external circuit.

Earlier in this chapter we discussed a zinc-copper cell. This was made up of a zinc **half-cell**, containing a zinc electrode and a zinc(II) sulphate ($ZnSO_4$) electrolyte solution, and a copper **half-cell**, containing a copper electrode and a copper(II) sulphate solution.

DEFINITION: *Half-cell*

A half-cell is a structure that consists of a conductive electrode surrounded by a conductive electrolyte.

In each half-cell a half-reaction takes place:

- **Copper plate**
At the copper plate, there was an *increase* in mass. This means that Cu^{2+} ions

from the copper(II) sulfate solution were deposited onto the plate as atoms of copper metal. The half-reaction that takes place at the copper plate is:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

As electrons are gained by the copper ions this is the **reduction half-reaction**.

- **Zinc plate**

  At the zinc plate, there was a *decrease* in mass. This means that some of the solid zinc goes into solution as Zn^{2+} ions. The electrons remain on the zinc plate, giving it a negative charge. The half-reaction that takes place at the zinc plate is:

  $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

  As electrons are lost by the zinc atoms this is the **oxidation half-reaction**.

- **The overall reaction**

  You can then combine the two half-reactions from these two half-cells to get the overall reaction:

  $Zn(s) + Cu^{2+}(aq) + 2e^- \rightarrow Zn^{2+}(aq) + Cu(s) + 2e^-$ or, if we cancel the electrons:

  $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

It is possible to look at the half-reaction taking place in a half-cell and determine which electrode is the anode and which is the cathode.

- **Oxidation** is loss at the **anode**, therefore the oxidation half-reaction occurs in the half-cell containing the anode.
- **Reduction** is gain at the **cathode** so the reduction half-reaction occurs in the half-cell containing the cathode.

Remember that for an electrochemical cell the standard cell notation is:

$\mathbf{Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)}$

$|$ = a phase boundary (solid/aqueous) $||$ = the salt bridge

Worked example 4: Understanding galvanic cells

QUESTION

For the following cell: $Zn(s)|Zn^{2+}(aq)||Ag^+(aq)|Ag(s)$

1. Give the anode and cathode half-reactions.
2. Write the overall equation for the chemical reaction.
3. Give the direction of the current in the external circuit.

SOLUTION

Step 1: Identify the oxidation and reduction reactions

By convention in standard cell notation, the anode is written on the left and the cathode is written on the right. So, in this cell:

Zinc is the anode (solid zinc is oxidised).

Silver is the cathode (silver ions are reduced).

Step 2: Write the two half-reactions

Oxidation is loss of electrons at the anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Reduction is gain of electrons at the cathode: $Ag^+(aq) + e^- \rightarrow Ag(s)$

Step 3: Combine the half-reactions to get the overall equation

Balance the charge by multiplying the reduction half-reaction by 2.

$2Ag^{+}(aq) + 2e^{-} \rightarrow 2Ag(s)$ $Zn(s) + 2Ag^{+}(aq) \rightarrow Zn^{2+}(aq) + Ag(s)$

Step 4: Determine the direction of current flow

The solid zinc is oxidised to form zinc ions. These electrons are left on the zinc electrode (anode) making it negative.

The silver ions take electrons and are reduced to form solid silver. This makes the silver electrode (cathode) positive.

Electron flow is from negative to positive, so from the anode to the cathode. Conventional current is in the opposite direction to electron flow. Therefore current will flow from the cathode (silver) to the anode (zinc).

Exercise 13 – 5: Galvanic cells

1. The following half-reactions take place in an electrochemical cell:

   $Fe(s) \rightarrow Fe^{3+}(aq) + 3e^{-}$ $Fe^{2+}(aq) + 2e^{-} \rightarrow Fe(s)$

   a) Which is the oxidation half-reaction?
   b) Which is the reduction half-reaction?
   c) Name the oxidising agent.
   d) Name the reducing agent.
   e) Use standard notation to represent this electrochemical cell.

2. For the following cell: $Mg(s)|Mg^{2+}(aq)||Mn^{2+}(aq)|Mn(s)$

   a) Give the cathode half-reaction.
   b) Give the anode half-reaction.
   c) Give the overall equation for the electrochemical cell.
   d) What metals could be used for the electrodes in this electrochemical cell?
   e) Suggest two electrolytes for this electrochemical cell.
   f) In which direction will the current flow?
   g) Draw a simple sketch of the complete cell.

3. For the following cell: $Sn(s)|Sn^{2+}(aq)||Ag^{+}(aq)|Ag(s)$

   a) Give the cathode half-reaction.
   b) Give the anode half-reaction.
   c) Give the overall equation for the electrochemical cell.
   d) Draw a simple sketch of the complete cell.

4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

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Predictions in half-cells ESCR9

Worked example 5: Reactions at the anode and cathode

QUESTION

A cell contains a silver anode and a copper cathode. Give the half-cell reactions occurring at the anode and cathode, as well as standard cell notation for this cell.

SOLUTION

Step 1: What type of reaction occurs at the anode, and what type occurs at the cathode?

Oxidation is loss of electrons at the anode. So the silver anode will be oxidised.

Reduction is gain of electrons at the cathode. So the copper cathode will be reduced.

Step 2: Write down the half-reactions as they would occur in the cell

$Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$ (oxidation half-reaction)

$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ (reduction half-reaction)

Step 3: Give the standard cell notation for this reaction

The anode is always written first (on the left): $Ag(s)|Ag^{+}(aq)$

The cathode is always written second (on the right): $Cu^{2+}(aq)|Cu(s)$

Therefore the standard cell notation is: $Ag(s)|Ag^{+}(aq)||Cu^{2+}(aq)|Cu(s)$

Worked example 6: Determining overall reactions

QUESTION

The following half-reactions take place in a cell:

$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^{-}$ $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$

Determine the overall reaction that takes place as a balanced chemical equation and in standard cell notation.

SOLUTION

Step 1: Identify the oxidation and reduction half-reactions

$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^{-}$ (oxidation half-reaction)

$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$ (reduction half-reaction)

Step 2: Which metal is the anode and which is the cathode?

Oxidation is loss of electrons at the anode, therefore Fe is the anode. Reduction is gain of electrons at the cathode, therefore Cu is the cathode.

Step 3: Compare the number of electrons in each equation

There are 2 electrons in both equations, so the charges are balanced.

Step 4: Combine the equations into one equation

$Cu^{2+}(aq) + 2e^{-} + Fe(s) \rightarrow Cu(s) + Fe^{2+}(aq) + 2e^{-}$

Step 5: Clean reaction up by combining appropriate ions and molecules

$Cu^{2+}(aq) + Fe(s) \rightarrow Cu(s) + Fe^{2+}(aq)$

Step 6: Give the standard cell notation for this reaction

The anode is always written first (on the left): $Fe(s)|Fe^{2+}(aq)$

The cathode is always written second (on the right): $Cu^{2+}(aq)|Cu(s)$

Therefore the standard cell notation is: $Fe(s)|Fe^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$

13.5 The effects of current and potential on rate and equilibrium ESCRB

Current and rate of reaction ESCRC

A galvanic cell

Let's think back to the Zn-Cu(s) electrochemical cell. This cell is made up of two half-cells and the reactions that take place at each of the electrodes are as follows:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}$ $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$

- At the zinc electrode, the zinc metal loses electrons and forms Zn^{2+} ions. The electrons are concentrated on the zinc metal while the Zn^{2+} ions are in solution.
- At the copper electrode, the copper ions gain electrons and forms solid copper.
- This means that there is an excess of electrons on the zinc anode, and a deficit of electrons on the copper cathode.
- Electrons will flow from an area of high concentration to an area of low concentration. Therefore the electrons on the zinc anode flow through the external circuit towards the more positive copper cathode.
- The larger the difference between the excess and the deficit of electrons, the faster the electrons will flow and the greater the current will be.
- The faster the electrons will flow, the greater the rate of the reaction must be.
- Therefore the larger the current, the faster the rate of the reaction.

An electrolytic cell

- Conversely, in an electrolytic cell a redox reaction takes place when a current is applied.
- This redox reaction is the decomposition of a chemical compound (electrolysis).
- The rate of this decomposition (into ions) is increased when the current applied is increased.

Potential difference, equilibrium and concentration ESCRD

Again we can use the zinc-copper cell as an example.

When the chemical reaction between the zinc and the copper slows, the increase in product concentration, and the decrease in reactant concentration, slow too. This means that the electron transfer rate will decrease.

When the chemical reaction in the cell stops:

- The reaction is no longer converting chemical potential energy to electrical potential energy.
- The concentrations of the reactants and products have become constant and equilibrium has been reached.
- There is no excess or deficit of electrons on the electrodes, and the potential difference of the cell is 0.
- A potential difference of 0 means that the current is 0.

So the potential difference across a cell is related to the extent to which the cell reaction has reached equilibrium. When equilibrium is reached, the potential difference of the cell is zero and the cell is said to be 'flat'. There is no longer a potential difference between the two half-cells, and no more current will flow.

13.6 Standard electrode potentials ESCRF

Standard conditions ESCRG

Standard electrode potentials are a measurement of equilibrium potentials. The position of this equilibrium can change if you change some of the conditions (e.g. concentration, temperature). It is therefore important that standard conditions be used:

- pressure = 101,3 kPa (1 atm)
- temperature = 298 K (25 °C)
- concentration = 1 $mol.dm^{-3}$

FACT
The standard hydrogen electrode used now is actually the potential of a platinum electrode in a theoretical acidic solution.
The electrode potential of the hydrogen electrode at 25 °C is estimated to be 4,4 V. However, in order to use this as a reference electrode this value is set to **zero at all temperatures** so that it can be compared with other electrodes.

The standard hydrogen electrode ESCRH

It is the **potential difference** (recorded as a voltage) between the two electrodes that causes electrons to flow from the **anode** to the **cathode** through the external circuit of a galvanic cell (remember, conventional current goes in the opposite direction).

It is possible to measure the potential of an electrode and electrolyte. It is not a simple process however, and the value obtained will depend on the concentration of the electrolyte solution, the temperature and the pressure.

A way to remove these inconsistencies is to compare all electrode potentials to a **standard reference electrode**. These comparisons are all done with the same concentrations, temperature and pressure. This means that these values can be used to calculate the potential difference between two electrodes. It also means that electrode potentials can be compared without the need to construct the specific cell being studied.

This reference electrode can be used to calculate the *relative* electrode potential for a substance. The reference electrode that is used is the **standard hydrogen electrode** (Figure 13.8).

DEFINITION: *Standard hydrogen electrode*

The standard hydrogen electrode is a redox electrode which forms the basis of the scale of oxidation-reduction potentials.

The standard hydrogen electrode consists of a platinum electrode in a solution containing H^+ ions. The solution (e.g. H_2SO_4) has a concentration of 1 $mol.dm^{-3}$. As the hydrogen gas bubbles over the platinum electrode, the reaction is as follows:

$2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$

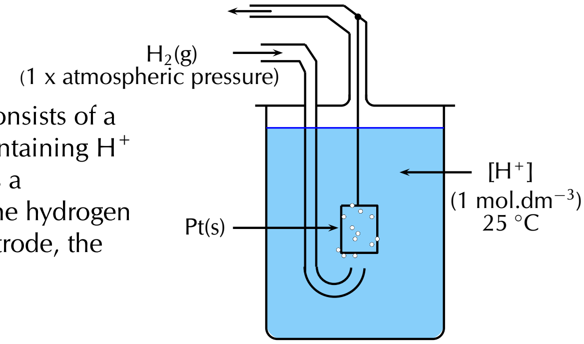


Figure 13.8: A simplified version of the standard hydrogen electrode.

FACT
When determining standard electrode reduction potentials the standard hydrogen electrode is considered to be *on the left* ($Pt(s)|H_2(g), H^+(aq)||$). So a negative value means that the other element or compound has a greater tendency to oxidise, and a positive value means that the other element or compound has a greater tendency to be reduced.

TIP
By convention, we always write the reduction half-reaction when giving the standard electrode potential.

TIP
In the examples we used earlier, zinc's electrode reduction potential is $-0{,}76$ and copper's is $+0{,}34$. So, if an element or compound has a *negative* standard electrode reduction potential, it means it forms ions easily. The more negative the value, the easier it is for that element or compound to form ions (be oxidised, and be a reducing agent). If an element or compound has a *positive* standard electrode potential, it means it does not form ions as easily.

Standard electrode potentials ESCRJ

In order to *use* the hydrogen electrode, it needs to be attached to the electrode system that you are investigating. For example, if you are trying to determine the electrode potential of copper, you will need to connect the copper half-cell to the hydrogen electrode; if you are trying to determine the electrode potential of zinc, you will need to connect the zinc half-cell to the hydrogen electrode and so on. Let's look at the examples of zinc and copper in more detail.

Zinc

Zinc has a greater tendency than hydrogen to form ions (to be oxidised), so if the standard hydrogen electrode is connected to the zinc half-cell, the **zinc** will be relatively **more negative** because the electrons that are released when zinc is oxidised will accumulate on the metal.

$Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$

$2H^+(aq) + 2e^- \rightarrow H_2(g)$

The solid zinc is more likely to form zinc ions than the hydrogen gas is to form ions. A simplified representation of the cell is shown in Figure 13.9.

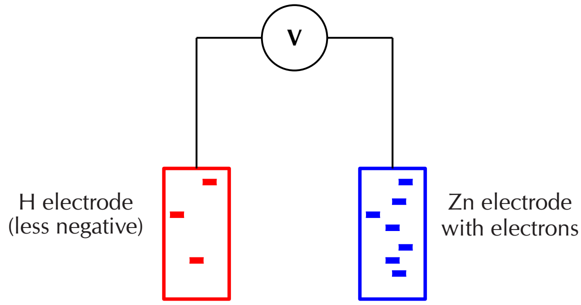


Figure 13.9: When zinc is connected to the standard hydrogen electrode, relatively few electrons build up on the platinum (hydrogen) electrode. There are lots of electrons on the zinc electrode.

The voltmeter measures the potential difference between the charge on these electrodes. In this case, the voltmeter would read $-0{,}76$ V as the Zn electrode has a relatively higher number of electrons.

Copper

Copper has a lower tendency than hydrogen to form ions, so if the standard hydrogen electrode is connected to the copper half-cell, the **copper** will be relatively **less negative**.

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

$2H^+(aq) + 2e^- \rightarrow H_2(g)$

The copper ions are more likely to form solid copper than the hydrogen ions are to form hydrogen gas. A simplified representation of the cell is shown in Figure 13.10.

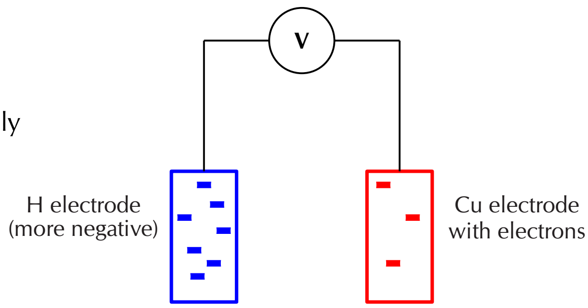


Figure 13.10: When copper is connected to the standard hydrogen electrode, relatively few electrons build up on the copper electrode. There are lots of electrons on the hydrogen electrode.

The voltmeter measures the potential difference between the charge on these electrodes. In this case, the voltmeter would read $+0{,}34$ V as the Cu electrode has a relatively lower number of electrons.

The voltages recorded when zinc and copper were connected to a standard hydrogen electrode are in fact the **standard electrode potentials** for these two metals. It is important to remember that these are not *absolute* values, but are potentials that have been measured *relative* to the potential of hydrogen if the standard hydrogen electrode is taken to be zero.

Luckily for us, we do not have to determine the standard electrode potential for every metal. This has been done already and the results are recorded in a table of standard electrode potentials. Table 13.2 is presented as the standard electrode reduction potentials.

Half-Reaction	**E° V**
$Li^{+} + e^{-} \rightleftharpoons Li$	$-3,04$
$K^{+} + e^{-} \rightleftharpoons K$	$-2,92$
$Ba^{2+} + 2e^{-} \rightleftharpoons Ba$	$-2,90$
$Ca^{2+} + 2e^{-} \rightleftharpoons Ca$	$-2,87$
$Na^{+} + e^{-} \rightleftharpoons Na$	$-2,71$
$Mg^{2+} + 2e^{-} \rightleftharpoons Mg$	$-2,37$
$Al^{3+} + 3e^{-} \rightleftharpoons Al$	$-1,66$
$Mn^{2+} + 2e^{-} \rightleftharpoons Mn$	$-1,18$
$2H_2O + 2e^{-} \rightleftharpoons H_2(g) + 2OH^{-}$	$-0,83$
$Zn^{2+} + 2e^{-} \rightleftharpoons Zn$	$-0,76$
$Cr^{2+} + 2e^{-} \rightleftharpoons Cr$	$-0,74$
$Fe^{2+} + 2e^{-} \rightleftharpoons Fe$	$-0,44$
$Cr^{3+} + 3e^{-} \rightleftharpoons Cr$	$-0,41$
$Cd^{2+} + 2e^{-} \rightleftharpoons Cd$	$-0,40$
$Co^{2+} + 2e^{-} \rightleftharpoons Co$	$-0,28$
$Ni^{2+} + 2e^{-} \rightleftharpoons Ni$	$-0,25$
$Sn^{2+} + 2e^{-} \rightleftharpoons Sn$	$-0,14$
$Pb^{2+} + 2e^{-} \rightleftharpoons Pb$	$-0,13$
$Fe^{3+} + 3e^{-} \rightleftharpoons Fe$	$-0,04$
$2H^{+} + 2e^{-} \rightleftharpoons H_2(g)$	**0,00**
$S + 2H^{+} + 2e^{-} \rightleftharpoons H_2S(g)$	$+0,14$
$Sn^{4+} + 2e^{-} \rightleftharpoons Sn^{2+}$	$+0,15$
$Cu^{2+} + e^{-} \rightleftharpoons Cu^{+}$	$+0,16$
$SO_4^{2-} + 4H^{+} + 2e^{-} \rightleftharpoons SO_2(g) + 2H_2O$	$+0,17$
$Cu^{2+} + 2e^{-} \rightleftharpoons Cu$	$+0,34$
$2H_2O + O_2 + 4e^{-} \rightleftharpoons 4OH^{-}$	$+0,40$
$Cu^{+} + e^{-} \rightleftharpoons Cu$	$+0,52$
$I_2 + 2e^{-} \rightleftharpoons 2I^{-}$	$+0,54$
$O_2(g) + 2H^{+} + 2e^{-} \rightleftharpoons H_2O_2$	$+0,68$
$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$	$+0,77$
$NO_3^{-} + 2H^{+} + e^{-} \rightleftharpoons NO_2(g) + H_2O$	$+0,78$
$Hg^{2+} + 2e^{-} \rightleftharpoons Hg(\ell)$	$+0,78$
$Ag^{+} + e^{-} \rightleftharpoons Ag$	$+0,80$
$NO_3^{-} + 4H^{+} + 3e^{-} \rightleftharpoons NO(g) + 2H_2O$	$+0,96$
$Br_2 + 2e^{-} \rightleftharpoons 2Br^{-}$	$+1,06$
$O_2(g) + 4H^{+} + 4e^{-} \rightleftharpoons 2H_2O$	$+1,23$
$MnO_2 + 4H^{+} + 2e^{-} \rightleftharpoons Mn^{2+} + 2H_2O$	$+1,28$
$Cr_2O_7^{2-} + 14H^{+} + 6e^{-} \rightleftharpoons 2Cr^{3+} + 7H_2O$	$+1,33$
$Cl_2 + 2e^{-} \rightleftharpoons 2Cl^{-}$	$+1,36$
$Au^{3+} + 3e^{-} \rightleftharpoons Au$	$+1,50$
$MnO_4^{-} + 8H^{+} + 5e^{-} \rightleftharpoons Mn^{2+} + 4H_2O$	$+1,52$
$Co^{3+} + e^{-} \rightleftharpoons Co^{2+}$	$+1,82$
$F_2 + 2e^{-} \rightleftharpoons 2F^{-}$	$+2,87$

Table 13.2: Table of standard electrode (reduction) potentials.

A few examples from the table are shown in Table 13.3. These will be used to explain some of the trends in the table of electrode potentials.

Half-Reaction	E° V
$Li^{+} + e^{-} \rightleftharpoons Li$	−3,04
$Mg^{2+} + 2e^{-} \rightleftharpoons Mg$	−2,37
$Zn^{2+} + 2e^{-} \rightleftharpoons Zn$	−0,76
$Fe^{3+} + 3e^{-} \rightleftharpoons Fe$	−0,4
$Pb^{2+} + 2e^{-} \rightleftharpoons Pb$	−0,13
$2H^{+} + 2e^{-} \rightleftharpoons H_2(g)$	**0,00**
$Cu^{2+} + 2e^{-} \rightleftharpoons Cu$	+0,34
$Ag^{+} + e^{-} \rightleftharpoons Ag$	+0,80
$Au^{3+} + 3e^{-} \rightleftharpoons Au$	+1,50

Table 13.3: A few examples of standard electrode potentials.

- A **large negative value** (e.g. $Li^{+} + e^{-} \rightleftharpoons Li$) means that the element or compound ionises easily, in other words, it releases electrons easily. This element or compound is **easily oxidised** and is therefore a good **reducing agent**.

- A **large positive value** (e.g. $Au^{3+} + e^{-} \rightleftharpoons Au$) means that the element or compound gains electrons easily. This element or compound is **easily reduced** and is therefore a good **oxidising agent**.
- The **reducing ability** (i.e. the ability to act as a reducing agent) of the elements or compounds in the table **decreases** as you move **down** in the table.
- The **oxidising ability** of elements or compounds **increases** as you move **down** in the table.

Experiment: Ice cube tray redox experiment

Aim:

To determine relative reactivity of the respective metals and to gain understanding of the workings of a simple electrochemical cell.

Apparatus:

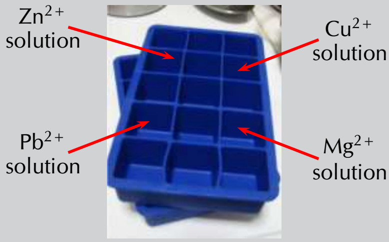
- Ice cube tray, voltmeter and connecting wires.
- Lead (Pb), magnesium (Mg), zinc (Zn), copper (Cu) strips.
- 1 $mol.dm^{-3}$ solutions of lead (e.g. $PbSO_4$), magnesium (e.g. $MgSO_4$), zinc (e.g. $ZnSO_4$) and copper (e.g. $CuSO_4$).
- String soaked in a sodium nitrate ($NaNO_3$) solution.

Pre-knowlege:

Electrons move from the anode to the cathode. Conventional current moves from the cathode to the anode - therefore the positive terminal of the voltmeter will be on the cathode and the negative terminal will be on the anode.

Method:

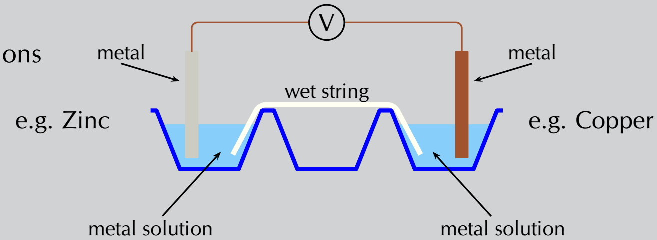
1. Place approximately 15 cm^3 of the Pb, Zn, Cu and Mg solutions into four different ice cube depressions. These should not be next to each other to avoid mixing of solutions.



2. Attach two different metals to the crocodile clips. Drape the wet string across the two solutions being used (so that each end of the string is in a solution). Then place the metal into its respective ion solution.
   i.e. zinc electrode **must** go into the Zn^{2+} solution, copper electrode **must** go into the Cu^{2+} solution.

Use the cell combinations in the following order:

Pb/Zn; Pb/Cu; Pb/Mg; Zn/Cu; Zn/Mg; Cu/Mg



3. Determine the combinations of metals that give a positive reading.
4. Draw up a table that shows:
   - the combination of the metals
   - which metal is the anode in that pair of metals
   - which metal is the cathode in that pair of metals

Metal combination	Anode	Cathode

5. Use this table to rank the metals from the strongest **reducing agent** (rank from the strongest to the weakest).
6. For each combination - write down the reduction half-reaction and oxidation half-reaction and then the overall cell reaction.
7. Note all observations for each cell.

Questions:

1. Explain why the voltages appear to be lower/higher than expected.
2. What is the purpose of the string?

Conclusions:

Depending on the electrode potential of each metal, the same metal could be the anode in one reaction and the cathode in another reaction. This can be seen by the positive, or negative, reading on the voltmeter.

For example, lead is more likely to be reduced than zinc, therefore in that pair lead will be the cathode and zinc will be the anode. However, lead is more likely to be oxidised than copper, therefore in that pair copper will be the cathode and lead will be the anode.

Exercise 13 – 6: Table of standard electrode potentials

1. Give the standard electrode potential for each of the following metals:
   a) magnesium
   b) lead
   c) nickel
2. Refer to the electrode potentials in Table 13.2.
   a) Which of the metals is most likely to be oxidised?
   b) Which metal ion is most likely to be reduced?

TIP
In this worked example you are given two half-reactions. Both are presented as shown in the table of standard reduction potentials, but in reality only one metal is being reduced, the other is being oxidised. The question being asked is: Which metal is being oxidised and which is being reduced?

c) Which metal is the strongest reducing agent?
d) If the other electrode is magnesium, is reduction or oxidation more likely to take place in the copper half-reaction? Explain your answer.
e) If the other electrode is tin, is reduction or oxidation more likely to take place in the mercury half-reaction? Explain your answer.

3. Use the table of standard electrode potentials to put the following in order from the **strongest oxidising agent** to the **weakest oxidising agent**.
   - Cu^{2+}
   - MnO_4^-
   - Br_2
   - Zn^{2+}
4. Look at the following half-reactions:
   - $Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$
   - $Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$
   - $Fe^{3+}(aq) + 3e^- \rightarrow Fe(s)$
   - $I_2(s) + 2e^- \rightarrow 2I^-(aq)$

   a) Which substance is the strongest oxidising agent?
   b) Which substance is the strongest reducing agent?
5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

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Uses of the standard electrode potentials ESCRK

So now that you understand this useful table of reduction potentials, it is important that you can use these values to calculate the potential energy differences. The following worked examples will help you do this. In all of these cases it is important that you understand what the question is asking.

Worked example 7: Using the table of standard electrode potentials

QUESTION

The following reactions take place in an electrochemical cell:

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $Ag^+(aq) + e^- \rightleftharpoons Ag(s)$

Determine which reaction is the oxidation half-reaction and which is the reduction half-reaction in this cell.

SOLUTION

Step 1: Determine the electrode potential for each metal

From the table of standard electrode potentials:

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ ($E^\circ = +0{,}34$ V) $Ag^+(aq) + e^- \rightleftharpoons Ag(s)$ ($E^\circ = +0{,}80$ V)

Step 2: Use the electrode potential values to determine which metal is oxidised and which is reduced

Both values are positive, but silver has a larger, positive electrode potential than copper. Therefore silver is more easily reduced than copper, and copper is more easily oxidised than silver.

Step 3: Write the reduction and oxidation half-reactions

The E° values are taken from the table of standard **reduction** potentials. Therefore the reduction half-reaction is as seen in the table.

The reduction half-reaction: $Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$ (reduction is a gain of electrons)
Oxidation is a loss of electrons, and so would be written in reverse.
The oxidation half-reaction: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$.

TIP
If magnesium is able to displace silver from a solution of silver nitrate, this means that magnesium metal will form magnesium ions and the silver ions will become silver metal. In other words, there will now be **silver** metal and a solution of **magnesium nitrate**. This will only happen if magnesium has a greater tendency than silver to form ions. In other words, what this worked example is asking is whether magnesium or silver will form ions more easily.

TIP
Remember that solid metal will not always be formed when an ion is reduced. For example $Sn^{4+} + 2e^{-} \rightarrow Sn^{2+}$.

Worked example 8: Using the table of standard electrode potentials

QUESTION

Is magnesium able to displace silver from a solution of silver nitrate?

SOLUTION

Step 1: Find appropriate reactions on the table of standard electrode potentials

The reaction involves magnesium and silver.

$Mg^{2+}(aq) + 2e^{-} \rightleftharpoons Mg(s)$ $\qquad$ $Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s)$

Step 2: Determine the electrode potential for each metal

From the table of standard electrode potentials:

$Mg^{2+}(aq) + 2e^{-} \rightleftharpoons Mg(s)\ (E^{\circ} = -2,37\ V)$ $\qquad$ $Ag^{+}(aq) + e^{-} \rightleftharpoons Ag(s)\ (E^{\circ} = +0,80\ V)$

Step 3: Which metal is more likely to be reduced, which is more likely to be oxidised?

Silver has a positive E°, while magnesium has a negative E°. Therefore silver is more easily reduced than magnesium, and magnesium is more easily oxidised than silver. The following reactions would occur:

Reduction half-reaction: $Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$
Oxidation half-reaction: $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-}$

It can be concluded that magnesium **will** displace silver from a silver nitrate solution so that there will be silver metal and magnesium ions in the solution.

Worked example 9: Determining overall reactions

QUESTION

For a zinc (Zn) and gold(III) oxide (Au_2O_3) cell in solution of KOH determine the:

- oxidation and reduction half-reactions
- overall balanced chemical equation
- standard cell notation for the cell

SOLUTION

Step 1: Find appropriate reactions on the table of standard electrode potentials

The reaction involves zinc and gold.

$Zn^{2+}(aq) + 2e^{-} \rightleftharpoons Zn(s)$ $\qquad$ $Au^{3+}(aq) + 3e^{-} \rightleftharpoons Au(s)$

Step 2: Determine the electrode potential for each metal

From the table of standard electrode potentials:

$Zn^{2+}(aq) + 2e^{-} \rightleftharpoons Zn(s)\ (E^{\circ} = -0,76\ V)$ $\qquad$ $Au^{3+}(aq) + 3e^{-} \rightleftharpoons Au(s)\ (E^{\circ} = +1,50\ V)$

Step 3: Which metal is more likely to be reduced, which is more likely to be oxidised?

Zinc has a negative E°, while gold has a positive E°. Therefore gold is more easily reduced than zinc, and zinc is more easily oxidised than gold. The following reactions would occur:

Reduction half-reaction: $Au^{3+}(aq) + 3e^{-} \rightarrow Au(s)$

Oxidation half-reaction: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-}$

Step 4: Compare the number of electrons in each equation

There are 3 electrons in the reduction half-reaction, and 2 electrons in the oxidation half-reaction.

Oxidation: $\times 3$: $3Zn(s) \rightarrow 3Zn^{2+}(aq) + \mathbf{6e^{-}}$

Reduction: $\times 2$: $2Au^{3+}(aq) + \mathbf{6e^{-}} \rightarrow 2Au(s)$

Step 5: Combine the equations into one equation

$2Au^{3+}(aq) + 6e^{-} + 3Zn(s) \rightarrow 2Au(s) + 3Zn^{2+}(aq) + 6e^{-}$

Step 6: Add the spectator ions and remove the electrons from the equation

$Au_2O_3(aq) + 3Zn(s) \rightarrow 2Au(s) + 3ZnO(aq)$

Step 7: Which material is the anode and which is the cathode?

Oxidation is loss of electrons at the anode, therefore Zn(s) is the anode.

Reduction is gain of electrons at the cathode, therefore Au_2O_3 is the cathode.

Step 8: Give the standard cell notation for this reaction

The anode is always written first (on the left): $Zn(s)|ZnO(aq)$

The cathode is always written second (on the right): $Au_2O_3(s), Au(s)$

Therefore the standard cell notation is: $Zn(s)|ZnO(aq)||Au_2O_3(s), Au(s)$

Experiment: Displacement experiment

Aim:

To demonstrate the effect of reacting a halogen with a halide.

Apparatus:

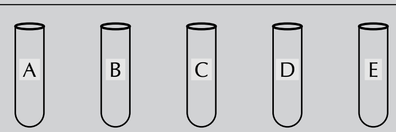
- Bleach (approximately 10 cm^3), bromine water, aqueous solutions of sodium chloride (NaCl), sodium bromide (NaBr) and sodium iodide (NaI), paraffin, concentrated HCl.
- 5 test tubes, 2 plastic droppers.

Method:

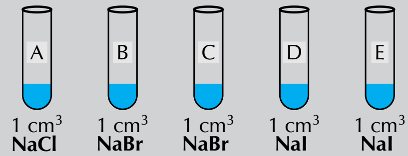
WARNING!

Concentrated HCl can cause serious burns. We suggest using gloves and safety glasses whenever you work with an acid. Remember to add the acid to the water and to avoid sniffing the acid. Handle all chemicals with care.

1. Label the test tubes A, B, C, D and E.



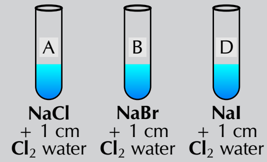
2. Place 1 cm^3 NaCl solution into test tube A.
3. Place 1 cm^3 NaBr solution into both test tubes B and C.
4. Place 1 cm^3 NaI solution into both test tubes D and E.



5. Activate 10 cm^3 of the bleach by adding 2 cm^3 of the concentrated HCl.

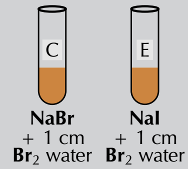
   Observe the liquid and note what happens on adding HCl, record your observations. *You have formed a solution of chlorine in water*.
6. Using a plastic dropper transfer approximately 1 cm height of the chlorine water into the test tubes labelled A, B and D.

   Note any changes to the test tube. Record all observations.

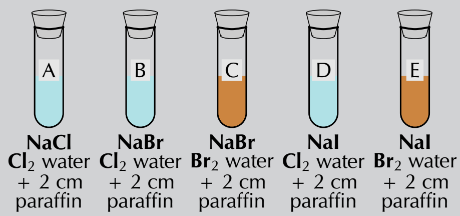


7. Pour 1 cm^3 of bromine water into the test tubes labelled C and E.

   Note any changes to the test tube. Record all observations.



8. Using a plastic dropper transfer approximately 2 cm height of the paraffin into each test tube. Use a cork or rubber stopper to close the test tube, hold it firmly in place with your thumb, and shake the mixture.



9. Use the redox table to write overall net ionic equations for the reactions in test tubes B, D and E.
10. Using your understanding of solubility rules (*like dissolves like*) explain why the layer of paraffin became coloured in test tubes B, D and E. Explain what caused the paraffin to become coloured in test tube C.
11. Why was there no reaction and no colour change in test tube A?

Results:

- In test tube A Cl_2 is present, but is not coloured and so no change in the colour of the paraffin is observed.
- In test tubes B and D the chlorine displaces the Br^- and I^- ions. Br_2 and I_2 are formed.

  Br_2 is a brown colour in paraffin, while I_2 is a purple colour in paraffin.
- In test tube C the Br_2 was already present and coloured the paraffin a brown colour.
- In test tube E the Br_2 displaced the I^- ions to form I_2 which would turn the paraffin a purple colour.

Conclusion:

Halogen molecules are non-polar. They will therefore dissolve in a non-polar solvent such as paraffin. The paraffin layer will become the colour of the halogen present in the solution. Chlorine is the most likely of these three halogens to be reduced, followed by bromine, and then iodine. This can be seen on the standard electrode potential table as chlorine has the largest, positive electrode potential of the three halogens.

FACT

The EMF of a cell is a the same as the **voltage** across a disconnected cell (*electric circuit theory*). A voltmeter is effectively a high resistance ammeter, so a very small current will flow when a voltmeter reading is taken (although this is too small to be noticeable).

TIP

Remember:

Standard conditions are:

$p = 101{,}3$ kPa

$C = 1$ mol.dm^{-3}

$T = 298$ K.

In standard cell notation the **anode half-cell** is always written on the left and the **cathode half-cell** is always written on the right.

The reducing agent is being oxidised. Oxidation is a loss of electrons at the anode.

The oxidising agent is being reduced. Reduction is a gain of electrons at the cathode.

Exercise 13 – 7: Using standard electrode potentials

1. If silver was added to a solution of copper(II) sulfate, would it displace the copper from the copper(II) sulfate solution? Explain your answer.
2. If zinc is added to a solution of magnesium sulfate, will the zinc displace the magnesium from the solution? Give a detailed explanation for your answer.
3. If aluminium is added to a solution of cobalt sulfate, will the aluminium displace the cobalt from the solution? Give a detailed explanation for your answer.
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 282H 2. 282J 3. 282K

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EMF of a cell ESCRM

Using the example of the zinc and copper half-cells, we know that when these two half-cells are combined, zinc will be the oxidation half-reaction and copper will be the reduction half-reaction. A voltmeter connected to this cell will show that the zinc electrode is more negative than the copper electrode.

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ ($E^\circ = +0{,}34$ V) $Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ ($E^\circ = -0{,}76$ V)

The reading on the meter will show the potential difference between the two half-cells. This is known as the **EMF** of the cell. The higher the EMF, the greater amount of energy released per unit charge.

DEFINITION: *EMF of a cell*

The EMF of a cell is defined as the maximum potential difference between two electrodes or half-cells in a galvanic cell.

It is important to be able to calculate the EMF of an electrochemical cell. To calculate the EMF of a cell:

- take the E° of the atom that is being reduced
- subtract the E° of the atom that is being oxidised

The reason for defining a reference electrode becomes obvious now. Potential differences can be calculated from the electrode potentials (determined relative to the hydrogen half-cell) without having to construct the cells themselves each time.

You can use any one of the following equations:

- $E^\circ_{(cell)} = E^\circ$(reduction half-reaction) - E°(oxidation half-reaction)
- $E^\circ_{(cell)} = E^\circ$(oxidising agent) - E°(reducing agent)
- $E^\circ_{(cell)} = E^\circ$(cathode) - E°(anode)

So, for the Zn-Cu cell: $E^\circ_{(cell)} = 0{,}34 - (-0{,}76) = 0{,}34 + 0{,}76 = 1{,}10$ V

DEFINITION: *Standard EMF*

Standard EMF (E°_{cell}) is the EMF of a galvanic cell operating under standard conditions. The symbol ° denotes standard conditions.

Worked example 10: Calculating the EMF of a cell

QUESTION

A cell contains a solid lead anode in a gold ion solution.

1. Represent the cell using standard notation.
2. Calculate the cell potential (EMF) of the electrochemical cell.

SOLUTION

Step 1: Find appropriate reactions on the table of standard electrode potentials

The reaction involves lead and gold.

$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s)$ (E° V = $-0{,}13$ V) $Au^{3+}(aq) + 3e^- \rightleftharpoons Au(s)$ (E° V = $+1{,}50$ V)

Step 2: Which metal is more likely to be reduced, which is more likely to be oxidised?

The E° of lead is a small, negative value and the E° of gold is a large, positive number. Therefore lead is more easily oxidised than gold, and gold is more easily reduced than lead.

Step 3: Determine which metal is the cathode and which is the anode

Oxidation is loss at the anode, therefore lead is the anode.

Reduction is gain at the cathode, therefore gold is the cathode.

Step 4: Represent the cell using standard cell notation

The anode is always written on the left, the cathode on the right.

$Pb(s)|Pb^{2+}(aq)||Au^{3+}(aq)|Au(s)$

Step 5: Calculate the cell potential

$E^\circ_{(cell)} = E^\circ_{(cathode)} - E^\circ_{(anode)}$

$E^\circ_{(cell)} = E^\circ_{(gold)} - E^\circ_{(lead)} = +1{,}50 - (-0{,}13) = +1{,}63$ V

Worked example 11: Calculating the EMF of a cell

QUESTION

Calculate the cell potential of the electrochemical cell in which the following reaction takes place, and represent the cell using standard notation.

$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$

SOLUTION

Step 1: Find appropriate reactions on the table of standard electrode potentials

$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s)$ ($E^\circ = -2{,}37$ V) $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ ($E^\circ = 0{,}00$ V)

Step 2: Which element is more likely to be reduced, which is more likely to be oxidised?

The E° of magnesium is a more negative value than the E° of hydrogen. Therefore magnesium is more easily oxidised than hydrogen, and hydrogen is more easily reduced than magnesium.

Step 3: Determine which metal is the cathode and which is the anode

Oxidation is loss at the anode, therefore magnesium is the anode.

Reduction is gain at the cathode, therefore hydrogen is the cathode reaction.

Step 4: Represent the cell using standard notation

The anode is always written on the left, the cathode on the right. The hydrogen

TIP

Spontaneous
positive EMF
Non-spontaneous
negative EMF

Table 13.4: Using EMF to determine cell spontaneity.

FACT
One can perform experiments to predict whether a reaction will be spontaneous or not. It turns out that the sign of the EMF is equivalent to whether a cell reaction is spontaneous or not. Those reactions that are spontaneous have a positive EMF and those reactions that are non-spontaneous have a negative EMF.

electrode includes an inert platinum plate.
$Mg(s)|Mg^{2+}(aq)||H^{+}(aq), H_2(g)|Pt(s)$

Step 5: Calculate the cell potential

$E_{(cell)} = E^{\circ}_{(cathode)} - E^{\circ}_{(anode)}$

$E_{(cell)} = E^{\circ}_{(hydrogen)} - E^{\circ}_{(magnesium)} = 0{,}00 - (-2{,}37) = +2{,}37$ V

Exercise 13 – 8: Standard electrode potentials

1. In your own words, explain what is meant by the 'electrode potential' of a metal.
2. Calculate the EMF for each of the following standard electrochemical cells:
   a) $Mn(s)|Mn^{2+}(aq)||H^{+}(aq), H_2(g)|Pt(s)$
   b) $Fe(s)|Fe^{3+}(aq)||Fe^{2+}(aq)|Fe(s)$
   c) $Cr(s)|Cr^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$
   d) $Pb(s)|Pb^{2+}(aq)||Hg^{2+}(aq)|Hg(\ell)$
3. Given the following two half-reactions:
   $Fe^{3+}(aq) + e^{-} \rightleftharpoons Fe^{2+}(aq)$
   $MnO_4^{-}(aq) + 8H^{+}(aq) + 5e^{-} \rightleftharpoons Mn^{2+}(aq) + 4H_2O(\ell)$
   a) Give the standard electrode potential for each half-reaction.
   b) Which reaction takes place at the cathode and which reaction takes place at the anode?
   c) Represent the electrochemical cell using standard notation (the cathode and anode reactions take place at inert platinum electrodes).
   d) Calculate the EMF of the cell
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 282M 2a. 282N 2b. 282P 2c. 282Q 2d. 282R 3. 282S

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Spontaneity ESCRN

You can see from the table of reduction potentials (Table 13.2) that different metals have different reactivities. Some are reduced more easily than others. You can also say that some are oxidised more easily than others.

For example, copper ($E^{\circ} = +0{,}34$ V) is more easily reduced than zinc ($E^{\circ} = -0{,}76$ V). Therefore, if a reaction involves the reduction of copper and the oxidation of zinc, it will occur spontaneously. If, however, it requires the oxidation of copper and the reduction of zinc, it will not occur spontaneously.

To predict whether a reaction occurs spontaneously you can look at the sign of the EMF value for the cell. If the EMF is **positive** then the reaction is **spontaneous**. If the EMF is **negative** then the reaction is **not spontaneous**.

Look at the following example to help you to understand how to predict whether a reaction will take place spontaneously or not.

In the reaction $Pb^{2+}(aq) + 2Br^-(aq) \rightleftharpoons Br_2(\ell) + Pb(s)$ the two half-reactions are as follows:

$Pb^{2+}(aq) + 2e^- \rightleftharpoons Pb(s)$ ($E^\circ = -0{,}13$ V) $Br_2(\ell) + 2e^- \rightleftharpoons 2Br^-(aq)$ ($E^\circ = +1{,}06$ V)

EMF = E°(reduction half-reaction) - E°(oxidation half-reaction)

EMF = E°(lead) - E°(bromide)

EMF = $-0{,}13$ V - 1,06 = $-1{,}19$ V

The sign of the EMF is negative, therefore this reaction will not take place spontaneously. Let's look at the reasoning behind this in more detail.

Look at the electrode potential for the first half-reaction. The negative value shows that lead loses electrons more easily than bromine, in other words it is easily oxidised. However, in the original equation, lead ions (Pb^{2+}) are being reduced. This part of the reaction is not spontaneous.

For the second half-reaction the positive electrode potential value shows that bromine is more easily reduced than lead: bromine will more easily gain electrons to become Br^-. This is not what is happening in the original equation and therefore this is also not spontaneous.

It therefore makes sense that the reaction will not proceed spontaneously.

Worked example 12: Determining spontaneity

QUESTION

Will copper react with dilute sulfuric acid (H_2SO_4)? You are given the following half-reactions:

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ ($E^\circ = +0{,}34$ V) $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ ($E^\circ = 0{,}00$ V)

SOLUTION

Step 1: Determine the overall equation for this reaction

The question asked is if copper metal will react with dilute sulfuric acid, therefore the reactants are Cu(s) and $H^+(aq)$:

$Cu(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2(g)$

Step 2: Which reactant should be oxidised, and which should be reduced?

$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$, therefore copper needs to be oxidised.

$2H^+(aq) + 2e^- \rightarrow H_2(g)$, therefore hydrogen ions needs to be reduced.

Step 3: Predict whether the reaction will be spontaneous or non-spontaneous

Copper has a larger, positive E° than hydrogen. Therefore copper is more easily reduced than hydrogen, and hydrogen is more easily oxidised than copper.

The reaction is non-spontaneous.

Step 4: Calculate the EMF of the cell

$E^\circ_{(cell)} = E_{(reduction)} - E_{(oxidation)}$

$E^\circ_{(cell)} = E_{(hydrogen\ ions)} - E_{(copper)} = 0{,}00\text{ V} - (+0{,}34\text{ V}) = -0{,}34\text{ V}$

Step 5: Is the reaction spontaneous?

The EMF is negative, therefore the reaction is non-spontaneous.

Worked example 13: Determining spontaneity

QUESTION

Will zinc react with dilute hydrochloric acid (HCl)? You are given the following half-reactions:

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ ($E^\circ = -0,76$ V) $2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ ($E^\circ = 0,00$ V)

SOLUTION

Step 1: Determine the overall equation for this reaction

The question asked is if zinc metal will react with dilute hydrochloric acid, therefore the reactants are Zn(s) and H^+:

$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

Step 2: Which reactant should be oxidised, and which should be reduced?

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$, therefore zinc needs to be oxidised.

$2H^+(aq) + 2e^- \rightarrow H_2(g)$, therefore hydrogen ions need to be reduced.

Step 3: Predict whether the reaction will be spontaneous or non-spontaneous

Zinc has a larger, negative E° than hydrogen. Therefore zinc is more easily oxidised than hydrogen, and hydrogen is more easily reduced than copper.

The reaction is spontaneous.

Step 4: Calculate the EMF of the cell

$E^\circ_{(cell)} = E_{(reduction)} - E_{(oxidation)}$

$E^\circ_{(cell)} = E_{(hydrogen\ ions)} - E_{(zinc)} = 0,00 \text{ V} - (-0,76 \text{ V}) = +0,76 \text{ V}$

Step 5: Is the reaction spontaneous?

The EMF is positive, therefore the reaction is spontaneous.

Exercise 13 – 9: Predicting spontaneity

1. Will the following reaction take place spontaneously or not? Show all your working.
   $2Ag(s) + Cu^{2+}(aq) \rightarrow Cu(s) + 2Ag^+(aq)$
2. Nickel metal reacts with an acid, H^+ (aq) to produce hydrogen gas.
   a) Write an equation for the reaction, using the table of electrode potentials.
   b) Predict whether the reaction will take place spontaneously. Show your working.
3. Zinc is added to a beaker containing a solution. Will a spontaneous reaction take place if the solution is:
   a) $Mg(NO_3)_2$ b) $Ba(NO_3)_2$ c) $Cu(NO_3)_2$ d) $Cd(NO_3)_2$
4. State whether the following solutions can be stored in an aluminium container.
   a) $CuSO_4$ b) $ZnSO_4$ c) NaCl d) $Pb(NO_3)_2$
5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 282T 2. 282V 3a. 282W 3b. 282X 3c. 282Y 3d. 282Z
4a. 2832 4b. 2833 4c. 2834 4d. 2835

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13.7 Applications of electrochemistry ESCRP

Electrochemistry has a number of different uses, particularly in industry. The principles of cells are used to make **electrical batteries**. In science and technology, a battery is a device that stores chemical energy and makes it available in an electrical form. Batteries are made of electrochemical devices such as one or more galvanic cells or fuel cells. Batteries have many uses including in:

- torches
- electrical appliances such as cellphones (long-life alkaline batteries)
- digital cameras (lithium batteries)
- hearing aids (silver-oxide batteries)
- digital watches (mercury/silver-oxide batteries)
- military applications (thermal batteries)

In this section we are going to look at a few examples of the uses of electrochemistry in industry.

FACT
A **fuel cell** converts the chemical potential energy produced by the oxidation of fuels (e.g. hydrogen gas, hydrocarbons, alcohols) into electrical energy.

Electroplating ESCRQ

The electrolytic cell can be used for **electroplating**.

DEFINITION: *Electroplating*

The process of coating an electrically conductive object with a thin layer of metal using an electrical current.

Electroplating occurs when an electrically conductive object is coated with a layer of metal using electrical current. Sometimes, electroplating is used to give a metal particular properties or for aesthetic reasons:

- corrosion protection
- abrasion and wear resistance
- the production of jewellery

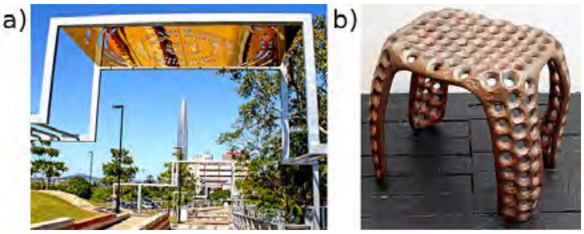


Figure 13.11: **a)** An electroplated piece of aluminium artwork and **b)** a wax stool electroplated in copper.

Electro-refining (also sometimes called **electrowinning**) is electroplating on a large scale. Copper plays a major role in the electrical industry as it is very conductive and is used in electric cables. One of the problems though is that copper must be pure if it is to be an effective current carrier. One of the methods used to purify copper, is electrowinning (copper ore is processed into impure *blister copper*, which is then deposited as pure copper through electroplating). The copper electrowinning process is as follows:

1. A bar of **impure** copper containing other metallic impurities acts as the **anode**.
2. The **cathode** is made up of **pure** copper with few impurities.
3. The electrolyte is a solution of aqueous $CuSO_4$ and H_2SO_4.
4. When current passes through the cell, electrolysis takes place:

FACT
Remember that electrolytic cells are used to transform reactants into products by turning electric current into chemical potential energy.

- The **impure copper anode** oxidises to form Cu^{2+} ions in solution. The **anode decreases in mass**.
  Cu(s) $\rightarrow Cu^{2+}(aq) + 2e^-$
- At the cathode reduction of positive copper ions takes place to produce **pure copper metal**. The **cathode increases in mass**.
  $Cu^{2+}(aq) + 2e^- \rightarrow$ **Cu(s)** (>99% purity)

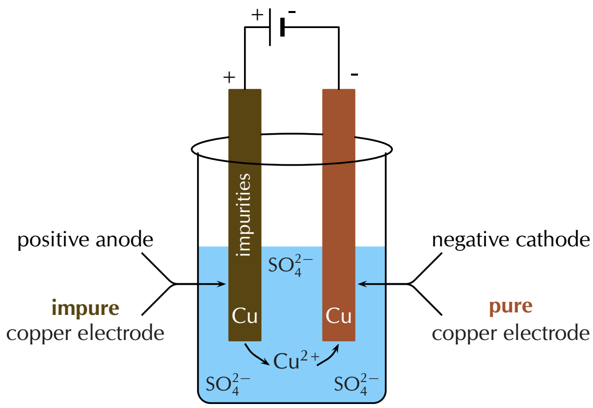


Figure 13.12: A simplified diagram to illustrate what happens during the electrowinning of copper.

5. The other metal impurities do not dissolve (Au(s), Ag(s)) and form a solid sludge at the bottom of the tank or remain in solution (Zn(aq), Fe(aq) and Pb(aq)) in the electrolyte.

The chloralkali industry ESCRR

The chlorine-alkali (chloralkali) industry is an important part of the chemical industry, which produces **chlorine** and **sodium hydroxide** through the electrolysis of the raw material **brine**. Brine is a saturated solution of sodium chloride (NaCl) that is obtained from natural salt deposits.

DEFINITION: *Brine*

A saturated aqueous solution of sodium chloride.

The products of the chloralkali industry have a number of important uses:

Chlorine is used:

- to purify water
- as a disinfectant
- in the production of:
  - hypochlorous acid (used to kill bacteria in drinking water)
  - paper, food
  - antiseptics, insecticides, medicines, textiles, laboratory chemicals
  - paints, petroleum products, solvents, plastics (such as polyvinyl chloride)

Sodium hydroxide (also known as 'caustic soda') is used to:

- make soap and other cleaning agents
- purify bauxite (the ore of aluminium)

- make paper
- make rayon (artificial silk)

One of the problems of producing chlorine and sodium hydroxide is that when they are produced together the chlorine combines with the sodium hydroxide to form chlorate (ClO^-) and chloride (Cl^-) ions. This leads to the production of sodium chlorate, NaClO, a component of household bleach.

To overcome this problem the chlorine and sodium hydroxide must be separated from each other so that they don't react. There are three industrial processes that have been designed to overcome this problem. All three methods involve **electrolytic cells**.

1. **The Mercury Cell**

   In the mercury-cell (Figure 13.13):

   - The anode is a carbon electrode suspended from the top of a chamber.
   - The cathode is liquid mercury that flows along the floor of this chamber.
   - The electrolyte is brine (NaCl solution) that is passed through the chamber.
   - When an electric current is applied to the circuit, chloride ions in the electrolyte are oxidised at the anode to form chlorine gas.

     $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$
   - At the same time sodium ions are reduced at the anode to solid sodium. The solid sodium dissolves in the mercury making a sodium/mercury amalgam.

     $Na^+(aq) + Hg(\ell) + e^- \rightarrow Na(Hg)$
   - The amalgam is poured into a separate vessel, where it decomposes into sodium and mercury.
   - The sodium reacts with water in the vessel and produces sodium hydroxide and hydrogen gas, while the mercury returns to the electrolytic cell to be used again.

     $2Na(Hg) + 2H_2O(\ell) \rightarrow 2NaOH(aq) + H_2(g) + Hg(\ell)$

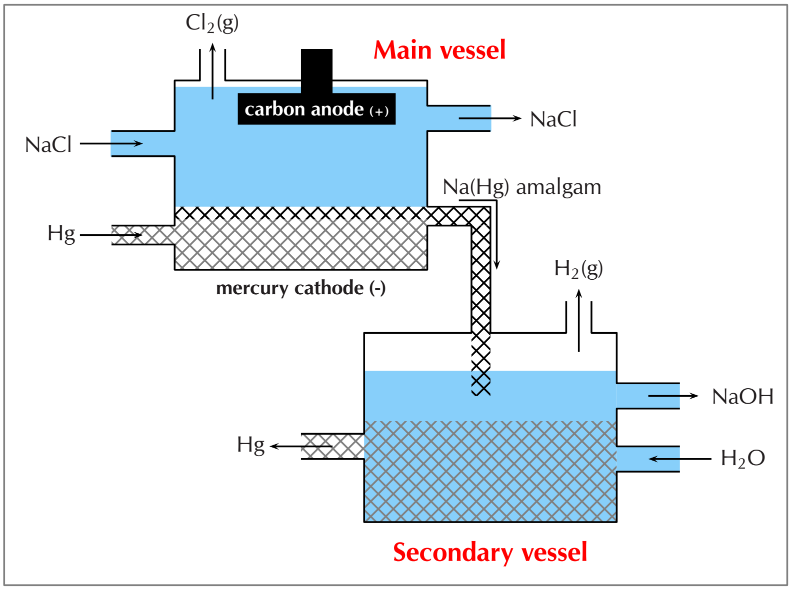


Figure 13.13: The mercury cell.

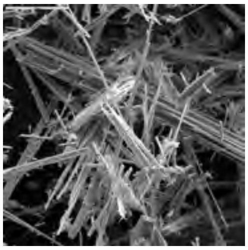
The following animation gives a good demonstration of how a mercury cell works.

See video: 2836 at www.everythingscience.co.za

FACT

In a mercury cell the sodium dissolves in the liquid mercury to form a liquid amalgam of the two metals. This separates the Cl^- and Na^+ ions.

FACT
To separate the chlorine from the sodium hydroxide, the two half-cells were traditionally separated by a porous asbestos diaphragm, which needed to be replaced every two months.



This was damaging to the environment, as large quantities of asbestos had to be disposed. Asbestos is toxic to humans, and causes cancer and lung problems. Today, the asbestos is being replaced by other polymers, which do not need to be replaced as often, and are not toxic.

This method only produces a fraction of the chlorine and sodium hydroxide that is used by industry as it has certain disadvantages:

- mercury is expensive and toxic
- some mercury always escapes with the brine that has been used
- mercury reacts with the brine to form mercury(II) chloride
- the mercury cell requires a lot of electricity
- although the chlorine gas produced is very pure, mercury has to be removed from the sodium hydroxide and hydrogen gas mixture.

In the past the effluent was released into lakes and rivers, causing mercury to accumulate in fish and other animals feeding on the fish. Today, the brine is treated before it is discharged so that the environmental impact is lower.

2. **The Diaphragm Cell**

In the diaphragm-cell (Figure 13.14):

- a porous diaphragm divides the electrolytic cell into an anode compartment and a cathode compartment
- brine is introduced into the anode compartment and flows through the diaphragm into the cathode compartment
- an electric current is passed through the brine causing the salt's chloride ions and sodium ions to move to the electrodes
- Chlorine gas is produced at the anode
  $2Cl^{-}(aq) + 2e^{-} \rightarrow Cl_2(g)$
- At the cathode, sodium ions react with water forming caustic soda (NaOH) and hydrogen gas.
  $2Na^{+}(aq) + 2H_2O(\ell) + e^{-} \rightarrow 2NaOH(aq) + H_2(g)$
- Some NaCl salt remains in the solution with the caustic soda and can be removed at a later stage.

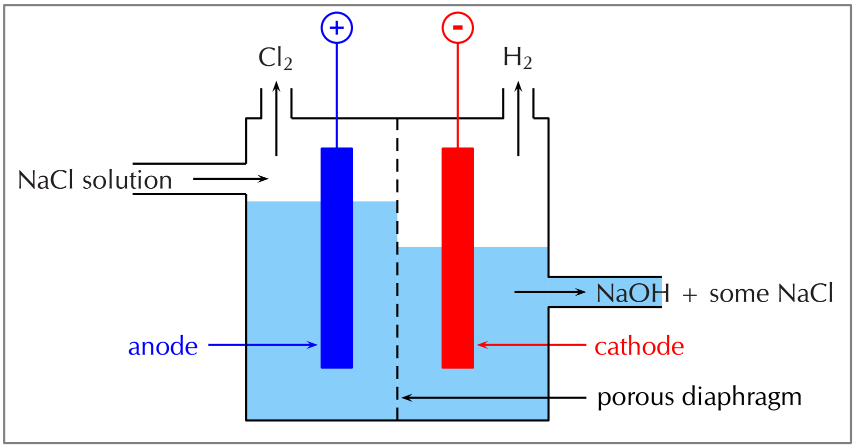


Figure 13.14: The diaphragm cell.

The following animation gives a good demonstration of how a diaphragm cell works.

See video: 2837 at www.everythingscience.co.za

The advantages of the diaphragm cell are:

- uses less energy than the mercury cell
- does not contain toxic mercury

It also has disadvantages however:

- the sodium hydroxide is much less concentrated and not as pure
- the chlorine gas often contains oxygen gas as well

- the process is less cost-effective as the sodium hydroxide solution needs to be concentrated and purified before it can be used

The Membrane Cell

The membrane cell (Figure 13.15) is very similar to the diaphragm cell, and the same reactions occur. The main differences are:

3.
- the two electrodes are separated by an ion-selective membrane, rather than by a diaphragm
- the membrane structure allows cations to pass through it between compartments of the cell but does not allow anions to pass through (this has nothing to do with the size of the pores, but rather with the **charge on the ions**)
- brine is pumped into the anode compartment, and only the positively charged sodium ions pass into the cathode compartment, which contains pure water

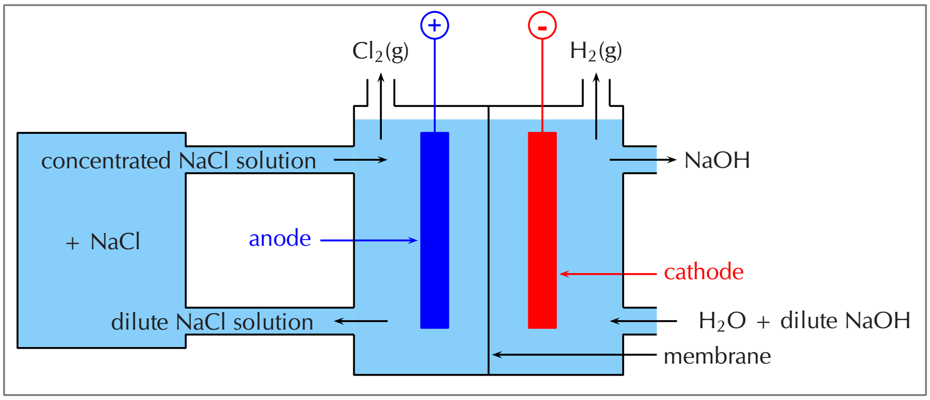


Figure 13.15: The membrane cell.

The following animation gives a good demonstration of how a membrane cell works.

See video: 2838 at www.everythingscience.co.za

- At the **positively charged anode**, Cl^- ions from the brine are oxidised to Cl_2 gas.

  $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$
- At the **negatively charged cathode**, hydrogen ions in the water are reduced to hydrogen gas.

  $2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-$
- The Na^+ ions flow through the membrane to the cathode compartment and react with the remaining hydroxide (OH^-) ions from the water to form sodium hydroxide (NaOH).

  $Na^+(aq) + OH^-(aq) \rightarrow NaOH(aq)$
- The chloride ions cannot pass through the membrane, so the chlorine does not come into contact with the sodium hydroxide in the cathode compartment. The sodium hydroxide is removed from the cell. The overall equation is as follows:

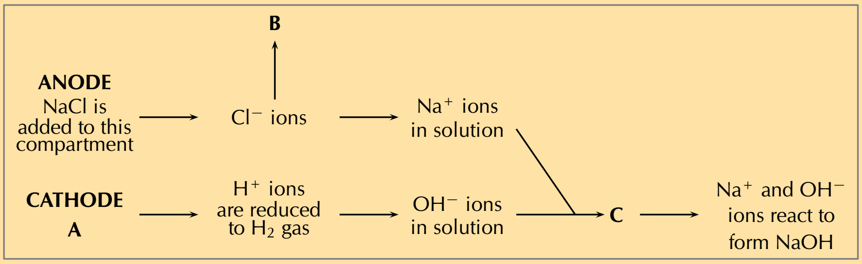
  $2NaCl(aq) + 2H_2O(\ell) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(g)$

The advantages of using this method are:

- the sodium hydroxide that is produced is very pure because it is kept separate from the sodium chloride solution
- the sodium hydroxide has a relatively high concentration
- this process uses the least electricity of all three cells
- the cell is cheaper to operate than the other two cells
- the cell does not contain toxic mercury or asbestos

Exercise 13 – 10: The chloralkali industry

1. Refer to the flow diagram, which shows the reactions that take place in the membrane cell, and then answer the questions that follow.



   a) What liquid is present in the cathode compartment at **A**?

   b) Identify the gas that is produced at **B**.

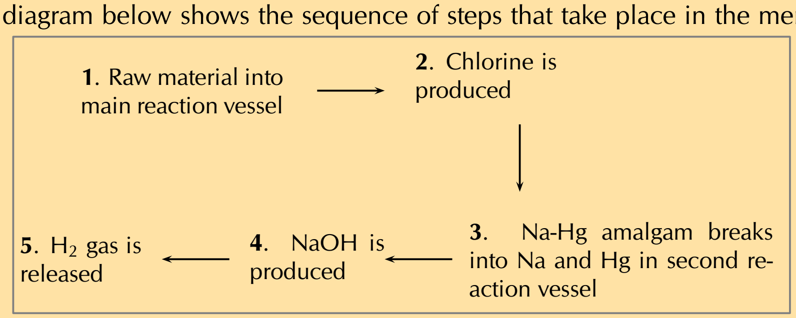
   c) Explain one feature of this cell that allows the Na^+ and OH^- ions to react at **C**.

   d) Give a balanced equation for the reaction that takes place at **C**.

2. Summarise what you have learnt about the three types of cells in the chloralkali industry by completing the table below:

	Mercury cell	**Diaphragm cell**	**Membrane cell**
Main raw material			
Mechanism of separating Cl_2 and NaOH			
Anode reaction			
Cathode reaction			
Purity of NaOH produced			
Purity of Cl_2 produced			
Energy consumption			
Environmental impact			
Cost of production			

3. The diagram below shows the sequence of steps that take place in the mercury cell.



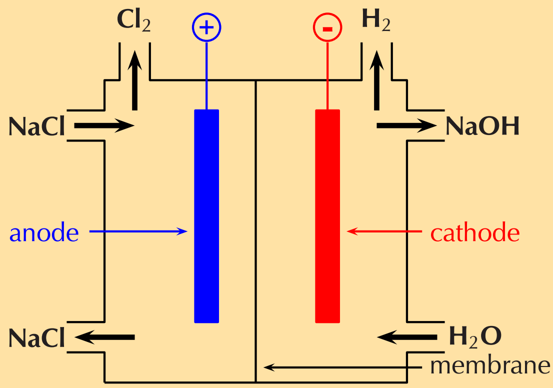
   a) Name the 'raw material' in step **1**.

   b) Give the chemical equation for the reaction that produces chlorine gas in step 2.

c) What other product is formed in step 2.

d) Name the reactants in step 4.

4. Approximately 30 million tonnes of chlorine are used throughout the world annually. Chlorine is produced industrially by the electrolysis of brine. The diagram represents a membrane cell used in the production of Cl_2 gas.



a) What ions are present in the electrolyte in the left hand compartment of the cell?

b) Give the equation for the reaction that takes place at the anode.

c) Give the equation for the reaction that takes place at the cathode and forms a gas.

d) What ion passes through the membrane while these reactions are taking place?

e) Chlorine is used to purify drinking water and swimming pool water. The substance responsible for this process is the weak acid, hypochlorous acid (HOCl). One way of putting HOCl into a pool is to bubble chlorine gas through the water. Give an equation showing how bubbling $Cl_2(g)$ through water produces HOCl.

f) A common way of treating pool water is by adding 'granular chlorine'. Granular chlorine consists of the salt calcium hypochlorite, $Ca(OCl)_2$. Give an equation showing how this salt dissociates in water. Indicate the phase of each substance in the equation.

(IEB Paper 2, 2003)

5. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 2839 2. 283B 3. 283C 4. 283D

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FACT

Bauxite is a rock that contains a large amount of aluminium oxide (Al_2O_3) and aluminium hydroxide ($Al(OH)_3$) as well as many other aluminium containing minerals. Bauxite is the richest source of aluminium when compared with any other common rock, and is the best aluminium ore.

The extraction of aluminium ESCRS

Aluminium is a commonly used metal in industry, where its properties of being both light and strong can be utilised. It is used in the manufacture of products such as aeroplanes and motor cars. The metal is present in deposits of bauxite. Bauxite is a mixture of silicas, iron oxides and hydrated alumina ($Al_2O_3.xH_2O$).

Electrolysis can be used to extract aluminium from bauxite. The process described below produces 99% pure aluminium:

1. Aluminium is melted along with cryolite (Na_3AlF_6) which acts as the electrolyte. Cryolite helps to lower the melting point and dissolve the ore.
2. The carbon rod anode provides a site for the oxidation of O^{2-} and F^- ions. Oxygen and fluorine gas are given off at the anode and also result in anode

consumption.

$2O^{2-}(aq) \rightarrow O_2(g) + 4e^-$ $2F^-(aq) \rightarrow F_2(g) + 2e^-$

3. At the cathode cell lining, the Al^{3+} ions are reduced and metal aluminium deposits on the lining. $Al^{3+}(aq) + 3e^- \rightarrow Al(s)$ (99% purity)
4. The AlF_6^{3-} electrolyte is stable and remains in its molten state.

The overall reaction is as follows: $2Al_2O_3(s) \rightarrow 4Al(s) + 3O_2(g)$

The only problem with this process is that the reaction is **endothermic** and large amounts of electricity are needed to drive the reaction. The process is therefore very expensive.

13.8 Chapter summary ESCRT

See presentation: 283F at www.everythingscience.co.za

- **Oxidation** is the loss of electrons and **reduction** is the gain of electrons.
- A **redox reaction** is one where there is always a change in the oxidation numbers of the elements that are involved in the reaction.
- It is possible to **balance** redox equations using the half-reactions that take place within the overall reaction.
- An **electrochemical reaction** is one where either a chemical reaction produces an electric current, or where an electric current causes a chemical reaction to take place.
- In a **galvanic cell** a chemical reaction produces a current in the external circuit. An example is the zinc-copper cell.
- An **electrolytic cell** is an electrochemical cell that uses electricity to drive a non-spontaneous reaction. In an electrolytic cell, **electrolysis** occurs, which is a process of separating elements and compounds using an electric current.
- Cells have a number of **components**. They consist of two **electrodes**, which are connected to each other by an external circuit wire.
- In a **galvanic** cell each electrode is placed in a separate container in an **electrolyte** solution. The two electrolytes are connected by a **salt bridge**.
- In an **electrolytic** cell both electrodes are placed in the same container in an **electrolyte** solution.
- One of the electrodes is the **anode**, where **oxidation** takes place. The **cathode** is the electrode where **reduction** takes place.
- In a galvanic cell, the build up of electrons at the anode sets up a potential difference between the two electrodes, and this causes a **current** to flow in the external circuit.
- Standard cell notation for a galvanic cell has the anode on the left and the cathode on the right. For example:
  $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$
  $|$ = a phase boundary (solid/aqueous) $||$ = the salt bridge
- Different metals have different **reaction potentials**. The reduction potential of metals (in other words, their ability to ionise), is recorded in a **table of standard electrode reduction potentials**. The more negative the value, the greater the tendency of the metal to be oxidised. The more positive the value, the greater the tendency of the metal to be reduced.
- The values on the table of standard electrode potentials are measured relative to the **standard hydrogen electrode**.

- The EMF of a cell can be calculated using one of the following equations:
  E°(cell) = E°(reduction half-reaction) - E°(oxidation half-reaction)
  E°(cell) = E°(oxidising agent) - E°(reducing agent)
  E°(cell) = E°(cathode) - E°(anode)
- It is possible to predict whether a reaction is **spontaneous** or not, either by looking at the sign of the cell EMF or by comparing the electrode potentials of the two half-cells.
- Industrial applications of cells include electrolysis (the electrowinning of copper), in the chloralkali industry (**mercury**, **diaphragm** and **membrane** cells), as well as the extraction of metals from ores (e.g. aluminium from bauxite).

A comparison of galvanic and electrolytic cells

It should be much clearer now that there are a number of differences between *galvanic* and *electrolytic* cells. Some of these differences have been summarised in Table 13.5.

	Galvanic cell	Electrolytic cell
Chemical reactions	spontaneous reactions	non-spontaneous reactions
Energy changes	Chemical potential energy from chemical reactions is converted to electrical energy	An external supply of electrical energy causes a chemical reaction to occur
Anode	is **negative**, oxidation occurs at anode	is **positive**, oxidation occurs at anode
Cathode	is **positive**, reduction occurs at cathode	is **negative**, reduction occurs at cathode
Cell set-up	two half-cells, one electrode in each, connected by a salt-bridge	one cell, both electrodes in cell, no salt-bridge
Electrolyte solution(s)	The electrolyte solutions are kept separate from one another, and are connected by a salt bridge	The cathode and anode are in the same electrolyte
Applications	batteries	Electrolysis e.g. of water, NaCl, electroplating

Table 13.5: A comparison of galvanic and electrolytic cells.

See video: 283G at www.everythingscience.co.za

Exercise 13 – 11:

1. A cell is set up. There is a negative charge on the anode.
   a) What type of cell is this?
   b) What type of reaction is occurring at the cathode?
   c) Can the electrodes have the same reaction potential?
   d) Draw and label a diagram of this cell.
2. For each of the following, say whether the statement is **true** or **false**. If it is false, re-write the statement correctly.
   a) The anode in an electrolytic cell has a negative charge.
   b) The reaction $2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$ is an example of a redox reaction.

c) Lead is a stronger oxidising agent than nickel.

3. Sulfur dioxide gas can be prepared in the laboratory by heating a mixture of copper turnings and concentrated sulfuric acid in a suitable flask.
   a) Derive a balanced ionic equation for this reaction using the half-reactions that take place.
   b) Give the E° value for the overall reaction.
   c) Explain why it is necessary to heat the reaction mixture.

4. For each of the following questions, choose the **one correct** answer.
   a) Which one of the following reactions is a redox reaction?
      i. $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(\ell)$
      ii. $AgNO_3(s) + NaI(s) \rightarrow AgI(s) + NaNO_3(s)$
      iii. $2FeCl_3(aq) + 2H_2O(\ell) + SO_2(aq) \rightarrow H_2SO_4(aq) + 2HCl(aq) + 2FeCl_2(aq)$
      iv. $BaCl_2(s) + MgSO_4(s) \rightarrow MgCl_2(s) + BaSO_4(s)$

      (IEB Paper 2, 2003)
   b) Consider the reaction represented by the following equation:
      $Br_2(\ell) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(s)$
      Which one of the following statements about this reaction is correct?
      i. bromine is oxidised
      ii. bromine acts as a reducing agent
      iii. the iodide ions are oxidised
      iv. iodine acts as a reducing agent

      (IEB Paper 2, 2002)
   c) The following equations represent two hypothetical half-reactions:
      $X_2 + 2e^- \rightleftharpoons 2X^-$ ($E° = +1{,}09$ V)
      and $Y^+ + e^- \rightleftharpoons Y$ ($E° = -2{,}80$ V)
      Which one of the following substances from these half-reactions has the greatest tendency to lose electrons?
      i. X^-
      ii. X_2
      iii. Y
      iv. Y^+
   d) Which one of the following redox reactions will **not** occur spontaneously at room temperature?
      i. $Mn(s) + Cu^{2+}(aq) \rightarrow Mn^{2+}(aq) + Cu(s)$
      ii. $Fe^{3+}(aq) + 3NO_2(g) + 3H_2O(\ell) \rightarrow Fe(s) + 3NO_3^-(aq) + 6H^+(aq)$
      iii. $Zn(s) + SO_4^{2-}(aq) + 4H^+(aq) \rightarrow Zn^{2+}(aq) + SO_2(g) + 2H_2O(\ell)$
      iv. $5H_2S(g) + 2MnO_4^-(aq) + 6H^+(aq) \rightarrow 5S(s) + 2Mn^{2+}(aq) + 8H_2O(\ell)$
      (DoE Exemplar Paper 2, 2008)

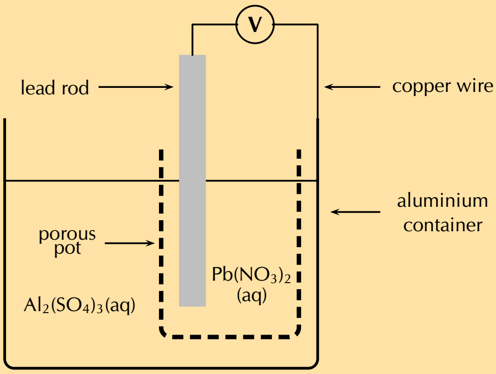
5. In order to investigate the rate at which a reaction proceeds, a learner places a beaker containing concentrated nitric acid on a sensitive balance. A few pieces of copper metal are dropped into the nitric acid.
   a) Use the relevant half-reactions from the table of standard electrode potentials to derive the balanced net ionic equation for the reaction that takes place in the beaker.
   b) What chemical property of nitric acid is illustrated by this reaction?
   c) List three observations that this learner would make during the investigation. (IEB Paper 2, 2005)

6. The following reaction takes place in an electrochemical cell:
   $Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$
   a) Give an equation for the oxidation half-reaction.
   b) Which metal is used as the anode?

c) Determine the EMF of the cell under standard conditions.

(IEB Paper 2, 2003)

7. A galvanic cell is constructed by placing a lead rod in a porous pot containing a solution of lead nitrate. The porous pot is then placed in a large aluminium container filled with a solution of aluminium sulfate. The lead rod is then connected to the aluminium container by a copper wire and voltmeter as shown.



a) Define the term *reduction*.

b) Write balanced equations for the reactions that take place at...

i. the cathode

ii. the anode

c) Write a balanced net ionic equation for the reaction which takes place in this cell.

d) In which direction do electrons flow in the copper wire? (Al to Pb or Pb to Al)

e) What are the two functions of the porous pot?

f) Calculate the EMF of this cell under standard conditions.

(IEB Paper 2, 2005)

8. An electrochemical cell is made up of a copper electrode in contact with a copper nitrate solution and an electrode made of an unknown metal M in contact with a solution of MNO_3. A salt bridge containing a KNO_3 solution joins the two half-cells. A voltmeter is connected across the electrodes. Under standard conditions the reading on the voltmeter is +0,46 V.

a) Sketch the electrochemical cell.

b) Write down the standard conditions which apply to this electrochemical cell.

c) $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$. Identify the metal M. Show calculations.

d) Use the standard electrode potentials to write down equations for the:

i. anode half-reaction

ii. cathode half-reaction

iii. overall cell reaction

e) What is the purpose of the salt bridge?

(IEB Paper 2, 2004)

9. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 283H   2a. 283J   2b. 283K   2c. 283M   3. 283N   4a. 283P
4b. 283Q   4c. 283R   4d. 283S   5. 283T   6. 283V   7. 283W
8. 283X

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CHAPTER 14

The chemical industry

14.1 Introduction ESCRV

South Africa has a population of over 50 million people, and this number is increasing every year. Therefore, maintaining healthy crops plays an important role in providing enough food for the nation. Fertilisers are used to provide sufficient nutrients to the soil in order to sustain optimum crop yields. The fertiliser industry is therefore an important chemical industry in South Africa.

Figure 14.1: Farming beetroot in South Africa.

In this chapter we will investigate what fertilisers are, why they are important, how they are produced and what their impact on the environment is.

14.2 Nutrients ESCRW

The importance of nutrients ESCRX

Nutrients are very important for life to exist. An **essential nutrient** is a chemical that a plant needs to be able to grow from a seed and complete its life cycle, but that it cannot produce itself. The same is true for animals. A **macronutrient** is a chemical element that is required in *large quantities* by the plant or animal, whereas a **micronutrient** is only needed in *small amounts* for a plant or an animal to function properly.

DEFINITION: *Nutrient*

A nutrient is a chemical substance used for the metabolism and the physiology of an organism and is absorbed from the environment.

In plants, examples of *macronutrients* include carbon (C), hydrogen (H), oxygen (O), nitrogen (N), phosphorus (P) and potassium (K), while *micronutrients* include iron (Fe), chlorine (Cl), copper (Cu) and zinc (Zn).

Nutrients that plants absorb **from the soil** are called *mineral nutrients*. Mineral nutrients have to dissolve in the water in the soil before plants can absorb them. **Non-mineral nutrients** are *not provided by the soil* itself, but from the environment. For example oxygen and hydrogen can be obtained from rain water, while carbon, in the form of carbon dioxide (CO_2) is obtained from the air. The source of each of these nutrients for plants, and their function, is summarised in Table 14.1 (for non-mineral nutrients) and Table 14.2 (for mineral nutrients).

Non-mineral nutrients	Where the nutrient is found (source)	Why the nutrient is needed (function)
Carbon (C)	Carbon dioxide in the air	Component of organic molecules such as carbohydrates, lipids and proteins
Hydrogen (H)	Water	Component of organic molecules
Oxygen (O)	Water	Component of organic molecules

Table 14.1: The source and function of the non-mineral macronutrients in plants.

Mineral nutrients	Where the nutrient is found (source)	Why the nutrient is needed (function)
Nitrogen (N)	Nitrogen compounds in the soil	Part of plant proteins and chlorophyll, also boosts plant growth
Phosphorus (P)	Phosphate compounds in the soil	Needed for photosynthesis, blooming and root growth
Potassium (K)	Potassium compounds in the soil	Cell building, part of chlorophyll, and reduces diseases in plants

Table 14.2: The source and function of the mineral macronutrients in plants.

Animals need similar nutrients in order to survive. However, since animals do not photosynthesise, they rely on plants to supply them with the nutrients they need. Think of the human diet - we cannot synthesise our own food and so we either need to eat vegetables, fruits and seeds (all of which are direct plant products) or the meat of other animals which would have fed on plants. It is important therefore that plants are always able to access the nutrients that they need so that they will grow and provide food for other forms of life.

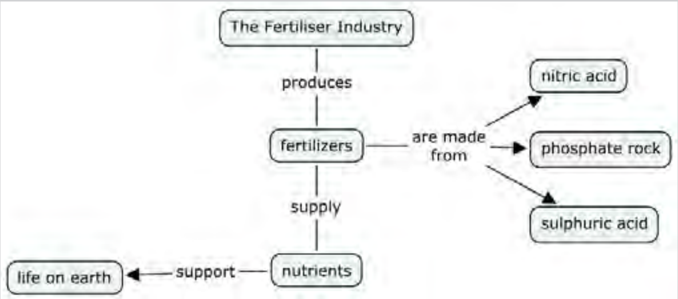
FACT
The chemical elements mentioned in this chapter as nutrients really form part of larger nutrient molecules such as proteins or amino acids, carbohydrates, fats, and vitamins.

TIP
Blooming refers to plants forming flowers. Phosphorus is particularly good for flowering plants.

Activity: Concept map

In this activity you are going to make a concept map of this section as a summary that you can use when you study.

1. Read through the content on the first two pages of this chapter.
2. Highlight the most important concepts and words.
3. Use the **words that you have highlighted** to make a concept map of this section. Each concept must be linked to another concept using *linking words*. Below you will find an example that you can use as a starting point for your own concept map.



4. Add to this concept map as you progress through this chapter.

14.3 Fertilisers ESCRY

The role of fertilisers ESCRZ

Refer to the Chemical Industries Resource Pack for more information.

Plants are only able to absorb nutrients from the soil when they are *dissolved* in water so that their root systems can absorb the nutrients. Nitrogen gas (N_2) for example, cannot be absorbed in the gas form, and needs to be changed into an ion that is soluble in water, for example the nitrate ion (NO_3^-). In the same way phosphorus is absorbed as phosphate ions (PO_4^{3-}).

Plants will grow best in soil that is able to provide sufficient nutrients to them. Natural processes, like the nitrogen cycle, replace nutrients in the soil. However, these natural processes of maintaining soil nutrients take a long time. As populations grow and the demand for food increases, the soil has to supply nutrients faster than the natural processes can sustain. This places more and more strain on the soil to be able to produce a crop.

Often, cultivation practices don't give the soil enough time to recover and to replace the nutrients that have been lost. Today, fertilisers play a very important role in *restoring soil nutrients* so that crop yields can stay high. Some of these fertilisers are **organic** (e.g. compost, manure and fishmeal), which means that they started off as part of something living. Industrial (commercial) fertilisers are **inorganic** (for example ammonium nitrate (NH_4NO_3) or super phosphates ($Ca(H_2PO_4)_2$) and have the advantage of being in a *soluble form* that can be absorbed by a plant immediately.

DEFINITION: *Fertiliser*

A fertiliser is a chemical compound that is given to a plant to promote growth.

Fertilisers usually provide the three major plant nutrients (nitrogen, phosphorus and potassium). Fertilisers are in general applied to the soil so that the nutrients are absorbed by plants through their roots.

DEFINITION: *Organic and inorganic fertilisers*

Organic fertilisers are made from *natural products*, like manure or compost. **Inorganic** fertilisers refer to *industrially produced compounds*.

Figure 14.2: Examples of **(a)** organic (compost) and **(b)** inorganic (industrially made) fertilisers indicating the NPK ratio that will be discussed later.

Photos by Kessner Photography on wikipedia and Elvera Viljoen in the UCT Chemical Engineering Chemical Industries Resource pack.

The NPK ratio ESCS2

Fertiliser packaging contains a set of numbers, for example 6:1:5 (see Figure 14.2). These numbers are called the **NPK ratio**, and they give the mass ratio of nitrogen, phosphorus and potassium in the fertiliser.

The NPK ratio expresses the content of each nutrient as a percentage of N, P and K in this order. A number in brackets after this ratio indicates the percentage by mass of N, P and K that is present in the fertiliser (what percentage of the total fertiliser is N, P and K). For example:

N	P	K	
3	1	5	(38)

38% of the total fertiliser is N, P or K.

% **N**: 3 in every 9 parts of the 38% contains nitrogen (N)

% **P**: 1 in every 9 parts of the 38% contains phosphorus (P)

% **K**: 5 in every 9 parts of the 38% contains potassium (K)

FACT
A starter fertiliser is usually used when seeds or young plants are planted. It has high concentrations of nitrogen, to boost plant growth, and phosphorus to boost root growth.

Table 14.3 below gives an idea of the amounts of nitrogen, phosphorus and potassium there are in different types of fertilisers.

Description	Grade (NPK ratio)
Ammonium nitrate	34:0:0
Urea	46:0:0
Bone meal	4:21:1
Seaweed	1:1:5
Starter fertilisers	18:24:6
Equal NPK fertilisers	12:12:12
High N, low P and medium K fertilisers	25:5:15

Table 14.3: Common grades of some fertiliser materials.

Depending on the types of plants you are growing, and the growth stage they are in, you may need to use a fertiliser with a slightly different ratio. For example, if you want to **encourage root growth** in your plant you might choose a fertiliser with a **greater ratio of phosphorus** in it. The main functions of each nutrient (N, P, K) are given in Table 14.4. Remember to refer to Tables 14.1 and 14.2 as well.

Nutrient	Promotes	When to use
Nitrogen (N)	leafy plant growth, faster plant growth	on lawns and other plants with lots of green leaves
Phosphorus (P)	strong roots, healthy fruit, blooming	on flowers and flower beds
Potassium (K)	disease resistance, growth of fruit	on fruit bearing plants

Table 14.4: The types of plant growth different nutrients promote.

Some countries express the phosphorus content as P_2O_5 and potassium content as K_2O. South Africa expresses the NPK ratio in terms of the elements present as explained above. The rest of the fertiliser (62%) is made up of fillers, such as gypsum, lime and sand. Other *micronutrients*, such as calcium (Ca), sulfur (S) and magnesium (Mg), are often added to the mixture.

Worked example 1: NPK ratios

QUESTION

Calculate the mass of nitrogen (N), phosphorus (P) and potassium (K) that is present in 500 g of industrial fertiliser with a NPK ratio of 5:2:3 (40).

SOLUTION

Step 1: Determine the mass of the total nutrients in the fertiliser sample.

40% of the sample contains nutrients, therefore:

40% of 500 g = 0,4 x 500 g = 200 g which contains nutrients.

TIP
If the percentage purity is not given (the number in brackets – e.g. 40 in the worked example above), then you can assume that it is a 100% pure fertiliser and no fillers were added.

Step 2: Determine the mass of the specific component in the sample

For every 5 units of nitrogen there are 2 units of phosphorus and 3 units of potassium so the total number of units is 10.

5 of the 10 units are nitrogen: $\frac{5}{10} \times 200 \text{ g} = 100 \text{ g}$ will be nitrogen (N).

2 of the 10 units are phosphorus: $\frac{2}{10} \times 200 \text{ g} = 40 \text{ g}$ will be phosphorus (P).

3 of the 10 units are potassium: $\frac{3}{10} \times 200 \text{ g} = 60 \text{ g}$ will be potassium (K).

Worked example 2: NPK ratios

QUESTION

Calculate the mole ratio of an industrial fertiliser with the NPK ratio of 5:2:3 (40).

SOLUTION

Step 1: Calculate the mass of N, P and K in a 500 g sample of the fertiliser

As we are calculating the mole ratio, any amount of fertiliser can be assumed.

As in the previous worked example the ratio is 5:2:3 (40), so the mass of nutrients will be 200 g.

The mass of N $= \frac{5}{10} \times 200 \text{ g} = 100 \text{ g}$.

The mass of P $= \frac{2}{10} \times 200 \text{ g} = 40 \text{ g}$.

The mass of K $= \frac{3}{10} \times 200 \text{ g} = 60 \text{ g}$.

Step 2: Calculate the number of moles for each element in this sample of fertiliser

$n = \frac{m}{M}$ The molar mass (M) for N $= 14{,}0 \text{ g.mol}^{-1}$

Therefore n for nitrogen $= \frac{100 \text{ g}}{14{,}0 \text{ g.mol}^{-1}} = 7{,}14 \text{ mol}$

M for P $= 31{,}0 \text{ g.mol}^{-1}$

Therefore n for phosphorus $= \frac{40 \text{ g}}{31{,}0 \text{ g.mol}^{-1}} = 1{,}29 \text{ mol}$

M for K $= 39{,}1 \text{ g.mol}^{-1}$

Therefore n for potassium $= \frac{60 \text{ g}}{39{,}1 \text{ g.mol}^{-1}} = 1{,}53 \text{ mol}$

Step 3: Determine the mole ratio

N:P:K mole ratio = 7,14 : 1,29 : 1,53

To get the mole ratio divide all the numbers by the smallest number: 5,5 : 1 : 1,2
The ratio should be expressed as whole numbers: 55 : 10 : 12

Extension

N : P_2O_5 : K_2O

As mentioned previously, in South Africa the NPK ratio is expressed in elemental form, for example the P in the ratio refers to the mass of the element phosphorus (P). However, in some countries the ratios are expressed in compound form: $N:P_2O_5:K_2O$. Nitrogen still refers to the element nitrogen, but phosphorus and potassium refer to the compounds phosphorus pentoxide (P_2O_5) and potassium oxide (K_2O). This example will show you how to calculate the mass or moles when a compound ratio is given.

Worked example 3: NPK ratios in other countries

QUESTION

Calculate the moles of phosphorus (P) in 120 g of a fertiliser with the $N:P_2O_5:K_2O$ mass ratio of 4:3:8 (50).

SOLUTION

Step 1: Calculate the mass of nutrients in the sample of fertiliser

50% of the sample contains nutrients.

50% of 120 g = 0,5 x 120 g = 60 g which contains nutrients.

Step 2: Calculate the mass of P_2O_5

The $N:P_2O_5:K_2O$ ratio is 4 : 3 : 8. 4 + 3 + 8 = 15.

3 of the 15 units are P_2O_5. The mass of $P_2O_5 = \frac{3}{15} \times 60\text{ g} = 12\text{ g}$.

Step 3: Calculate the number of moles of P_2O_5

$n = \frac{m}{M}$ $M(P_2O_5) = 2 \times 31{,}0\text{ g.mol}^{-1} + 5 \times 16{,}0\text{ g.mol}^{-1} = 142{,}0\text{ g.mol}^{-1}$

Therefore $n(P_2O_5) = \frac{12\text{ g}}{142{,}0\text{ g.mol}^{-1}} = 0{,}0845\text{ mol}$

Step 4: Calculate the number of moles of P

For every 1 mole of P_2O_5 there are 2 moles of P atoms.

Therefore, n(P) = 2 x 0,0845 mol = 0,17 mol

Activity: NPK ratios

Use Table 14.3, if needed, to answer the following questions:

1. Calculate the mass of phosphorus that is present in 250 g of bone meal.
2. Calculate the moles of nitrogen that are present in 200 g of ammonium nitrate (NH_4NO_3).
3. Calculate the mass of potassium that is present in 100 g of seaweed.
4. Calculate the number of moles of potassium in 100 g of a commercial fertiliser with the $N:P_2O_5:K_2O$ mass ratio of 5:1:2 (40).

Exercise 14 – 1: The role of fertilisers

1. A starter fertiliser is usually used when seeds or young plants are planted. Use Table 14.3 to answer the following questions.
   a) What is the NPK ratio of a starter fertiliser?
   b) Which nutrient is present in the lowest concentration?

c) Do you think an equal NPK fertiliser will be a good starter fertiliser for new plants? Explain your answer using information from Tables 14.1 - 14.4.

2. Do you think bone meal is a good fertiliser to use for flowering plants? Explain your answer using Tables 14.1 - 14.4.
3. Determine the mass of each of the nutrient components in a 600 g bag of the fertiliser shown in Figure 14.2 (it is produced in South Africa).
4. You have determined that your vegetable garden will need 12 g of nitrogen. The fertiliser you have has 4:2:3 (40) written on the packet. How many grams of fertiliser will you need for your garden?
5. How many moles of potassium is present in 150 g of commercial fertiliser with the N:P_2O_5:K_2O mass ratio of 1:2:5.
6. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 283Y 2. 283Z 3. 2842 4. 2843 5. 2844

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14.4 The fertiliser industry ESCS3

The industrial production of fertilisers ESCS4

For more information on this section refer to the Chemical Industries Resource Pack.

The industrial production of fertilisers involves several processes. Figure 14.3 summarises how a number of different industrial processes are used to manufacture a variety of fertilisers. In the following sections we will discuss the various processes indicated on this diagram.

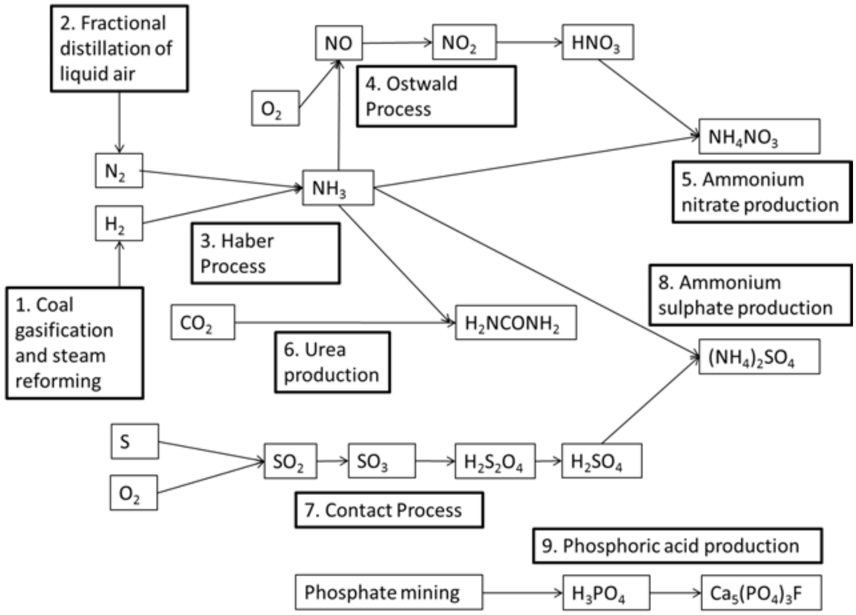

Figure 14.3: The industrial manufacturing of fertilisers.

Producing hydrogen: Coal gasification and steam reforming at Sasol ESCS5

Fossil fuels are the main source of industrial hydrogen. Hydrogen can be generated from **natural gas** or **coal**. These processes are used by Sasol at their Gas-to-Liquid (GTL) and Coal-to-Liquid (CTL) facilities. Hydrogen is usually produced by the **steam reforming** of methane gas (natural gas). At high temperatures (700 – 1100°C), steam (H_2O) reacts with **methane** (CH_4) in an endothermic reaction to yield **syngas**, a mixture of *carbon monoxide* (CO) and *hydrogen* (H_2).

$CH_4(g) + H_2O(g) \rightarrow CO(g) + 3H_2(g)$

During a second stage, which takes place at a lower temperature of about 130°C, the exothermic reaction generates additional hydrogen. This is called a **water gas shift reaction**.

$CO(g) + H_2O(g) \rightarrow CO_2(g) + H_2(g)$

Essentially, the **oxygen** (O) atom is stripped from the additional water (steam) to oxidise CO to CO_2. This oxidation also provides energy to maintain the reaction.

Coal can also be used to produce syngas in a similar way to natural gas. The reactions are shown here:

$C(s) + H_2O(g) \rightarrow CO(g) + H_2(g)$

$CO(g) + H_2O(g) \rightarrow CO_2(g) + H_2(g)$

Remember that **yield** describes the quantity of product in a container relative to the maximum product possible.

FACT
Sasol is an international company that was founded in Sasolburg, South Africa, in 1950. It employs over 34 000 people in at least 38 countries and has interests in synthetic fuels, mining, oil, gas and chemistry.

TIP
You are not required to know this information in as much depth as is provided here, but the extra information should help your understanding of the subject.

Obtaining nitrogen: Fractional distillation of liquefied air ESCS6

Fractional distillation is a separation method. It uses the difference in boiling temperatures of the components of a mixture to separate those components.

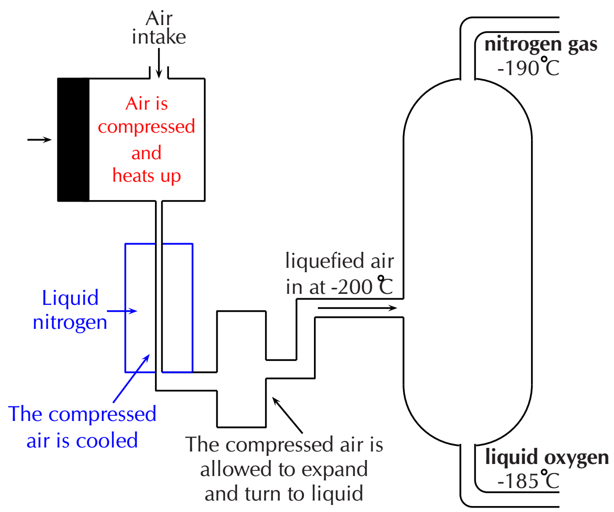

Figure 14.4: A fractional distillation column for the separation of the components of air.

- The mixture is heated to convert the components into the vapour (gas) phase.
- The vapour mixture is then pumped into a tall separation column (called a fractional distillation column), usually at the bottom of the column.
- As the vapour mixture moves up the column and cools, the different components (called fractions) condense as the temperature drops below the various boiling

FACT
1 atm = 101,3 kPa

FACT
The forward reaction of the Haber process is exothermic, so the forward reaction is favoured by low temperatures. However, low temperatures slow all chemical reactions. So, the Haber process requires high temperatures, and the ammonia is removed as soon as it is formed to prevent it being used in the reverse reaction.

point temperatures.

- These fractions are collected using collection trays.
- The fractions can be removed from the mixture in this way and thus the components are separated.

This process is used to separate the components of air or crude oil. Air is a mixture of gases, mainly nitrogen (N_2) and oxygen (O_2). Liquefied air (compressed and cooled to $-200°C$) is pumped into the fractional distillation column. Nitrogen gas has the lowest boiling point temperature and is collected at the top of the column (Figure 14.4).

Producing ammonia: The Haber process ESCS7

Ammonia (NH_3) plays an important role in the manufacturing process of fertilisers. The industrial process used to produce ammonia is called the **Haber process**. In this reaction nitrogen gas and hydrogen gas react to produce ammonia gas. The equation for the Haber process is:

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

This reaction takes place in the presence of an iron (Fe) catalyst under high pressure (200 atmospheres (atm)) and temperature (450 – 500°C) conditions.

By the 20th century, a number of methods had been developed to *fix* atmospheric nitrogen, in other words to make it usable for plants. Sources of nitrogen for fertilisers included *saltpetre* ($NaNO_3$) from Chile, and guano. Guano is the droppings of seabirds, bats and also seals. In the early 20th century, before the start of the First World War, the Haber process was developed by two German chemists, Fritz Haber and Karl Bosch. They determined what the best conditions were in order to get a high yield of ammonia, and found these to be **high temperature** and **high pressure**. During World War I, the ammonia produced by the Haber process was used to make explosives.

Producing nitric acid: The Ostwald process ESCS8

The **Ostwald process** is used to produce nitric acid from ammonia. Nitric acid can then be used in reactions that produce fertilisers. Ammonia is converted to nitric acid in a three-step process.

Firstly ammonia is **oxidised** by heating it with oxygen, in the presence of a platinum (Pt) catalyst, to form nitrogen monoxide (NO) and water. This step is strongly **exothermic**, which makes it a useful heat source. The reaction that takes place is:

$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$

In the second step, nitrogen monoxide is oxidised again to yield nitrogen dioxide (NO_2) according to the following reaction:

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

In step three, nitrogen dioxide is absorbed by water to produce nitric acid (HNO_3) as follows:

$3NO_2(g) + H_2O(\ell) \rightarrow 2HNO_3(aq) + NO(g)$

Nitrogen monoxide, also known as nitrogen oxide or nitric oxide, is a by-product of this reaction. The nitrogen monoxide is recycled and the acid is concentrated to the required strength (e.g., for use in further chemical processes).

Producing ammonium nitrate ESCS9

Nitric acid and ammonia can react together in an acid-base process to form the salt, ammonium nitrate (NH_4NO_3). Ammonium nitrate is soluble in water and is often used in fertilisers. The reaction is as follows:

$NH_3(\ell) + HNO_3(\ell) \rightarrow NH_4NO_3(s)$

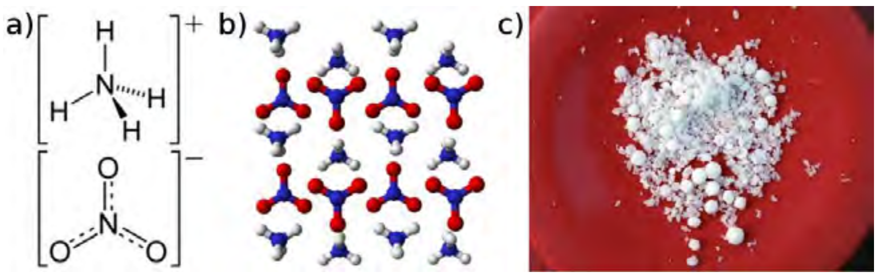

Figure 14.5: Ammonium nitrate presented as **a)** the structural formula, **b)** a three-dimensional molecular representation and **c)** solid crystals.

FACT

In 1828, the German chemist Friedrich Wöhler obtained urea by treating silver cyanate (AgOCN) with ammonium chloride (NH_4Cl):

$AgOCN + NH_4Cl \rightarrow (NH_2)_2CO + AgCl$

This was the first time an organic compound was artificially synthesised from inorganic starting materials, without the use of living organisms.

Producing urea ESCSB

Urea ($(NH_2)_2CO$) is a nitrogen-containing compound that is produced on a large scale worldwide. It has the highest nitrogen content (46,4%) of all solid nitrogen-containing fertilisers that are commonly used, and is produced by the reaction of *ammonia* with *carbon dioxide*. Two reactions are involved in producing urea and ammonium carbamate ($H_2NCOONH_4$) is an intermediate product.

$2NH_3(g) + CO_2(g) \rightleftharpoons H_2NCOONH_4(s)$

$H_2NCOONH_4(s) \rightleftharpoons (NH_2)_2CO(aq) + H_2O(\ell)$

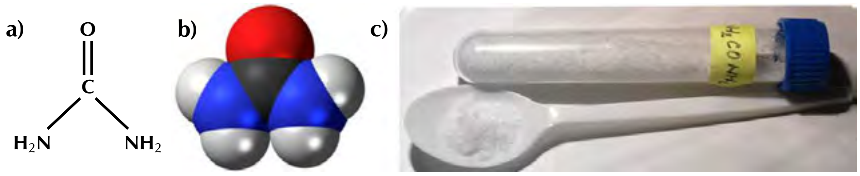

Figure 14.6: Urea presented as **a)** the structural formula, **b)** a space-filling three-dimensional model and **c)** solid crystals.

Producing sulfuric acid: The Contact process ESCSC

Sulfuric acid (H_2SO_4) is produced from sulfur, oxygen and water through the **Contact process**. In the first step, sulfur is burned to produce sulfur dioxide (SO_2):

$S(s) + O_2(g) \rightarrow SO_2(g)$

This is then oxidised to sulfur trioxide (SO_3) using oxygen in the presence of a vanadium(V) oxide catalyst:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

Finally the sulfur trioxide is treated with water to produce sulfuric acid with a purity of 98 - 99%:

$SO_3(g) + H_2O(\ell) \rightarrow H_2SO_4(\ell)$

Producing ammonium sulfate ESCSD

Ammonium sulfate ($(NH_4)_2SO_4$) can be produced industrially through a number of processes, one of which is the reaction of ammonia with sulfuric acid to produce a solution of ammonium sulfate according to the acid-base reaction below:

$$2NH_3(g) + H_2SO_4(\ell) \rightleftharpoons (NH_4)_2SO_4(s)$$

The ammonium sulfate solution is concentrated by evaporating the water until ammonium sulfate crystals are formed.

Producing phosphoric acid and super phosphates ESCSF

The production of phosphate fertilisers also includes a number of processes. The first is the production of sulfuric acid through the **Contact process**. Sulfuric acid is then used in a reaction with phosphate rock (e.g. fluorapatite ($Ca_5(PO_4)_3F$)) to produce phosphoric acid (H_3PO_4). For more information refer to the Chemical Industries Resource Pack.

$$Ca_5(PO_4)_3F(s) + 5H_2SO_4(\ell) \rightleftharpoons 5CaSO_4(s) + HF(\ell) + 3H_3PO_4(\ell)$$

Sulfuric acid can be reacted further with phosphate rock to produce single super phosphates (SSP):

$$2Ca_5(PO_4)_3F(s) + 7H_2SO_4(\ell) \rightleftharpoons 3Ca(H_2PO_4)_2(s) + 7CaSO_4(s) + 2HF(\ell)$$

If phosphoric acid reacts with phosphate rock, triple super phosphates (TSP) are produced:

$$Ca_5(PO_4)_3F(s) + 7H_3PO_4(\ell) \rightleftharpoons 5Ca(H_2PO_4)_2(s) + HF(\ell)$$

Producing compound fertilisers: The nitrophosphate process ESCSG

The nitrophosphate process is one of the processes used to make complex fertilisers. It involves acidifying calcium phosphate ($Ca_3(PO_4)_2$) with nitric acid to produce a mixture of phosphoric acid and calcium nitrate ($Ca(NO_3)_2$):

$$Ca_3(PO_4)_2(s) + 6HNO_3(\ell) + 12H_2O(\ell) \rightarrow 2H_3PO_4(aq) + 3Ca(NO_3)_2(aq) + 12H_2O(\ell)$$

When calcium nitrate and phosphoric acid react with ammonia, a compound fertiliser is produced.

$$Ca(NO_3)_2(s) + 4H_3PO_4(\ell) + 8NH_3(\ell) \rightarrow CaHPO_4(s) + 2NH_4NO_3(s) + 8(NH_4)_2HPO_4(s)$$

If potassium chloride or potassium sulfate is added, the result will be a NPK fertiliser.

Advantages and disadvantages of inorganic fertilisers ESCSH

- **Advantages**
  Organic fertilisers generally do not contain high levels of nutrients and might not be suitable to sustain high intensity crop production. **Large scale** agricultural facilities **prefer inorganic** fertilisers as they provide a more accurate control over their nutrient supply. Inorganic fertilisers are supplied in a water-soluble form which ensures that they are easily absorbed by plants. Much less inorganic fertiliser can therefore be applied to have the same result as organic fertilisers.

- **Disadvantages**

  The two major disadvantages of inorganic fertilisers are that:

  - Inorganic fertilisers must be manufactured industrially. This involves cost in terms of both chemicals and the energy involved in the production. Air pollution is also a result of these industrial processes.
  - Nutrients which are **not taken up by plants**, will either accumulate in the soil therefore **poisoning the soil**, or **leach into the ground water** where they will be washed away and accumulate in water sources like dams or underground rivers. This is discussed in more detail in Section 14.6.

Activity: The industrial production of fertilisers

Now that you have gone through this section, watch the following animation and go through the worksheet that accompanies it.

See video: 2845 at www.everythingscience.co.za

Reproductions of the worksheets are given in the teachers guide.

Exercise 14 – 2: The industrial production of fertilisers

1. Use the following processes to create your own mind map of the manufacturing of fertilisers:
   - Fractional distillation
   - Steam reforming
   - Haber process
   - Contact process
   - Ostwald process
   - you can, and should, add to this list

   Include what the reactants and the products are for each process and how these link with the other processes involved.
2. Research the process of fractional distillation (from old school science textbooks, from your teacher, or the internet) and write a paragraph on how this process works.
3. The reaction of hydrogen and nitrogen is an exothermic reversible reaction.
   a) Write a balanced equation for this reaction.
   b) Use your knowledge of chemical equilibrium and explain what effect the following will have on this equilibrium reaction:
      i. raising the temperature
      ii. raising the pressure
4. Ammonia and nitric acid react to form the compound A. Compound A is soluble in water and can be used as a fertiliser.
   a) Write down the name and formula for compound A.
   b) What type of reaction takes place to form compound A.
   c) Write a balanced equation for the reaction to form compound A.
   d) Compound A is dissolved in water. Write a balanced ionic equation for this dissolution reaction. Include the state symbols in your equation.
   e) Why is it important for fertilisers to be soluble in water?
   f) Compound A can also be used as an explosive. What precautions need to be taken when this compound is packaged and transported?

5. The following reaction can be used to manufacture hydrogen gas. It is an exothermic redox reaction.
   $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$
   a) Write down the reduction half reaction for this reaction.
   b) Which compound is the reducing agent?
   c) Which compound is the oxidising agent?
   d) Identify two conditions that will ensure a high yield of hydrogen.
6. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 2846 2. 2847 3. 2848 4. 2849 5. 284B

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14.5 Alternative sources of fertilisers ESCSJ

Industrial fertilisers have not always been used, and even today many people prefer to use natural fertilisers rather than man-made ones. For more information on alternative sources of fertiliser see the Chemical Industries Resource Pack worksheets here and here.

Organic fertilisers ESCSK

Organic fertilisers are not manufactured by man, but come from natural sources. Examples are manure, blood and bones, guano, compost and kelp products. Organic fertilisers contain lower levels of nutrients and might take longer than inorganic fertilisers to be absorbed as they generally have to degrade first. The advantage of organic fertilisers is that they increase the organic component of the soil. This improves the physical structure of the soil, which in turn increases the soil's water-holding capacity. The nutrients also tend to be released slower than those of inorganic fertilisers, decreasing their contribution to water pollution.

Manure ESCSM

Figure 14.7: Horse dung, a type of manure.

Manure is a solid waste product from animals that is widely used as an organic fertiliser in agriculture. It contains **high levels** of *nitrogen, phosphorus, potassium* and other nutrients. Manure decomposes over time through bacterial and microbial action and in the process releases these nutrients into the soil. This **slow release mechanism** is a great benefit to farmers as it limits the leaching of nutrients into the ground water, making it available to plants over a longer period of time. Manure also adds organic matter to the soil, *increasing the quality of the soil itself*.

Plants can only absorb nutrients that are dissolved in water. When manure decomposes, the nitrogen compounds are converted to a form that is soluble in water, for example nitrates (NO_3^-). The nitrates are dissolved in the moisture in the soil and plants are then able to absorb the nitrogen compounds through their root system.

Extension

Kraal manure

Phosphorus and nitrogen are often the most deficient mineral nutrients in South African soil. The average NPK formula for kraal manure found in the Eastern Cape is 3:1:4 (3). Unfortunately this is not as high in phosphorus, and has a much lower nutrient concentration, than many inorganic fertilisers. However, kraal manure is a much cheaper alternative, is readily available and does not need to be transported as far.

More information on the use of kraal manure can be found on the Agriculture, Forestry and Fisheries website.

Guano ESCSN

Figure 14.8: A cliff covered in guano.

Figure 14.9: A guano mine off the coast of Peru.

Guano is the excretion of seabirds, bats and seals. Guano consists of *ammonia, uric acid, phosphoric acid, oxalic acid* and *carbonic acid* and also has a high concentration of nitrates. The particularly high levels of phosphorus make this an **effective phosphorus fertiliser**. Guano was mined off the West Coast of South Africa as early as 1666. Since the 1840's, large scale mining of guano caused the reserves to be depleted and mining was stopped by the turn of the century.

Crop rotation ESCSP

Figure 14.10: A soy bean field.

Figure 14.11: A maize field.

Crop rotation is a farming method that is used to manage the nutrients in soil naturally. When the **same crop** is grown repeatedly in the same place, it eventually **depletes the soil** of specific nutrients. With crop rotation, one type of crop that depletes the soil of a particular kind of nutrient, is rotated with another type of crop which replaces the depleted nutrient. For example, legumes, like beans or peas, have nodules on their roots which contain nitrogen-fixing bacteria. These bacteria help 'fix' or change nitrogen into a soluble form. Legumes are therefore often alternated with plants requiring nitrogen, and soy beans can therefore be followed by maize.

Lime ESCSQ

Figure 14.12: Limestone.

Agricultural lime, or crushed limestone, can be used as an alternative fertiliser. Lime **increases the pH of soil**, making the soil less acidic, and **more soluble** for *nitrogen, potassium* and *phosphorus* compounds. These nutrients will therefore be more readily available for absorption by plants.

Potash ESCSR

Figure 14.13: Potash.

Potash is the **common name** for various mined and manufactured **salts** that contain **potassium** in the water-soluble form. The name is derived from *pot ash*, which was the main method of manufacturing potassium salts before the industrial era. Ashes from a fire were soaked in a pot of water after which the water was filtered out. The water, containing the leached potassium salts, was then evaporated to obtain a white powder known as potash.

Potash is a rich source of potassium and could contain **potassium carbonate** (K_2CO_3), **potassium chloride** (KCl), **potassium sulfate** (K_2SO_4) or **potassium nitrate** (KNO_3). Up to the 19th century potash was manufactured in asheries using wood ashes, but these declined in the late 19th century when large-scale production of potash from mineral salts was established in Germany.

Activity: Alternative sources of fertilisers

1. Work on your own to summarise one of the sections on the alternative fertiliser sources. Remember to summarise in your own words.
2. Find all the other learners in your class who have summarised the same topic as you and form a group. Share with your group members what you have summarised.
3. As a group, prepare 3 - 4 sentences on your alternative source of fertiliser. Share your information with the class in 1 - 2 minutes in a debate format.

14.6 Fertilisers and the environment ESCSS

The fertiliser industry is a very important industry in South Africa and in the world. It helps provide the nutrients that we need to ensure healthy crops to sustain life on Earth. However, fertilisers can also harm the environment if they are not used in a responsible manner. This section will focus on some of the environmental problems that excessive use of fertilisers can cause.

Eutrophication ESCST

Eutrophication (Figure 14.14) is the enrichment of an ecosystem with large quantities of chemical compounds, mostly containing the nutrients **nitrogen** and **phosphorus**. Eutrophication is considered a form of **pollution** because it promotes excessive plant growth, favouring certain species over others. In aquatic environments, the rapid growth of certain types of plants can **disrupt the normal functioning of an ecosystem**, causing a variety of problems. Human society is impacted as well, as health-related problems can occur if eutrophic conditions interfere with the treatment of drinking water. Eutrophication can also decrease the resource value of rivers, lakes, and estuaries.

Figure 14.14: An example of the effect of eutrophication (algal bloom).

DEFINITION: *Eutrophication*

Eutrophication refers to an over-supply in chemical nutrients in an ecosystem, leading to the depletion of oxygen in a water system through excessive plant growth. These chemical nutrients usually contain nitrogen or phosphorus.

In some cases, eutrophication can be a natural process that occurs very slowly over time. However, it can also be accelerated by certain human activities. **Agricultural runoff**, where excess fertilisers are washed off fields and into ground water, and **sewage**, are two of the major causes of eutrophication. The impacts of eutrophication are the following:

- **A decrease in biodiversity (the number of plant and animal species in an ecosystem)**

  When a system is enriched with nitrogen, plant growth is accelerated. When the number of plants increases in an aquatic system, it can block light from reaching deeper water. Plants also consume oxygen for respiration, depleting the oxygen content of the water, which can cause other organisms, such as fish, to die.

- **Toxicity**

  In certain instances the plants that flourish during eutrophication can be toxic and these toxins may accumulate in the food chain.

Despite these impacts, there are a number of ways to prevent eutrophication from taking place.

FACT
A buffer zone is an area that lies between two other areas, for example, a farm and a river. It is an area of land designated for environmental protection.

FACT
South Africa's Department of Water Affairs and Forestry has a *National Eutrophication Monitoring Programme*, which was set up to monitor eutrophication in impoundments such as dams, where no monitoring was taking place.

Prevention of eutrophication:

- Clean-up measures can directly remove the excess nutrients such as nitrogen and phosphorus from the water.
- Creating buffer zones near farms, roads and rivers can also help. These act as filters and cause nutrients and sediments to be deposited there instead of in the aquatic system.
- Laws relating to the treatment and discharge of sewage can also help to control eutrophication.
- A final possible intervention is nitrogen testing and modelling. By assessing exactly how much fertiliser is needed by crops and other plants, farmers can make sure that they apply only the required amount of fertiliser. This means that there is no excess to run off into neighbouring streams when it rains. This includes a cost benefit for the farmer as they are less likely to waste fertiliser.

Activity: Dealing with the consequences of eutrophication

In many cases, the damage from eutrophication is already done. As a class, discuss the following:

1. List all the possible consequences of eutrophication that you can think of.
2. Suggest ways to solve the problems that arise because of eutrophication.
3. Discuss how the public can help to prevent eutrophication.

Investigation: Fertiliser in your area

For this investigation you will be working in a group. Your task is to find out what fertilisers are used in your area and whether people know about the impact of fertilisers on the environment, especially the water sources in the area.

1. **Design a survey to find out the following:**
    - Do people use fertilisers in their gardens or the areas around their homes?
    - What type of fertilisers are they using?
    - Why are they using fertilisers?
    - How often do they apply fertilisers?
    - How much fertiliser do they use in each application?
    - Do they use organic or inorganic fertilisers?
    - Which ones do they think are better to use?
    - How do they think fertilisers influence the quality of water in the area?
2. **Collect your data**
   Take the survey to at least 10 people in your area. Record their responses.
3. **Present your findings**
   Present your findings in tables or graphs and write a one-page summary of what you have found.
4. **Interpret your findings**
   Discuss your findings:
    - Answer all the questions that were posed in the beginning of the investigation.
    - Did you find what you thought you would?

- What was different?
- What was the same?
- Why do you think this is the case?

5. **Make recommendations**

   Include some recommendations to people in your neighbourhood regarding the use of fertilisers and their impact on the environment.

   Suggestion: Report your findings in your local newspaper to promote awareness.

14.7 Chapter summary ESCSV

- The growth of South Africa's **chemical industry** was largely due to the mining industry, which requires explosives for their operation. One of South Africa's major chemical companies is **Sasol**. Other important examples of chemical industries in the country are the **chloralkali** and **fertiliser** industries.
- The **fertiliser industry** is very important in providing fertilisers with the correct nutrients in the correct quantities to ensure maximum growth for various plants and crops.
- All plants need certain **macronutrients** (e.g. carbon, hydrogen, oxygen, potassium, nitrogen and phosphorus) and **micronutrients** (e.g. iron, chlorine, copper and zinc) in order to survive. Fertilisers provide these nutrients.
- In plants, essential nutrients are obtained from the atmosphere or from the soil.
- Animals also need similar nutrients, but they obtain most of these from plants or plant products. They may also obtain them from other animals, which may have fed on plants during their life.
- **Fertilisers** can be produced industrially using a number of chemical processes: the **Haber process** reacts nitrogen and hydrogen to produce **ammonia**; the **Ostwald process** reacts oxygen and ammonia to produce **nitric acid**; the **Contact process** produces **sulfuric acid**; sulfuric acid then reacts with phosphate rock to produce **phosphoric acid**, after which phosphoric acid reacts with ground phosphate rock to produce fertilisers such as **triple superphosphate**.
- Potassium is obtained from **potash** through a mineral salt extraction process.
- Fertilisers can have a damaging effect on the environment when they are present in high quantities in ecosystems. This can lead to **eutrophication**. A number of preventative actions can be taken to reduce these impacts.

Exercise 14 – 3:

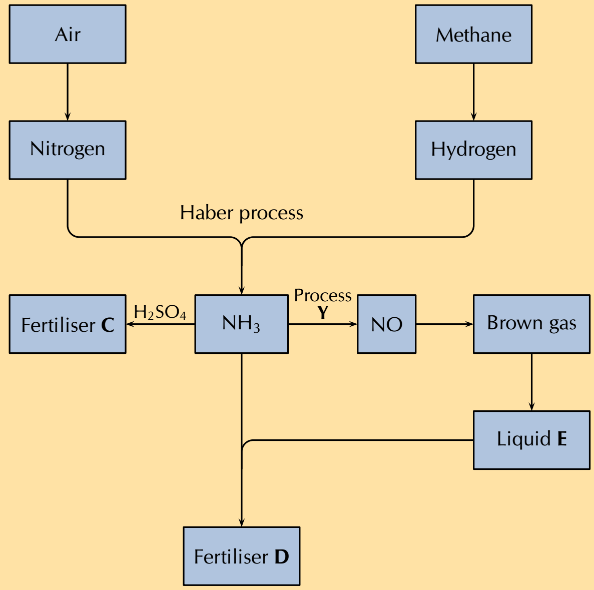
1. The following extract comes from an article on fertilisers:

   A world without food for its people and a world with an environment poisoned through the actions of man are two contributing factors towards a disaster scenario.

   Do you agree with this statement? Write down at least three arguments to substantiate your answer.
2. There is likely to be a gap between food production and demand in several parts of the world by 2020. Demand is influenced by population growth and urbanisation, as well as income levels and changes in dietary preferences.

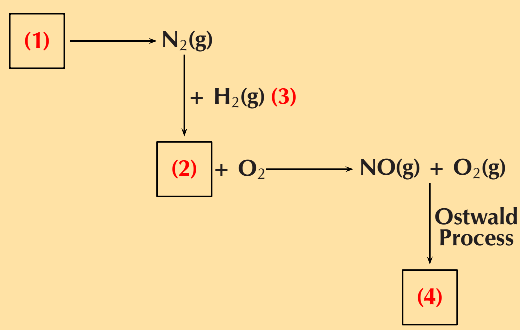
While there is an increasing world population to feed, most soils in the world used for large-scale, intensive production of crops lack the necessary nutrients for the crops. This is where fertilisers can play a role.

The flow diagram shows the main steps in the industrial preparation of two important solid fertilisers.



a) How can fertilisers play a role in increasing the soil's ability to produce crops?
b) Are the processes illustrated by the flow diagram above that of organic or inorganic fertilisers? Give a reason for your answer.
c) Write down the balanced chemical equation for the formation of the brown gas.
d) Write down the name of process **Y**.
e) Write down the chemical formula of liquid **E**.
f) Write down the chemical formulae of fertilisers **C** and **D** respectively.

3. The production of nitric acid is very important in the manufacture of fertilisers. Look at the diagram, which shows part of the fertiliser production process, and then answer the questions that follow.



   a) Name the process at (1).
   b) Name the gas at (2).
   c) Name the process at (3) that produces gas (2).
   d) Name the immediate product at (4).
   e) Name the final product of the Ostwald process (product (4) + H_2O).
   f) Name two fertilisers that can be produced from nitric acid.
4. More questions. Sign in at Everything Science online and click 'Practise Science'.

Check answers online with the exercise code below or click on 'show me the answer'.

1. 284D 2. 284C 3. 284F

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2 Momentum and impulse

Exercise 2 – 1:

1. a) $p = 0,16\ \text{kg} \cdot 44,81\text{m} \cdot \text{s}^{-1} = 7,17\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   b) $p = mv = 0,058\ \text{kg} \cdot 68,39\text{m} \cdot \text{s}^{-1} = 3,97\text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   c) $p = mv = 0,058\text{kg} \cdot 56,94\text{m} \cdot \text{s}^{-1} = 3,30\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   d) The ball with the smallest momentum gives you the least chance of being hurt and so you would choose to face Venus.

Exercise 2 – 2:

1. $\vec{F} \cdot \Delta t$
2. $\Delta p = 0,9\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ upwards

Exercise 2 – 3:

1. $p = -0,06\text{kg.m/s}$
2. $p_{\text{total}} = -279\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
3. $p_{\text{total}} = 6113\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$ to the south

Exercise 2 – 4: Collisions

1. a) $v = 19,09\ \text{m} \cdot \text{s}^{-1}$
   b) $KE_{Ti} = 1\,012\,500$ J
      $KE_{Tf} = 1\,002\,177$ J
   c) The energy difference is permanently transferred into non-elastic deformation during the collision.
   d) Inelastic. Kinetic energy was not conserved in the collision.
2. $\vec{v} = 2,5$ m·s^{-1} in the same direction as the car that was originally travelling at 20 m·s^{-1}

Exercise 2 – 5:

1. J·m·s^{-1}
2. a) Take Eastwards as positive.
      For the car:
      $\Delta p = mv_f - mv_i$
      $\Delta p = (1)(-3.4) - (1)(2) = -5.4\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
      For the train:
      $\Delta p = mv_f - mv_i$
      $\Delta p = (2)(1.2) - (2)(-1.5) = 5.4\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   b) For the car:
      Impulse $= -5.4\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
      For the train:
      Impulse $= 5.4\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   c) $\Delta t = \frac{5.4}{8} = 0.675\ \text{s}$
3. a) $\Delta p = (0.02)(200) - (0.02)(300) = -2\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   b) Impulse $= -2\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   c) $F_{\text{net}} = \frac{2}{0.0001} = 20000\ \text{N}$
4. a) If the bullet leaves the target its total momentum is: $\Delta\vec{p}$ = (0,02)(200) − (0,02)(300) = −2 kg·m·s^{-1}
   b) If the bullet remains in the target its total momentum is: $\Delta\vec{p}$ = (0,02)(0) − (0,02)(300) = −6 kg·m·s^{-1}
   c) If the bullet rebounds from the target its total momentum is: $\Delta\vec{p}$ = (0,02)(−200) − (0,02)(300) = −10 kg·m·s^{-1}
      So the bullet experiences the greatest change in momentum and hence impulse, when it rebounds from the target.
5. a) $\Delta p = mv_f - mv_i = (0.2)(-9) - (0.2)(12) = -4.2\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   b) $-4.2\ \text{kg} \cdot \text{m} \cdot \text{s}^{-1}$
   c) $F_{\text{net}}\Delta t = \Delta p$
      $F_{\text{net}} = \frac{-4.2}{0.02} = -210\ \text{N}$
6. a) impulse = 2,4 kg·m·s^{-1} into the wall
   b)

$$F = \frac{\Delta\vec{p}}{\Delta t}$$
$$F = \frac{2,4}{0,5}$$
$$= 4,8 \text{ N into the wall}$$

Exercise 2 – 6:

1. The canon recoils at 1 m·s^{-1} towards the left.
2. We calculate the momentum of the system before and after the collision.
   Before the collision the velocity of the block is 0. The momentum is:
   $\vec{p}_i = m_1\vec{v}_1 + m_2\vec{v}_2 = 0 + (1)(3) = 3$ kg·m·s^{-1}
   After the collision the momentum is:
   $\vec{p}_f = (m_1 + m_2)\vec{v} = (1 + 0,5)(2) = 3$ kg·m·s^{-1}
   Since the momentum before the collision is the same as the momentum after the collision, momentum is conserved.
3. a) $v_{f1} = 4\ \text{m} \cdot \text{s}^{-1}$ to the left
   b) $\Delta x = 6,02\ \text{m}$ to the left
4. Since the momentum before the ball hits the floor is not equal to the momentum after the ball hits the floor the law of conservation of momentum does not apply to this situation.
   We say that the system is not isolated and that there is a force acting on the ball from outside the system.
5. a) Conservation of momentum
   b) $\vec{v}_f$ = 14,63 m·s^{-1}
   c) $\vec{F}$ = 58 540 N

Exercise 2 – 7: Momentum

1. be projected downwards at the same initial speed as the first piece.
2. The change in momentum of the wall is equal to the change in momentum of the ball.
3. $3\vec{v}$
4. Since their momenta are the same, and the stopping force applied to them is the same, it will take the same time to stop each of the balls.
5. V
6. resultant force
7. t
8. $2mv$
9. The units of momentum are: $mv = \text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
   Impulse can be defined as force over total time:
   $F\Delta t = \text{N} \cdot \text{s} = \text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
   This is the same as the units for momentum.
10. a) $\Delta p = F_{\text{net}}\Delta t = \left(3 \times 10^{3}\right)\left(5 \times 10^{-4}\right) = 1.5\ \text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
    b) $\Delta p = mv$
    $1.5 = 0.06v$
    $v = 25\ \text{m}\cdot\text{s}^{-1}$
11. a) Impulse
    b) 250 m·s^{-1}
    c) The area will remain the same because the final velocity and the mass are the same. The duration of the contact between the bat and the ball will be longer as the ball is soft, so the base will be wider. In order for the area to be the same, the height must be lower. Therefore, the player can hit the softer ball with less force to impart the same velocity on the ball.
12. If the front crumples then the force of the collision is reduced. The energy of the collision would go into making the front of the car crumple and so the passengers in the car would feel less force.
13. a) $v_{f2} = -0.6\ \text{m}\cdot\text{s}^{-1} = 0.6\ \text{m}\cdot\text{s}^{-1}$ west
    b) The principle of conservation of linear momentum. The total linear momentum of an isolated system is constant.
14. a) The total linear momentum of an isolated system is constant.
    b) 0,4 m·s^{-1} to the left
15. The magnitude of the change in momentum of the ball is equal to the magnitude of the change in momentum of the Earth.
16. a) $p = mv = (2)(5) = 10\ \text{kg}\cdot\text{m}\cdot\text{s}^{-1}$
    b)

$$\begin{aligned} 0 &= m_1\vec{v}_{1f} + m_2\vec{v}_{2f} \\ -10 &= (50)\vec{v}_{2f} \\ \vec{v}_{2f} &= \frac{-10}{50} \\ &= -0{,}2\ \text{m}\cdot\text{s}^{-1} \end{aligned}$$

0,2 m·s^{-1} in the opposite direction to the jetty

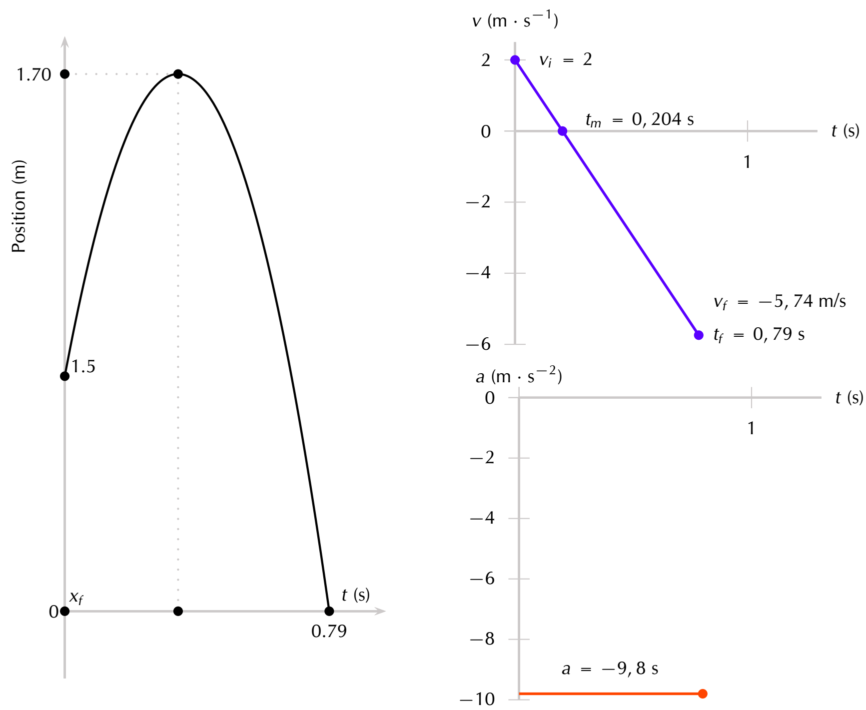
3 Vertical projectile motion in one dimension

Exercise 3 – 1: Equations of motion

1. a) 20,41 m
   b) 2,04 s
2. a) 3,96 m·s^{-1} upwards
   b) 0,4 s
3. 20,2 m·s^{-1}
4. 125 m

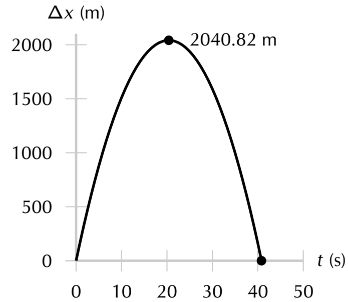
Exercise 3 – 2: Graphs of vertical projectile motion

1.

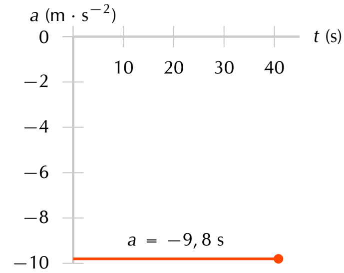


2. a) The bullet moves at a decreasing velocity upwards for 20,4 s. It then stops and drops back down for another 20,4 s until it reaches a speed of 200 m·s^{-1} which is the same speed with which it started.

b)



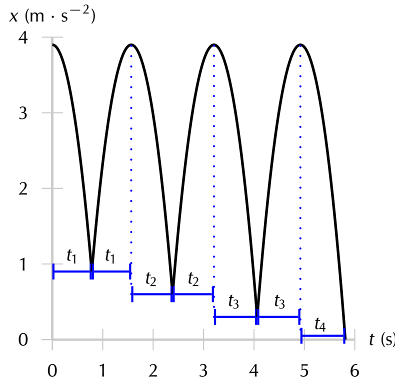
c)

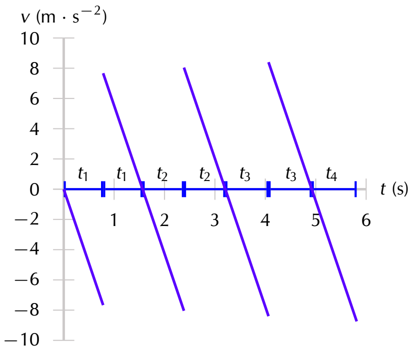


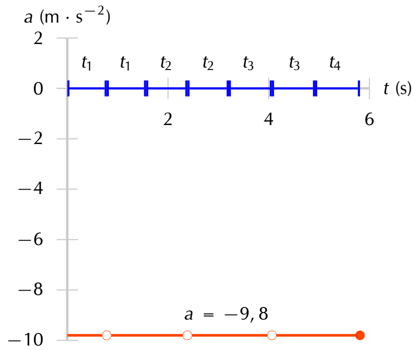
3. The object starts with an initial height of 0 m, moves upwards for 1 s, then slows and falls back to its starting position where it bounces upwards at a higher velocity, and travels upwards (higher than before) for 2 s, then stops and falls back to its starting position where it bounces upwards again at an even higher velocity, travelling over 3 s to an even higher position, before finally falling back to the ground and stopping.

5.

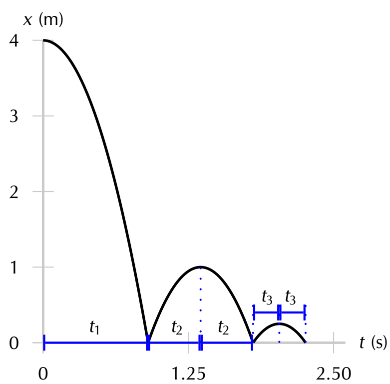
- $x_1 = 3$ m ; $t_1 = 0{,}782460$ s
- $x_2 = 3{,}3$ m ; $t_2 = 0{,}820651$ s
- $x_3 = 3{,}6$ m ; $t_3 = 0{,}857142$ s
- $x_4 = 3{,}9$ m ; $t_4 = 0{,}892142$ s

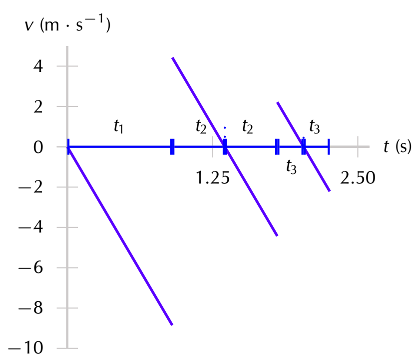


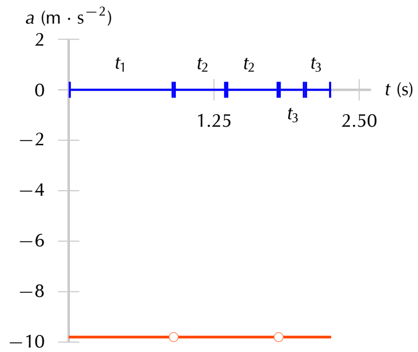




6.







Exercise 3 – 3:

1. is always downwards
2. A. The same velocity and the same acceleration.
3. b) $u_x = u_y$
4. d) displacement
5. c) the same as when it left the throwers hand.
6. C - the force is constant since the acceleration is constant.
7. The ball will travel upwards, stop at its highest point, and then fall back down to the ground. As it reaches the initial position of 5 metres above the ground, it will be travelling once again at 10 m·s^{-1} but this time in the downwards direction. Taking downwards as the +ve direction:

$$
\begin{aligned}
v_f^2 &= v_i^2 + 2a\Delta x \\
&= (10)^2 + 2(9{,}8)(5) \\
&= 198 \\
v_f &= \sqrt{198} = 14{,}07 \text{ m·s}^{-1}
\end{aligned}
$$

8. a) 5 m·s^{-1}
   b) the acceleration of the ball
   c)

$$
\begin{aligned}
\text{gradient} &= \frac{\Delta y}{\Delta x} \\
&= \frac{(5-(-15))}{(2-0)} \\
&= 10
\end{aligned}
$$

   10 m·s^{-2}

   d) According to the graph, the ball reaches the boy's hand again after 1 second (when the downward speed of 5 m·s^{-1} is the same as its initial upward speed.) Therefore the area under the graph between 1 and 2 seconds (when it reaches its maximum negative velocity) is equal to the remaining distance the ball falls to the ground. The area is equal to 10.

   Distance = 10 m.

   e) It takes 2 s until the ball bounces. Calculate the additional time from bouncing to reach maximum height. **Assume the acceleration is the same as before,** i.e. 10 m·s^{-2}.

$$
\begin{aligned}
v_f &= v_i + at \\
t &= \frac{v_f - v_i}{a} \\
&= \frac{0-10}{-10} \\
&= 1 \text{ s}
\end{aligned}
$$

   Therefore the total time taken is 3 s.

   f) Calculate the area under the graph for the final 1 second or use equation of motion to find maximum height after the bounce:

$$
\begin{aligned}
\Delta x &= v_i t + \frac{1}{2}at^2 \\
&= (10)(1) + \frac{1}{2}(-10)(1)^2 \\
&= 5 \text{ m}
\end{aligned}
$$

   Therefore the position of the ball, measured from the boy's hand is5 m-5 m = 0 m. In other words, it is the same height as the boy's hand.

9. a) 300 m
   b) 5 m·s^{-1}
   c) 600 N
   d) 61,22 kg
10. a) 8,62 m
    b) 1,63 m·s^{-2}

4 Organic molecules

Exercise 4 – 1: Representing organic compounds

1. a) Molecular formula: C_3H_8
   b) Molecular formula: C_5H_{12}
   c) Molecular formula: C_2H_6
2. a) Condensed Structural: $CH_3CHCHCH_3$
   Molecular: C_4H_8
   b) Condensed Structural: $CH_2CHCH(CH_3)CH_3$
   Molecular: C_5H_{10}

Exercise 4 – 2: The hydrocarbons

1. b) C_nH_{2n-2}
   c) i. the alkanes
   ii. the alkenes and alkynes
   d) The alkanes

Exercise 4 – 3: The alcohols

1. primary
2. tertiary
3. secondary

Exercise 4 – 4: Haloalkanes

1. a) $C_nH_{2n+1}X$
   b) Yes

Exercise 4 – 5: Carbonyl compounds

1. a) hydroxyl group
   b) carboxylic acid
   c) alcohol and carboxylic acid
2. a) ketone
   c) aldehyde
   d) ester

Exercise 4 – 6: Functional groups

1. a) alkane
   b) C_nH_{2n}
   c) esters
3. a) Esters
   b) alcohol and carboxylic acid
4. a) alkene
   b) alkane
   c) The product

Exercise 4 – 8: Naming alkanes

2. a) 3-methyl**pentane**
   b) 2,4-dimethyl**pentane**
   c) hexane
   d) ethane
   e) 2,3-dimethyl**butane**
   f) 2-methyl**butane**

Exercise 4 – 9: Naming alkenes

1. a) pent-1-ene
   b) but-2-ene
   c) but-1,3-diene

Exercise 4 – 10: Naming alkynes

2. a) 2,2-dimethylhex-3-yne
   b) prop-1-yne
   c) but-2-yne

Exercise 4 – 11: Naming hydrocarbons

2. a) pent-2-ene
   b) 2-ethylbut-1-ene
   c) hexane
   d) but-1-yne
   e) methane
   f) ethyne
   g) propene

Exercise 4 – 12: Naming haloalkanes

2. a) 2-iodopropane
   b) 2-bromo-3-methylhexane
   c) 1,4-dichlorooctane
   d) 1-fluoro-2,2-diiodobutane

Exercise 4 – 13: Naming alcohols

2. a) butan-2-ol
   b) pentan-2-ol
   c) 4-methylheptan-3,3-diol

Exercise 4 – 14: Naming aldehydes and ketones

1. a) pentanal
   b) butan-2-one
   c) hexanal
   d) methanal
   e) pentan-3-one

Exercise 4 – 15: Naming carboxylic acids

2. a) ethanoic acid
   b) 4,4-dimethylpentanoic acid
   c) propanoic acid
   d) 2,4-diethylheptanoic acid

Exercise 4 – 16: Naming esters

1. a) ethyl ethanoate
   b) pentyl methanoate
   c) propyl hexanoate

Exercise 4 – 17: Naming carbonyl compounds

2. a) 3-ethylhexan-2-one
   b) butanal
   c) propyl ethanoate
   d) methanoic acid

Exercise 4 – 19: Types of intermolecular forces

2. a) ethane
3. a) B
   b) i. CH_3CH_2COOH
      ii. $CH_2(OH)CH_2CH_2CH_3$

Exercise 4 – 20: Physical properties and functional groups

1. a) i. butane, but-1-ene
      ii. All except butane and but-1-ene

Exercise 4 – 21: Physical properties and chain length

1. c) $C_8H_{16}O_2$
2. a) i. butane, butene
      ii. pentane, pentene, hexane, hexene
4. d) hex-2-yne

Exercise 4 – 22: Physical properties and branched chains

1. a) 1-chlorobutane and 2-chloro-2-methylpropane
2. b) hexane, 3-methylpentane and 2,3-dimethylbutane

Exercise 4 – 23: Physical properties of organic compounds

1. a) Saturated
   b) aldehydes
   c) i. ethanal
      ii. ethanol
   e) **B**
2. a) butan-1-ol
   b) butan-2-ol
   c) 2-methylpropan-2-ol

Exercise 4 – 24: Alkanes as fossil fuels

3. exothermic

Exercise 4 – 25: Esters

1. a) methyl ethanoate
   b) octyl ethanoate
   c) hexyl ethanoate
   d) propyl ethanoate
2. acid-catalysed condensation
3. a) butyl ethanoate
   b) butyl pentanoate
   c) butyl heptanoate
   d) butyl methanoate

Exercise 4 – 26: Addition reactions

1. b) $CH_3CH_2(Cl)$
2. c) Hydration reaction
3. b) Unsaturated

Exercise 4 – 27: Elimination reactions

1. b) The more substituted carbon atom
2. a) saturated
   b) dehydration reaction

Exercise 4 – 28: Substitution reactions

1. tertiary alcohol

Exercise 4 – 29: Addition, elimination and substitution reactions

1. a) i. addition reaction
      ii. H_3PO_4
   c) i. No
      ii. light
   d) i. A. substitution
      ii. A. elimination
2. a) Elimination
4. a) Elimination

Exercise 4 – 30: Addition polymers

2. a) polypropene or polypropylene
   b) ethene or ethylene.
   c) chloroethene or vinyl chloride

Exercise 4 – 32: Polymers

1. a) propene
2. b) chloroethene or vinyl chloride
   d) Addition
3. b) polylactic acid
   c) Condensation

Exercise 4 – 33:

1. a) Alkenes
   b) Hydroxyl group
   c) Carboxylic acid
   d) Ethanoic acid
   e) Ethanol
   f) Ethanol
2. propanoic acid
3. 1,2-dichlorobutane
6. a) i. Alkanes
      ii. Alcohols
   e) 2-methylpropane
7. a) compound 2, ethanol
   b) iv) carboxylic acids
   c) ii) 4,4-dibromobut-1-yne
   d) iii) methyl ethanoate
8. b) ii. methyl ethanoate
   c) i. butane
9. b) 4,4-dimethylhexan-2-one
10. c) Substitution
11. b) C_nH_{2n+2}

5 Work, energy and power

Exercise 5 – 1: Work

1. Normal force:0 J, weight: 0 J, applied force: 50 J.
2. Normal force:0 J, weight: 0 J, frictional force: -50 J.
3. Normal force:0 J, weight: 0 J, frictional force: -50 J, applied force: 50 J.
4. No work is done as all the forces act perpendicular to the direction of displacement.
5. Work done by applied force is 100 J, work done by gravity is -100 J
6. An angle of 55° requires the least amount of work to be done.
7. 1800 J up stairs and 3000 J along passage.
8. 1,30 kJ
9. 50 000 J
10. 5 880 J

Exercise 5 – 2: Energy

1.

position	*EK*	*PE*	**v**
A	0	50 J	0
B	20	30 J	6,3
C	20	30 J	6,3
D	40	10 J	8,9
E	40	10 J	8,9
F	50	0 J	10
G	50	0 J	10

2. We can assume that the only force acting on the ball is gravity. The speed of the ball would be decreased when it hit the ground if air resistance were taken into account. Air resistance, just like friction on a surface, will be a force opposing the motion and will result in negative work done. By the work energy theorem, this will reduce the net work done on the object falling and therefore reduce the total change in Kinetic Energy.
3. 2,42 $m{\cdot}s^{-1}$

Exercise 5 – 3: Energy conservation

1. 9,46 $m{\cdot}s^{-1}$
2. a) 47,65 m
   b) -188 527,5 J
   c) 373,79 N
3. 141,4 $m{\cdot}s^{-1}$

Exercise 5 – 4: Power

1. V·A
2. The displacement is 0 and so there is no work done. Since power is the rate at which work is done, the power is also 0.
3. Jack did twice as much work as Jill but the same power.
4. 49 W
5. $1kWh = 1000Wh = 1000Wh \times \frac{3600}{h} =$ 3 600 000 J
6. 1143,33 W

Exercise 5 – 5:

1. 16 000 J
2. 2494,67 J
3. Power
4. $\frac{mgh}{t}$
5. Mgv
6. The best motor to use is the 3.5 kW motor. The 1.0 kW motor will not be fast enough, and the 5.5 kW motor will be too fast.
7. A box moves at constant velocity across a frictionless horizontal surface.

6 Doppler effect

Exercise 6 – 1: The Doppler effect with sound

1. The frequency of the sound gradually increases as the train moves towards you. The pitch increases. You would hear a higher pitched sound.
2. 798,73 Hz
3. 51 $m{\cdot}s^{-1}$
4. The speed of sound in air at -35 degrees celsius is309 $m{\cdot}s^{-1}$
5. The approach speed is 68 $m{\cdot}s^{-1}$.

   The frequency observed when moving away is 349,7 Hz.

   A speed of 68 $m{\cdot}s^{-1}$ or equivalently, 244,8 $km{\cdot}h^{-1}$, could be reached if the observer was in a high-speed train or a racing car.

Exercise 6 – 2:

1. a) The Doppler effect occurs when a source of waves and/or an observer move relative to each other, resulting in the observer measuring a different frequency of the waves than the frequency at which the source is emitting.
   b) Redshift is the shift in the position of spectral lines to longer wavelengths due to the relative motion of a source away from the observer.
   c) Ultrasound or ultra-sonic waves are sound waves with a frequency greater than 20 000 Hz (or 20 kHz).
2. a) 548,4 Hz
   b) 459,5 Hz
   c)
3. a) 15,5 $m{\cdot}s^{-1}$
   b) As the truck goes by, the frequency goes from higher to lower and the wavelength of the sound waves goes from shorter to longer.
4. 405,7 Hz
5. An instrument called a Doppler flow meter can be used to transmit ultrasonic waves into a person's body. The sound waves will be reflected by tissue, bone, blood etc., and measured by the flow meter. If blood flow is being measured in an artery for example, the moving blood cells will reflect the transmitted wave and due to the movement of the cells, the reflected sound waves will be Doppler shifted to higher frequency if the blood is moving towards the flow meter and to lower frequency if the blood is moving away from the flow meter.
6. 436,8 Hz

7 Rate and Extent of Reaction

Exercise 7 – 2: Reaction rates

1. b) 5
   c) 3
2. a) $1{,}99 \times 10^{-5}$ $mol.s^{-1}$
   b) $9{,}89 \times 10^{-6}$ $mol.s^{-1}$
   c) $2{,}01 \times 10^{-5}$ $mol.s^{-1}$
3. a) 0,021 mol
   b) 0,00035 $mol.s^{-1}$
   c) i. 0,021 mol
      ii. 0,48 g

Exercise 7 – 3: Rates of reaction

1. b) 1 minute
   d) 79 cm^3
   e) 15 - 25 min
   f) i. increase

Exercise 7 – 4: Reaction rates

1. a) False
   b) True
   c) True
3. b) i. decrease
      ii. increase

Exercise 7 – 5:

2. c)
4. b)
5. a) i. $7{,}5 \times 10^{-4}$ $mol.s^{-1}$
      ii. $7{,}5 \times 10^{-4}$ $mol.s^{-1}$
   b) i. $1{,}39 \times 10^{-6}$ $mol.s^{-1}$
      ii. $2{,}22 \times 10^{-6}$ $mol.s^{-1}$
      iii. $1{,}6 \times 10^{-6}$ $mol.s^{-1}$
7. e) 1,31 g
   f) 0,022 $g.s^{-1}$

8 Chemical equilibrium

Exercise 8 – 1: Chemical equilibrium

1. a) ii)
   b) i)
2. a) $A + B \rightarrow C + D$
   b) $C + D \rightarrow A + B$
   c) reversible

Exercise 8 – 2: The equilibrium constant

2. 0,28 $mol.dm^{-3}$
3. 2,25

Exercise 8 – 3: Equilibrium

1. b)
2. a) No
   b) lighter
   c) 67,19
3. a) 0,5 mol
   c) Exothermic
   d) decrease
4. a) 5 minutes
   c) 0,44
   g) endothermic
   h) i. no effect
      ii. no effect
      iii. increase

Exercise 8 – 4: Applying equilibrium principles

1. a) decreases
   b) reverse
   c) 25 °C

Exercise 8 – 5:

2. a)
   b) will change to blue
3. a) change to pink

9 Acids and bases

Exercise 9 – 1: Acids and bases

2. a) HCO_3^-
   b) H_2O
   c) HPO_4^{2-}
3. a) H_2O
   b) HCN
   c) HCO_3^-

Exercise 9 – 3: Types of acids and bases

1. a) Concentrated
   b) Dilute
2. a) strong
   b) weakly acidic
   c) dilute
3. a) 14,16 $mol.dm^{-3}$
   b) 0,186 $mol.dm^{-3}$
   c) 0,07 $mol.dm^{-3}$
   d) 4,52 $mol.dm^{-3}$

Exercise 9 – 4: Acids and bases

1. a) Weak acid
   b) Strong base
   c) Weak base
   d) Strong acid
2. a) Weak acid
   b) Weak base
   c) Strong acid

Exercise 9 – 5: Calculating pH

1. a) 13.3
   b) 0.31
2. 1×10^{-12} $mol.dm^{-3}$
3. a) basic
   b) 8

Exercise 9 – 6: Acids and bases

1. a) 9,00 g
   b) 0,46 $mol.dm^{-3}$

Exercise 9 – 7:

2. a) 0,019 mol
   b) $1,14 \times 10^{22}$ atoms
4. 0,26 dm^3
5. b) 3,37 g
   c) 13,3
   e) 0,075 $mol.dm^{-3}$

10 Electric circuits

Exercise 10 – 1: Series and parallel networks

1. a) 0
   b) 23.1
   c) 0.873
   d) 0.521 0.909 0.497
2. a) 2,5 A
   b) 3,75 V
   c) 1,25 A
3. 6 Ω; 2 Ω ; 12 Ω; 7,33 Ω
8. a) 0
   b) 14.4
   c) 0.350
   d) 0.242 0.317 0.292

Exercise 10 – 2:

1. So internal resistance is a measure of the resistance of the material that the battery is made of.
2. The emf of a battery is essentially constant because it only depends on the chemical reaction (that converts chemical energy into electrical energy) going on inside the battery. Therefore, we can see that the voltage across the terminals of the battery is dependent on the current drawn by the load. The higher the current, the lower the voltage across the terminals, because the emf is constant. By the same reasoning, the voltage only equals the emf when the current is very small.
3. 0,4 Ω
5. The emf is 10,0 V and the internal resistance is $1{,}1428 \times 10^{-2}$ Ω.
6. The emf is 21,11 V and the internal resistance is 0,55 Ω.
7. The emf is 1,5 V and the internal resistance is 0,6 Ω.

Exercise 10 – 3:

1. 20 Ω
4. 0,09 A; 2,25 V; 11,25 V ; 13,50 V
5. $9{,}6 \times 10^{-17}$ J
6. a) 0
   b) 20.8
   c) 1.717
   d) 2.521 1.790 1.571

11 Electrodynamics

Exercise 11 – 1: Generators and motors

1. An electrical generator is a mechanical device to convert energy from a source into electrical energy.

   An electrical motor is a mechanical device to convert electrical energy from a source into another form energy.
2. Faraday's law says that a changing magnetic flux can induce an emf, when the coil rotates in a magnetic field it is possible for the rotation to change the flux thereby inducing an emf.

   If the rotation of the coil is such that the flux doesn't change, i.e. the surface of the coil remains parallel to the magenetic field, then there will be no induced emf.
5. A current-carrying coil in a magnetic field experiences a force on both sides of the coil that are not parallel to the magnetics field, creating a twisting force (called a torque) which makes it turn. Any coil carrying current can feel a force in a magnetic field. The force is due to the magnetic component of the Lorentz force on the moving charges in the conductor, called Ampere's Law. The force on opposite sides of the coil will be in opposite directions because the charges are moving in opposite directions.
7. Cars (both AC and DC), electricity generation (AC only), anywhere where a power supply is needed.
8. Pumps, fans, appliances, power tools, household appliances, office equipment.

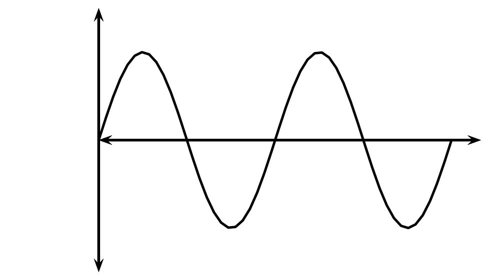
Exercise 11 – 2: Alternating current

1. - Easy to be transformed (step up or step down using a transformer).
   - Easier to convert from AC to DC than from DC to AC.
   - Easier to generate.
   - It can be transmitted at high voltage and low current over long distances with less energy lost.
   - High frequency used in AC makes it suitable for motors.
2. The correct answer is C.
3.
$$\begin{aligned} i &= I_{\max} \sin(2\pi f t + \phi) \\ v &= V_{\max} \sin(2\pi f t) \end{aligned}$$
4. The root mean square is the value that we use for AC and is what it's DC equivalent would be.

$$\begin{aligned} I_{rms} &= \frac{I_{\max}}{\sqrt{2}} \\ V_{rms} &= \frac{V_{\max}}{\sqrt{2}} \end{aligned}$$

5. In South Africa the frequency is 50 Hz
6.
$$\begin{aligned} V_{rms} &= \frac{V_{\max}}{\sqrt{2}} \\ &= \frac{340}{\sqrt{2}} \\ &= 240{,}42 \text{ V} \end{aligned}$$
7. 7.07
8. 188,09 V

9. The graph is the same for both voltage and for current:



Exercise 11 – 3:

1. Direct current (DC), which is electricity flowing in a constant direction. DC is the kind of electricity made by a battery, with definite positive and negative terminals. However, we have seen that the electricity produced by some generators alternates and is therefore known as alternating current (AC). So the main difference is that in AC the movement of electric charge periodically reverses direction while in DC the flow of electric charge is only in one direction.
3. While DC motors need brushes to make electrical contact with moving coils of wire, AC motors do not. The problems involved with making and breaking electrical contact with a moving coil are sparking and heat, especially if the motor is turning at high speed. If the atmosphere surrounding the machine contains flammable or explosive vapours, the practical problems of spark-producing brush contacts are even greater.
4. Instead of rotating the loops through a magnetic field to create electricity, as is done in a generator, a current is sent through the wires, creating electromagnets. The outer magnets will then repel the electromagnets and rotate the shaft as an electric motor. If the current is DC, split-ring commutators are required to create a DC motor.
5. In South Africa the frequency is 50 Hz
6. Mr Smith is not correct. He has misunderstood what power is and what Eskom is charging him for.

   AC voltage and current can be described as:

$$\begin{aligned} i &= I_{\text{max}} \sin(2\pi f t + \phi) \\ v &= V_{\text{max}} \sin(2\pi f t) \end{aligned}$$

   This means that for $\phi = 0$, i.e. if resistances have no complex component or if a student uses a standard resistor, the voltage and current waveforms are in-sync.

   Power can be calculated as $P = VI$. If there is no phase shift, i.e. if resistances have no complex component or if a student uses a standard resistor then power is always positive since:

   - when the voltage is negative ($-$), the current is negative ($-$), resulting in positive ($+$) power.
   - when the voltage is positive ($+$), the current is positive ($+$), resulting in positive ($+$) power.
7. $20{,}38 \times 10^{6}$ W

12 Optical phenomena and properties of matter

Exercise 12 – 1: The photoelectric effect

1. The photoelectric effect its the process whereby an electron is emitted by a metal when light shines on it, on the condition that the energy of the photons (light energy packets) are greater than or equal to the work function of the metal.
2. Two reasons why the observation of the photoelectric effect was significant are (1) that it provides evidence for the particle nature of light and (2) that it opened up a new branch for technological advancement e.g. photocathodes (like in old TVs) and night vision devices.
3. $6{,}25 \times 10^{-22}$ J

13 Electrochemical reactions

Exercise 13 – 1: Oxidation and reduction

2. a) oxidised
   b) reduced
   c) reduced
   d) oxidised
   e) reduced

Exercise 13 – 3: Electrochemical reactions

2. **d)** $Cr_2O_7^{2-}$

Exercise 13 – 4: Galvanic and electrolytic cells

1. a) Electrolysis
   b) Yes

Exercise 13 – 5: Galvanic cells

1. c) Fe^{2+}(aq)
   d) Fe(s)

Exercise 13 – 6: Table of standard electrode potentials

1. a) $-2,37$ V
   b) $-0,13$ V
   c) $-0,25$ V
2. a) lithium
   b) cobalt(III)
   c) lithium
   d) Reduction
   e) Reduction
4. a) Chlorine
   b) Calcium

Exercise 13 – 7: Using standard electrode potentials

1. No
2. No.
3. Yes.

Exercise 13 – 8: Standard electrode potentials

2. a) $+1,18$ V
   b) $-0,4$ V
   c) $+1,08$ V
   d) $+0,91$ V
3. d) $+0,75$ V

Exercise 13 – 9: Predicting spontaneity

1. not spontaneous
2. b) spontaneous
3. a) not spontaneous
   b) not spontaneous
   c) spontaneous
   d) spontaneous
4. a) cannot be stored
   b) cannot be stored
   c) can be stored
   d) cannot be stored

Exercise 13 – 10: The chloralkali industry

1. a) Water and dilute NaOH
   b) Chlorine gas
3. a) sodium chloride
   c) sodium mercury amalgam
   d) sodium mercury amalgam and water
4. d) Sodium ions

Exercise 13 – 11:

1. a) Galvanic cell
   b) Reduction
   c) No
2. a) False
   b) True
   c) True
3. b) $-0,17$ V
4. a) iii)
   b) **iii)**
   c) **iii)**
   d) **ii)**
6. b) copper
   c) 0,46 V
7. f) $+1,53$ V
8. c) silver

14 The chemical industry

Exercise 14 – 1: The role of fertilisers

1. a) 18:24:6
   b) potassium (K)
   c) No
2. Yes
4. 67,5 g
5. 1,99 mol

Exercise 14 – 2: The industrial production of fertilisers

4. a) ammonium nitrate
   b) Acid-base
5. a) $2H^{+} + 2e^{-} \rightarrow H_2$
   b) Carbon monoxide or CO
   c) Water or H_2O

Exercise 14 – 3:

1. b) Inorganic fertilisers
   c) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$
   d) Ostwald process
   e) HNO_3
   f) **C** - $(NH_4)_2SO_4$
      D - NH_4NO_3
3. a) Fractional distillation of air
   b) Ammonia
   c) Haber process
   d) nitrogen dioxide
   e) nitric acid

List of Definitions

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